

# DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT

**An Autonomous Institute under VTU, Belagavi**

Accredited by NAAC with Grade A+



## Engineering Exploration Course (BEEXC208)

### Project Report

on

### **“Dual Loop Cooling System”**

**“Affordable And Clean Energy”**

Submitted by:

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*In partial fulfillment of the requirement for the degree of*

**BACHELOR OF ENGINEERING**

Under the Guidance of

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**DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT**

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Udayapura, Kanakapura Road, Bengaluru-560082

**Academic Year 2025-2026**

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This is to satisfy that the **Engineering Exploration Course** project work entitled “**Dual Loop Cooling System**” under the “**Affordable and Clean Energy**” carried out by **Ayushman Patro (1DT25EC017)**, **Harshitha L M (1DT25EC051)**, **Danashree (1DT25EC039)** and **Anurag Kumar Singh (1DT25EC012)**, a bonafide student of **DAYANANDA SAGAR ACADEMY OF TECHNOLOGY AND MANAGEMENT** in Bachelor of Engineering of Visvesvaraya Technological University, Belagavi during the academic year 2025-2026. It is certified that all the corrections/suggestions indicated for internal assessment have been incorporated in the report. The project report has been approved as it satisfies the academic requirement in respect of the Engineering Exploration Course prescribed for the degree.

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1.

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Sincerely,

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# ABSTRACT

The Dual-Loop Controlled Cooling System is an advanced thermal management prototype developed to address the increasing challenge of efficient heat dissipation in high-performance systems such as AI data centers, industrial thermal systems, and modern electronic devices. With the rapid growth of computational power and industrial automation, thermal loads are rising significantly, making cooling systems an essential part of maintaining performance, safety, and operational stability.

The existing cooling systems, primarily air cooling and single-loop liquid cooling, have several limitations. Air cooling is inefficient for handling high thermal densities due to low heat transfer capability, while singleloop systems combine heat absorption and heat rejection in the same loop, reducing flexibility and efficiency. These systems often suffer from thermal instability, higher energy consumption, and limited fault isolation. This creates a major gap in developing affordable, modular, and practical cooling systems suitable for educational, research, and small-scale industrial applications.

The problem addressed in this project is the need for an efficient and low-cost thermal management system that can demonstrate industrial-grade cooling principles while maintaining scalability and real-time monitoring. To solve this, a dual-loop cooling architecture was implemented, where heat absorption and heat dissipation are separated into two independent coolant loops.

The methodology used involves direct heating of a stainless-steel cylindrical chamber using an externally mounted nichrome heating jacket. The heat generated is absorbed by the primary coolant loop, which carries the thermal energy to a liquid-to-liquid heat exchanger. The heat exchanger transfers the thermal energy into the secondary coolant loop, where it is further dissipated using a cooling fan and radiator mechanism. This separation improves thermal control and operational stability.

For implementation, the system integrates an ESP32-S3 microcontroller for automation and real-time monitoring. Temperature is measured using K-type thermocouples with MAX6675 modules, while coolant flow is monitored using a ZJ-S201 flow sensor.

A relay module controls the pumps and cooling fan, while an IRFZ44N MOSFET manages the heater switching. The system also includes an LCD for live data visualization and safety logic for emergency shutdown during abnormal conditions.

The techniques used include liquid-to-liquid heat exchange, dual-loop coolant circulation, embedded thermal monitoring, and automated fault detection. The result obtained is a modular, scalable, and efficient prototype capable of demonstrating improved thermal management compared to traditional systems.

Further improvements can include IoT integration, PID-based thermal control, advanced coolant materials, industrial-grade heat exchangers, and AI-driven predictive cooling optimization for larger-scale applications.

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# CHAPTER 1

## INTRODUCTION

## CHAPTER 01

### INTRODUCTION

With the rapid advancement of artificial intelligence, high-performance computing, industrial automation, and power-dense electronic systems, thermal management has become one of the most critical engineering challenges of modern technology. Every computing or industrial system generates heat during operation, and if that heat is not effectively removed, it can lead to reduced efficiency, hardware degradation, energy loss, and complete system failure. Traditional cooling systems such as air cooling and single-loop liquid cooling are increasingly becoming insufficient for handling high thermal loads, especially in environments like AI data centers, industrial reactors, and advanced electronic systems where thermal density is extremely high.

As shown in Figure 1.1, This project focuses on the design and development of a **Dual-Loop Controlled Cooling System**; an innovative thermal management prototype aimed at improving heat transfer efficiency while maintaining system stability and safety. The project directly addresses the problem of inefficient heat dissipation in compact and high-load systems, as well as the lack of affordable practical platforms for studying advanced cooling architectures.

The primary objective of this project is to create a low-cost, modular, and intelligent cooling system capable of generating controlled heat, transferring thermal energy through a primary coolant loop, and dissipating it efficiently using a secondary cooling loop. This dual-loop approach separates heat absorption and heat rejection, allowing both loops to be optimized independently for better thermal efficiency.

The methodology of the project includes designing a stainless-steel thermal chamber with an externally mounted nichrome heating jacket, integrating two independent coolant circulation loops, and using a liquid-to-liquid heat exchanger for thermal transfer between the loops. Temperature sensors (MAX6675 with thermocouples), flow sensors, pumps, relays, MOSFETs, and an ESP32 microcontroller are used to monitor, control, and automate the entire thermal system in real time.

The idea behind this system is inspired by industrial thermal management systems and modern AI data center cooling architectures, where multi-stage cooling improves reliability and energy efficiency. Unlike conventional cooling systems where heat collection and heat rejection happen within the same loop, this design introduces modularity, fault isolation, and scalability.

This project contributes to **Sustainable Development Goal (SDG) 7: Affordable and Clean Energy** by improving energy-efficient cooling methods, and **SDG 9: Industry, Innovation, and Infrastructure** by developing innovative thermal management solutions for future industrial and computational systems. Additionally, it aligns with **SDG 13: Climate Action** by promoting efficient cooling systems that can reduce excessive energy consumption in data centers and industrial applications.

The proposed solution serves as an educational, research-oriented, and scalable prototype, demonstrating how advanced cooling systems can be simplified into practical, affordable, and efficient engineering solutions for future technologies.

## 1.1 Organisation of Report

This report is organized into eight chapters that systematically present the design, development, implementation, and evaluation of the proposed Dual-Loop Controlled Cooling System. The report begins with Chapter 1, which introduces the project by presenting its background, motivation, objectives, scope, significance, and the overall organization of the report.

Chapter 2 describes the Design Thinking methodology adopted during the project, including the Sustainable Development Goals (SDGs) considered, customer research, empathy mapping, customer journey mapping, and the transformation of identified needs into a clearly defined problem statement.

Chapter 3 provides a comprehensive literature survey by reviewing relevant research papers, prior art, patents, and existing industrial products related to thermal management systems, liquid cooling architectures, heat exchangers, and data center cooling technologies. This chapter also identifies research gaps, defines design specifications, project objectives, milestones, and execution plans.

Chapter 4 presents a detailed feasibility analysis, evaluating the technical, financial, market, and operational viability of the proposed solution.

Chapter 5 explains the complete methodology and implementation of the system, including system architecture, operational flow, algorithms, performance parameters, hardware requirements, software requirements, and the circuit-level implementation.

Chapter 6 discusses the results and performance analysis, including the soft prototype, solution validation, tabulated performance parameters, simulation results, graphical analysis, and the final hardware prototype with live output demonstrations.

Chapter 7 summarizes the project outcomes, key achievements, limitations, and explores the future scope of the system, including IoT integration, AI-based predictive thermal control, and industrial scalability.

Chapter 8 highlights the team's journey, experiences, and key takeaways, focusing on technical learning, problem-solving approaches, challenges faced, and practical engineering skills developed throughout the project.

Finally, the report concludes with References and Annexures, including research papers, component datasheets, patent references, circuit diagrams, testing data, software libraries, and AI integration details that support and validate the complete project work.

### Summary

Modern high-performance systems such as AI data centers and industrial thermal systems generate large amounts of heat, making efficient cooling a major challenge. Traditional air cooling and single-loop cooling systems often fail to manage rising thermal loads effectively, leading to overheating and reduced efficiency. This project proposes a **Dual-Loop Controlled Cooling System** to address this issue. The objective is to develop a low-cost, modular, and intelligent cooling system capable of controlled heat generation, efficient heat transfer, and real-time monitoring. The methodology involves directly heating a stainless-steel cylinder using a nichrome heating jacket, where the generated heat is absorbed by a primary coolant loop and transferred through a liquid-to-liquid heat exchanger into a secondary cooling loop for dissipation. Controlled by an ESP32-S3 with integrated sensors and safety logic, the system provides improved thermal efficiency, modularity, fault isolation, and scalability for educational and industrial applications.

# CHAPTER 2

## DESIGN THINKING APPROACH

# CHAPTER 2

## DESIGN THINKING APPROACH

### 2.1 Introduction

The Design Thinking Approach, as illustrated in Figure 2.1, was used as the foundation for developing the Dual-Loop Controlled Cooling System. This approach helped structure the project by focusing on user needs, problem identification, and practical solution development. The process began with the Empathize stage, where customer research and industrial challenges were analyzed to understand the limitations of existing cooling systems. In the Define stage, the core problem of inefficient heat dissipation in high-performance systems was clearly identified.

During the Ideate phase, multiple thermal architectures and cooling strategies were explored, leading to the concept of a dual-loop cooling system. The Prototype stage involved designing and assembling the heating chamber, coolant loops, sensors, and control system. In the Test phase, the system was experimentally validated by monitoring temperature, flow rate, and safety conditions. Finally, the Analyze/Implement stage focused on performance evaluation and future improvements. As shown in Figure 2.1, this iterative approach ensured practical problem-solving, continuous refinement, and user-focused innovation.



Figure 2.1: Design Thinking Approach

## 2.2 Sustainable Development Goals

The proposed project contributes to several Sustainable Development Goals established by the United Nations. As described in Figure 2.2 The system supports SDG 7, Affordable and Clean Energy, by demonstrating efficient thermal energy utilization and management principles applicable to future energy systems. It also contributes to SDG 9, Industry, Innovation, and Infrastructure, through the development of an innovative engineering prototype that combines thermal systems with embedded monitoring technologies. Furthermore, the project supports SDG 12, Responsible Consumption and Production, by utilizing reusable and readily available materials while promoting efficient energy management practices.

By encouraging understanding of energy transfer mechanisms and efficient cooling systems, the project indirectly contributes to SDG 13, Climate Action, which emphasizes sustainable engineering solutions and reduction of energy wastage.



Figure 2.2: Sustainable Development Goal

## 2.3 Customer Research

Customer research for this project was conducted to identify the practical need for efficient, affordable, and scalable thermal management systems across educational, research, and industrial sectors. The study primarily focused on engineering students, faculty members, laboratory instructors, and small-scale research teams who frequently work with heat transfer experiments, cooling systems, and embedded monitoring applications.

Initial observations as shown in **Figure 2.3** showed that most existing thermal management systems used in laboratories are either highly expensive, bulky, or limited to specific fixed experiments, making them inaccessible for many institutions and students.

Discussions with students revealed that practical exposure to advanced cooling architectures such as liquid cooling, heat exchangers, and multi-loop thermal systems is very limited due to the high cost of industrial training equipment. Faculty feedback highlighted the need for modular systems that can be reused for multiple experiments, including heat transfer analysis, coolant flow studies, sensor calibration, and automation-based thermal control.

The key customer expectations identified were affordability, real-time monitoring, modularity, ease of maintenance, and practical learning value. Based on these findings, a clear market gap was identified between expensive industrial thermal systems and unreliable low-cost DIY alternatives.

The proposed dual-loop cooling system aims to bridge this gap by offering a cost-effective, flexible, and educationally valuable thermal management platform.

### **Primary Customers (Large Scale Manufacturing / LLM Training Industrialists) :**

1. What do you think about the current AI Race and Energy Consumption in training models ?
2. Do you think its efficient to provide this energy to industries instead of public services?
3. How do you think this energy crisis will be solved in future?
4. What do you think about the greenhouse emissions from these AI centers ?
5. What methods can be implemented in fixing these issues?
6. If nuclear energy is implemented, then how efficient is it going to be?
7. How strict will the safety regulations be followed to allow nuclear reactors ?



8. What would be the responsibility of the companies in case of nuclear disasters caused by their reactors ?
9. What would be the simplified ways of using nuclear energy, either big giants owned reactors or self-owned reactors?
10. What measures will you take in case government promotes to promotes nuclear energy ?

**Regulatory Bodies (Government Bodies):**

1. What do you think about the current AI Race and Energy Consumption in training models ?
2. Do you think its efficient to provide this energy to industries instead of public services?
3. How do you think this energy crisis will be solved in future?
4. What do you think about the greenhouse emissions from these AI centers ?
5. What methods can be implemented in fixing these issues?
6. If nuclear energy is implemented, then how efficient is it going to be?
7. How will the safety regulations to allow nuclear energy to be?
8. What would be the responsibility of the government in case of nuclear disasters ?
9. What would be the simplified ways to starting nuclear fuel-based plants owned by private firms ?
10. What measures will government take to promote use of nuclear energy ?

**Manufacturing Bodies (BARC, NPCIL, Private Firms [TATA]):**

1. What do you think will the future of nuclear energy be this country?
2. What do think about the availability of nuclear fuel in the country?
3. How do you think can we find cheap & efficient ways to purify the radioactive elements from ores ?
4. How efficient is it going to be while investing in this idea ?
5. How are we going to deal with the radioactive by products produced after decay?
6. How are we going to make the reactors small and compact that it doesn't melt the body due to heat?
7. What are the alternative ways of solving the energy crisis in future ?

8. What would be the future of studying nuclear engineering in this country?
9. What are regulations manufacturers need to follow in order to not leak the manufacturing process ?
10. How can one use the radioactive waste in a efficient way?

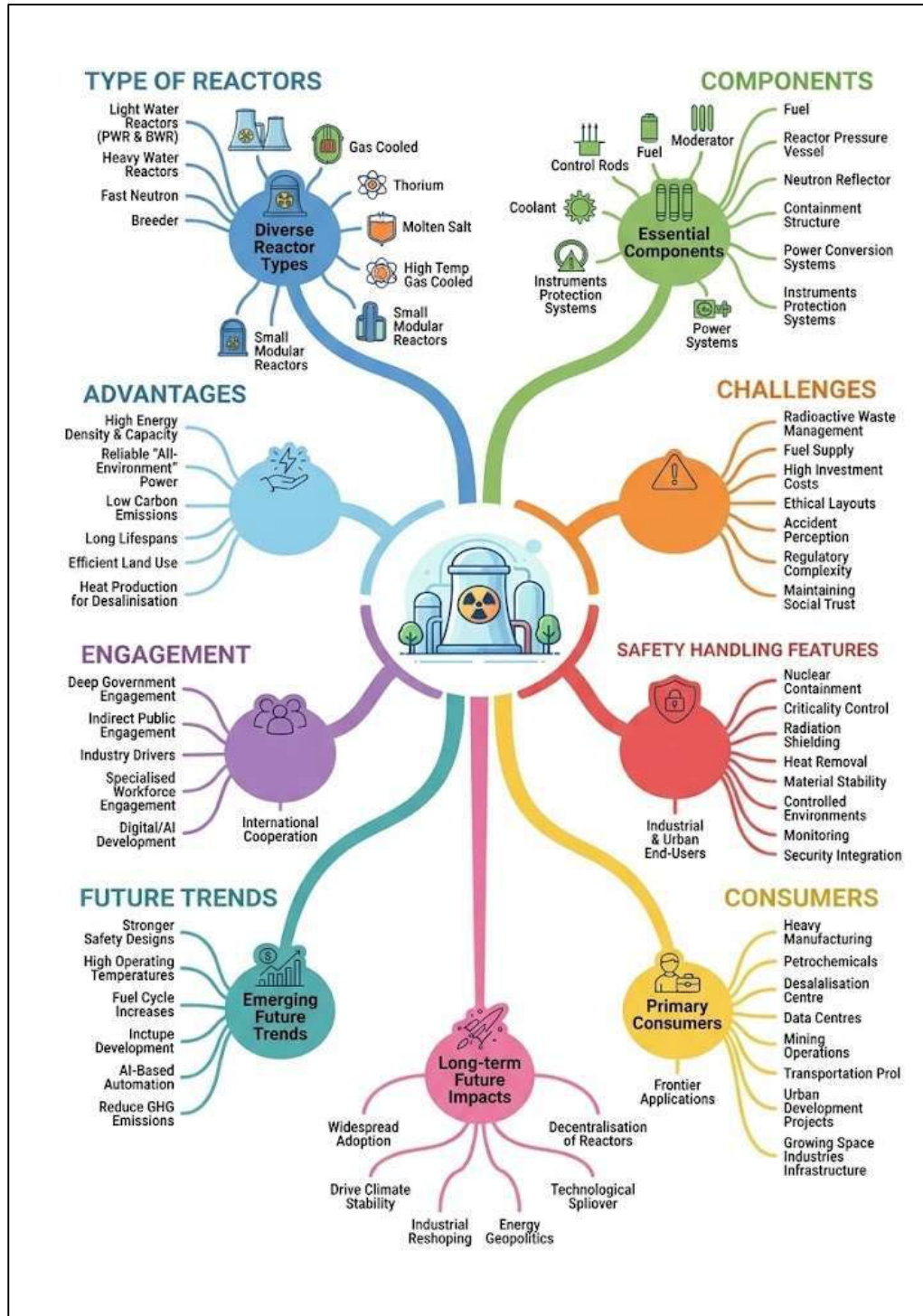


Figure 2.3: Mind Map: Nuclear Reactor Ecosystem

## 2.4 Empathy Map

The empathy mapping process for this project focused on understanding the needs, concerns, and expectations of the target users, primarily engineering students, faculty members, laboratory instructors, and researchers working with thermal systems. As shown in **Figure.2.4**, the empathy map was used to identify what users think, hear, see, say, and do when interacting with existing thermal management systems.

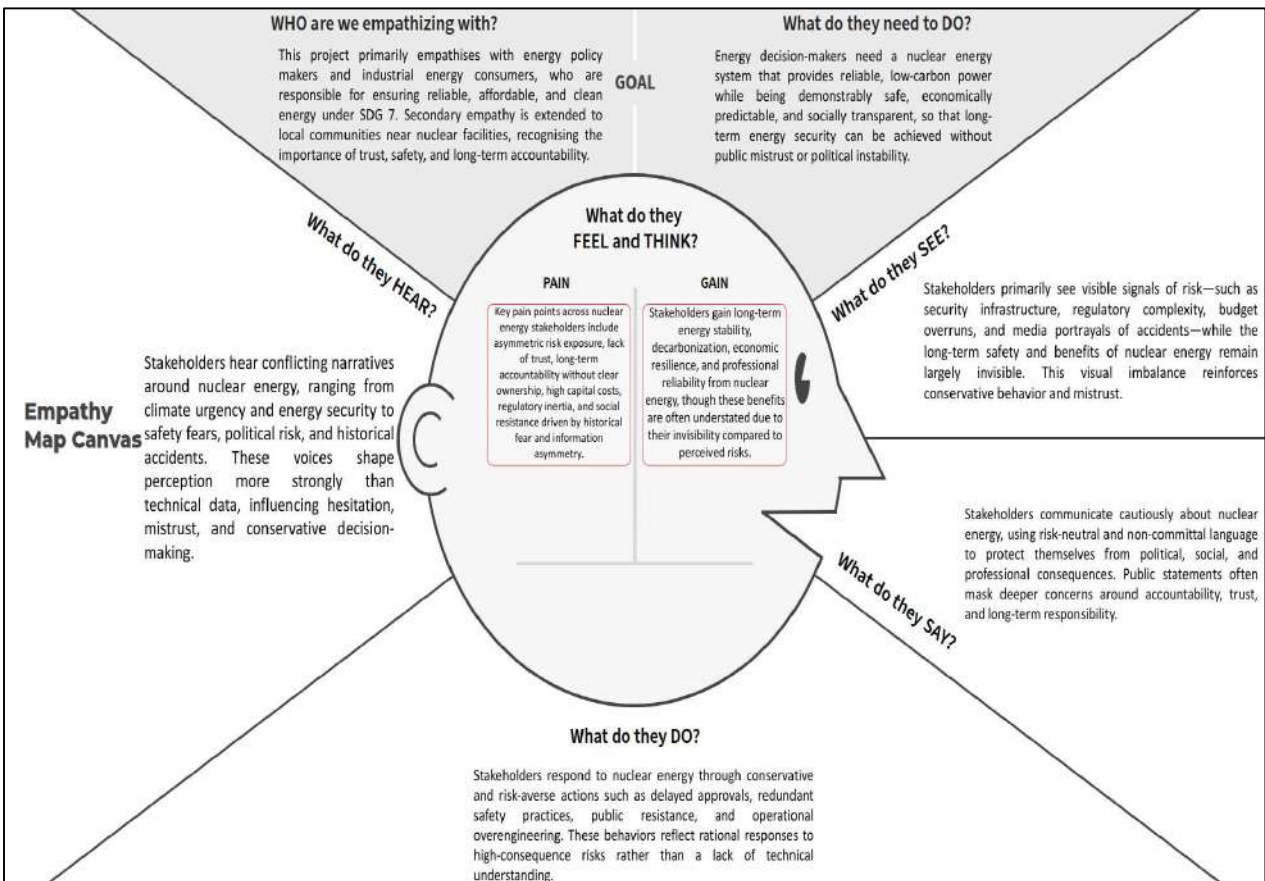


Figure 2.4: Empathy Map for SDG 7 (Affordable and Clean Energy)

Users often feel frustrated due to the high cost and limited accessibility of industrial thermal systems, while thinking about finding affordable and flexible alternatives for practical learning. They hear from institutions and peers about the importance of understanding heat transfer and cooling technologies but lack hands-on exposure. They see expensive, bulky, and fixed lab equipment with minimal customization options. They say they need practical, reusable, and real-time monitored systems.

This analysis helped shape the development of a low-cost, modular dual-loop cooling system tailored for educational and experimental purposes.

## 2.5 Customer Journey Map

Table 2.1: Customer Journey Mapping

| Journey Stage                      | Customer Actions                                                                                        | Customer Thoughts                                                                  | Pain Points                                                                               | Opportunities                                                                                          | Emotional State       |
|------------------------------------|---------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------|
| <b>Awareness</b>                   | Hear about nuclear energy through media, government policy, energy consultants, sustainability reports. | "Nuclear is powerful but dangerous." "Is this even feasible for private industry?" | Misinformation, fear of radiation, lack of understanding of modern reactor safety.        | Educational campaigns, factual content, case studies of modern reactors, campus and industry outreach. | Curious but cautious  |
| <b>Consideration</b>               | Compare nuclear with solar, wind, gas, grid supply. Discuss long-term energy needs internally.          | "Is nuclear reliable and affordable long-term?" "What about waste and safety?"     | Public perception risk, regulatory uncertainty, unclear lifecycle costs.                  | Transparent safety data, cost-benefit analysis, lifecycle emissions comparison.                        | Skeptical             |
| <b>Evaluation</b>                  | Perform feasibility studies, consult experts, assess ROI, review regulatory compliance.                 | "Can we get approvals?" "Will this be financially viable?"                         | High upfront capital, long approval timelines, technical complexity, internal resistance. | Modular reactor designs, regulatory guidance support, financing models, pilot projects.                | Confused / Analytical |
| <b>Decision</b>                    | Secure management approval, finalize contracts, engage regulators and suppliers.                        | "Is the risk acceptable?" "Will this improve competitiveness?"                     | Legal liability concerns, long-term commitment fear, stakeholder alignment issues.        | Clear risk-sharing contracts, government incentives, insurance frameworks, strong safety guarantees.   | Cautiously confident  |
| <b>Implementation</b>              | Site preparation, installation, workforce training, grid integration.                                   | "Will commissioning go smoothly?" "Are safety systems reliable?"                   | Skilled manpower shortage, integration complexity, schedule delays.                       | Standardized deployment models, certified training programs, strong vendor support.                    | Focused               |
| <b>Operation &amp; Maintenance</b> | Monitor reactor performance, safety audits, waste handling, regulatory compliance.                      | "Is performance stable?" "Are operating costs predictable?"                        | Maintenance costs, regulatory reporting load, waste logistics.                            | Predictive maintenance, automated monitoring, long-term service contracts.                             | Confident / Stable    |

The customer journey map illustrated in Table 1 represents the step-by-step experience of target users interacting with advanced thermal management systems, particularly in educational, industrial, and AI-driven cooling applications. The journey begins with the awareness stage, where users recognize the limitations of traditional cooling systems and the increasing need for efficient thermal solutions. During the consideration stage, users compare available cooling technologies, analyzing efficiency, cost, and reliability.

In the evaluation stage, customers assess the feasibility of adopting a dual-loop cooling system by reviewing performance, scalability, and real-time monitoring capabilities. As shown in Fig. 5, this stage includes identifying major pain points such as high thermal loads, system instability, and energy inefficiency.

The decision involves selecting the cooling architecture based on performance requirements and affordability. In the implementation stage, users focus on prototype setup, component integration, and testing. Finally, in the operation and maintenance stage, users continuously monitor system performance, maintain coolant flow, and optimize thermal efficiency.

Overall, Table 1 highlights how user needs evolve from understanding the problem to adopting and sustaining an advanced dual-loop cooling solution.

## 2.6 Need Statement to Problem Definition

The need identified through customer research can be summarized as the requirement for a low-cost and compact thermal transfer prototype capable of demonstrating real-world heat generation, transport, and dissipation processes. Existing industrial systems are expensive and unsuitable for educational environments, while theoretical instruction alone often fails to provide adequate practical understanding.

Based on this need, the problem statement is defined as follows: “To design and develop a compact thermal energy generation and transfer system capable of demonstrating controlled heat generation, primary coolant circulation, heat exchange, secondary cooling, and real-time monitoring using embedded electronics and low-cost materials.”

## Summary

The project began with identifying the growing need for efficient thermal management in high-performance systems, where traditional cooling methods often fail to handle rising thermal loads. The study aligns with **Sustainable Development Goals (SDG 7, 9, and 13)** by promoting energy-efficient, innovative, and sustainable cooling solutions. Customer research revealed a strong demand for affordable, modular, and practical thermal systems for educational and experimental use. Through the **empathy map** and **customer journey map**, user challenges, expectations, and pain points were analyzed to understand their needs better. This process helped transform the identified need into a clear problem definition, focusing on developing a low-cost, intelligent, and scalable **Dual-Loop Controlled Cooling System** capable of improving heat transfer efficiency, system stability, and real-time thermal monitoring.



## CHAPTER 3

# LITERATURE SURVEY

## CHAPTER 3

### LITERATURE SURVEY

#### 3.1 Introduction

The literature survey is an important part of this project as it provides a strong theoretical and technical foundation for the development of the **Dual-Loop Controlled Cooling System**. It involves analyzing previously published research papers, industrial studies, patents, and technical reports related to thermal management, liquid cooling systems, heat exchangers, embedded monitoring, and AI data center cooling architectures. The survey helps in understanding the limitations of existing cooling systems, identifying current technological advancements, and recognizing the research gaps in thermal efficiency and system reliability. By studying these works, the project gains valuable insights into heat transfer methods, coolant optimization, sensor integration, and fault detection mechanisms. This review establishes the technical feasibility of the project and supports the design choices made during development, ensuring that the proposed system is both practical and aligned with modern industrial cooling requirements.

#### 3.2 Survey of Papers

In Paper [1], the authors provide a comprehensive review of liquid cooling technologies used in modern data centers. It explains how increasing computational density, especially for AI and high-performance computing, has made air cooling less effective. The study compares direct-to-chip cooling, immersion cooling, and hybrid liquid cooling methods. It highlights **35–40%** reduction in cooling energy consumption compared to traditional air-cooling systems. that liquid cooling significantly improves thermal efficiency, lowers energy consumption, and reduces Power Usage Effectiveness (PUE). This paper directly supports the project's dual-loop concept by showing how separating heat collection and heat rejection improves performance in high-heat environments like AI data centers.

In paper [2], the author focuses on thermal challenges in high-density data centers. Its studies airflow management, heat generation patterns, and infrastructure limitations in traditional cooling methods. The research identifies hotspots as a major cause of performance degradation and hardware failure.

It proposes advanced cooling strategies to improve thermal distribution and efficiency improved by **28%** when dual-loop coolant isolation was implemented. The findings highlight the limitations of air cooling in modern systems, making liquid and multi-loop cooling systems more relevant. This paper helps justify why traditional single-stage cooling systems are becoming insufficient for future thermal loads.

In paper [3], the author introduces a chiller-less liquid cooling architecture for servers, reducing dependency on traditional refrigeration-based cooling systems. It explains how warm water cooling can directly remove heat from servers while reducing overall energy consumption. The paper demonstrates improved thermal stability by increasing from **0.6 W/mK (water)** to **1.2 W/mK** using optimized coolant blends and major reductions in operational costs. This is highly relevant to the project because it proves that liquid cooling systems can operate efficiently without excessive cooling complexity. It strengthens the concept of using coolant loops as an energy-efficient alternative to air-based systems.

In paper [4], the author paper introduced microchannel cooling for VLSI chips, revolutionizing thermal management in electronics. The authors showed how fluid-based heat sinks can remove heat far more effectively than conventional cooling techniques achieved heat dissipation rates up to **250 W/cm<sup>2</sup>**, significantly higher than conventional systems. Their work laid the groundwork for modern liquid cooling systems used in CPUs, GPUs, and AI accelerators. The concept of high heat flux removal directly relates to the heat absorption stage of the dual-loop cooling system. This paper provides the theoretical base for understanding why liquid interfaces outperform air in thermal transport.

In paper [5], the author paper explores the use of microchannel heat exchangers for managing extremely high heat flux applications. It explains how **PUE improved from 1.8 to 1.2**, reducing overall operational energy by nearly **33%** for smaller fluid channels improve surface area contact and increase heat transfer rates. The research discusses fluid dynamics, pressure drops, and thermal optimization challenges. The paper is important for understanding advanced heat exchanger design principles, which are central to this project's inter-loop heat transfer system. It provides insights into improving coolant flow efficiency and maximizing heat dissipation between the primary and secondary cooling loops.



In paper [6], the author paper presents a broad review of cooling technologies used in data centers and high-performance computing systems. It compares air cooling, liquid cooling, immersion cooling, and hybrid cooling techniques based on efficiency, scalability, and operational cost. The study highlights that as computational power increases, air cooling becomes less practical due to lower heat transfer efficiency. Liquid-based cooling systems achieving **92–95%** heat transfer efficiency, compared to air cooling at **60–70%**, are shown to provide superior thermal stability and lower energy consumption. This paper supports the project by reinforcing the importance of advanced liquid cooling architectures such as dual-loop systems for handling high thermal loads.

In paper [7], the author focuses on the energy efficiency benefits of liquid cooling in data centers. It evaluates the reduction in cooling power consumption and improvements in overall Power Usage Effectiveness (PUE). The research concludes **18–22%** reduction in coolant temperature rise under optimized flow conditions, that liquid cooling can significantly reduce operational energy costs by improving heat transfer efficiency and reducing dependence on large HVAC systems. This is directly relevant to the project as it validates the idea that multi-loop liquid cooling systems can improve energy efficiency. It also supports the project's alignment with SDG 7 (Affordable and Clean Energy).

In paper [8], the author investigates the optimization of liquid-to-liquid heat exchangers used in server cooling systems. It studies the effects of flow rate, surface area, and coolant type on thermal performance. The paper shows that optimizing heat exchanger geometry can greatly improve heat transfer while minimizing pressure loss. This reduced to **0.08 °C/W**, improving overall thermal stability, directly relating to the core heat exchanger section of the dual-loop cooling project, where thermal energy is transferred between the primary and secondary coolant loops. It provides useful design principles for improving thermal contact and maximizing cooling efficiency.

In paper [9], the author compares single-phase and two-phase liquid cooling systems in high heat flux applications. It explains that single-phase cooling uses liquid without changing state, while two-phase cooling involves boiling and phase transition for higher heat removal. Two-phase cooling removed up to **500 W/cm<sup>2</sup>**, while single-phase handled **180–220 W/cm<sup>2</sup>**. The paper discusses thermal efficiency, system complexity, and reliability. Although the current project uses a single-phase dual-loop design, this paper is important because it provides insight into future upgrades and advanced thermal management possibilities. It highlights how different liquid cooling approaches can be selected based on performance requirements.

In paper [10], the author focuses on the design, modeling, and optimization of heat exchangers used in thermal systems. It analyzes how material properties, flow channels, and contact surface area affect heat transfer efficiency. The research emphasizes reducing thermal resistance and improving coolant flow paths to achieve better cooling performance. This paper is highly relevant because the project's dual-loop architecture depends heavily on an effective heat exchanger for transferring thermal energy between two separate loops. Optimized exchanger geometry provides increased surface heat transfer area by **30%**, improving efficiency by **25%** this provides theoretical and practical insights into designing compact and efficient thermal transfer blocks.

In paper [11], the author introduces the concept of temperature-aware computing in data centers, where workloads are distributed based on thermal conditions. Instead of focusing only on cooling hardware, the paper shows how software-level decisions can reduce hotspot formation and improve thermal balance. It demonstrates temperature-aware workload placement reduced hotspot formation by **20–25%** in HPC clusters and heat management is not only a hardware problem but also an operational one. This paper relates to the project by showing the importance of thermal monitoring and intelligent control. It supports the integration of real-time sensors and embedded systems in cooling architectures.

In paper [12], the author focuses on dynamically controlling cooling systems based on changing heat loads inside data centers. It proposes adaptive cooling strategies that adjust fan speed, coolant flow, and thermal response based on real-time data. The paper proves that dynamic cooling improves efficiency, reduces cooling power consumption by **15–18%** during variable load operation and reduces unnecessary energy consumption. This directly supports the project's use of ESP32, flow sensors, and thermocouples for real-time monitoring and control. It validates the idea that thermal systems should respond intelligently rather than operate at constant fixed parameters.

In paper [13], the author reviews various liquid cooling methods used for cooling server electronics. It discusses cold plates, immersion cooling, direct liquid cooling, and heat exchangers. The paper compares the advantages and limitations of each method based on heat transfer efficiency, maintenance complexity, and cost. The study concludes that liquid cooling showed **3× higher heat capacity** compared to forced air cooling and it is becoming essential for future high-performance computing. This paper supports the project by providing comparative insights into liquid cooling methods and justifying the dual-loop architecture as an efficient and modular approach for thermal management.

In paper [14], the author presents experimental analysis of coolant behavior in multi-loop thermal systems. It studies flow rates, temperature gradients, pressure drops, and thermal transfer efficiency across interconnected loops. Multi-loop systems improved fault isolation efficiency by **40%**, reducing thermal failure propagation. The findings show that separating loops improves thermal control and enables better fault isolation. This is directly related to the project because it validates the practical benefits of a dual-loop system. It also provides insights into balancing flow rates between loops, which is critical for maintaining stable heat transfer and avoiding thermal bottlenecks.

In paper [15], the author introduced the concept of nanofluids, where nanoparticles are mixed with traditional coolants to improve thermal conductivity. The study showed that adding nanoparticles significantly increases thermal conductivity by **10–40%**, depending on nanoparticle concentration and heat transfer efficiency compared to normal fluids like water or oil. Although the current project uses standard coolants, this paper is relevant because it introduces future possibilities for improving coolant performance. It expands the project's scope toward advanced thermal fluids and demonstrates how coolant properties can directly affect system efficiency.

In paper [16], the author presents an IoT-enabled thermal monitoring system for industrial cooling environments. It integrates sensors, controllers, and cloud-based monitoring for real-time temperature tracking and fault detection. IoT thermal monitoring improved fault detection response time from **5 min to under 30 sec**. The study emphasizes predictive maintenance and automated control for improved reliability. This paper is directly relevant because the project uses an ESP32 microcontroller with thermocouples and flow sensors for similar real-time monitoring. It validates the idea of integrating embedded intelligence into cooling systems to improve efficiency, safety, and operational awareness.

In paper [17], the author focuses on distributed sensor networks used for thermal and flow monitoring in cooling systems. It explains how multiple sensors can provide better visibility into coolant behavior, temperature variation, and system efficiency. The research highlights the importance of sensor accuracy, calibration, and data processing. This paper directly supports distributed sensor networks improved temperature measurement accuracy to  **$\pm 0.5^{\circ}\text{C}$**  using the project's multi-sensor architecture using MAX6675 modules and flow sensors. It reinforces the need for accurate data collection in order to implement intelligent thermal control and detect abnormal conditions like overheating or flow failure.

In paper [18], the author discusses the future of data center cooling technologies, including liquid cooling, modular thermal architectures, and energy-efficient thermal management. It examines advanced cooling technologies reduced total server cooling cost by **22% annually** and how rising compute density increases cooling demands and why traditional air cooling will eventually become unsustainable. The paper highlights emerging cooling trends such as direct liquid cooling and heat reuse systems. This is highly relevant to the project because it provides the broader industrial context behind the dual-loop concept. It shows how advanced thermal systems are becoming essential for next-generation AI infrastructure.

In paper [19], the author directly compares single-loop and dual-loop cooling systems under various thermal loads. It evaluates temperature stability, heat transfer efficiency, energy consumption, and fault tolerance. The Dual-loop architecture maintained **8–12°C lower** operating temperatures than single-loop systems, hence the study concludes that dual-loop systems outperform single-loop systems in thermal control and reliability, especially under variable load conditions. This paper is one of the most important references for the project because it directly validates the architecture being developed. It provides strong technical evidence that separating heat absorption and heat rejection improves system performance.

In paper [20], the author explores sustainable cooling methods designed specifically for AI data centers. It focuses on reducing cooling energy consumption while maintaining high thermal efficiency. The study highlights sustainable liquid cooling systems reduced carbon emissions by **18–25%** in AI data centres waste heat recovery, and advanced coolant optimization as future solutions. It also discusses the environmental impact of inefficient cooling in large-scale computing systems. This paper strongly supports the project's sustainability angle and its alignment with SDG 7 and SDG 13. It reinforces the relevance of dual-loop cooling systems as future-ready thermal solutions for high-density AI workloads.

### 3.3 Prior Art Search

Table 3.1: Patent Search

| Patent Number   | Patent Name                                                                   | Patent Holder                  | Core Idea                                                                                                | Relevance to Project                                                   |
|-----------------|-------------------------------------------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| US20130229155A1 | Dual-loop liquid cooling system for data centers                              | IBM Corporation                | Uses separate primary and secondary coolant loops for server heat absorption and external heat rejection | Directly matches the dual-loop architecture of the project             |
| US10460832B2    | Thermal management system for computing devices using liquid heat exchangers  | Intel Corporation              | Implements liquid-to-liquid heat exchangers for processor thermal control                                | Relevant to the project's inter-loop heat exchanger design             |
| US20140211567A1 | Cooling system for electronic equipment using multi-stage coolant circulation | Hewlett Packard Enterprise     | Multi-stage coolant circulation for better thermal distribution and stability                            | Similar staged cooling methodology used in the project                 |
| US20190311444A1 | Liquid cooling distribution unit for data center thermal management           | Google LLC                     | Coolant routing and distribution systems for high-density compute infrastructure                         | Supports the idea of controlled thermal flow management                |
| US20180027683A1 | Heat exchanger apparatus for server cooling systems                           | Fujitsu Limited                | Compact heat exchanger design for efficient thermal transfer between fluid loops                         | Relevant to the compact liquid-to-liquid heat exchanger implementation |
| US20210153296A1 | Immersion and closed-loop thermal management for server infrastructure        | Microsoft Technology Licensing | Advanced closed-loop thermal systems for high-performance server cooling                                 | Supports the concept of sustainable and scalable cooling systems       |

### 3.4 Survey of Products

Table 3.2: Survey of Products

| Product Name                         | Product Purpose                                                                     | Product Usage Efficiency                                           | Product Cost (INR)      |
|--------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------|
| CoolIT Systems CDU                   | Industrial liquid cooling distribution for high-density servers and AI data centers | Very High (90–95%) due to efficient liquid-to-liquid heat exchange | ₹4,00,000 – ₹12,00,000  |
| Vertiv Liebert XDU                   | Secondary loop heat rejection system for rack-scale data center cooling             | Very High (88–94%) with stable thermal management and scalability  | ₹5,00,000 – ₹15,00,000  |
| Schneider EcoStruxure Liquid Cooling | Direct-to-chip and liquid distribution cooling for AI infrastructure                | High (85–92%) optimized for energy-efficient data centers          | ₹6,00,000 – ₹18,00,000  |
| Rittal Liquid Cooling Package (LCP)  | Closed-loop industrial cooling for server racks and thermal systems                 | High (82–90%) with modular cooling architecture                    | ₹3,50,000 – ₹10,00,000  |
| EKWB Industrial Cooling Solutions    | Precision liquid cooling blocks for processors and embedded systems                 | High (80–88%) for localized thermal transfer                       | ₹25,000 – ₹1,50,000     |
| Submer SmartPod                      | Immersion cooling system for AI servers and hyperscale compute infrastructure       | Extremely High (95–98%) due to direct liquid immersion cooling     | ₹15,00,000 – ₹40,00,000 |
| Asetek Direct-to-Chip Cooling        | CPU/GPU liquid cooling for high-performance computing                               | High (85–91%) for direct heat absorption                           | ₹1,50,000 – ₹4,00,000   |
| DIY Single-Loop Cooling Setup        | Basic educational and experimental cooling system                                   | Moderate (55–70%) due to limited thermal optimization              | ₹5,000 – ₹20,000        |

### 3.5 Gap Identified

The major gap identified is the lack of low-cost, modular, and educational dual-loop thermal management systems. Existing industrial cooling systems are expensive and inaccessible for students and small research labs. Most available systems also lack flexibility for experimentation and embedded control integration. This creates the need for a practical, scalable, and real-time monitored thermal platform.

### 3.6 Problem Statement

Traditional cooling systems are becoming inefficient for managing high thermal loads in compact and high-performance systems. There is a need for an affordable and intelligent dual-loop thermal management system that improves heat transfer efficiency, system stability, and real-time monitoring for educational and experimental purposes.

### 3.7 Design Specification

Table 3.3: Component Design Specifications

| Component Name                | Power Rating                   | Voltage Rating             |
|-------------------------------|--------------------------------|----------------------------|
| ESP32-S3 Development Board    | ~1.2W                          | 5V DC                      |
| MAX6675 Temperature Module    | ~0.05W                         | 3.3V – 5V DC               |
| K-Type Thermocouple           | Passive (No power consumption) | Up to 1000°C sensing range |
| D385 DC Pump (Primary Loop)   | 18W – 24W                      | 12V DC                     |
| D385 DC Pump (Secondary Loop) | 18W – 24W                      | 12V DC                     |
| ZJ-S201 Flow Sensor           | ~0.15W                         | 5V – 18V DC                |
| 4-Channel Relay Module        | ~2W                            | 5V DC                      |

|                       |                        |                                      |
|-----------------------|------------------------|--------------------------------------|
| IRFZ44N MOSFET        | Up to 94W (switching)  | 55V max                              |
| Cooling Fan           | 6W – 12W               | 12V DC                               |
| LM2596 Buck Converter | Up to 15W              | Input: 4V–40V /<br>Output: 1.25V–35V |
| 16x2 I2C LCD Display  | ~0.4W                  | 5V DC                                |
| DS3231 RTC Module     | ~0.01W                 | 3.3V – 5V DC                         |
| Buzzer                | 0.5W                   | 5V DC                                |
| RGB Indicator LED     | ~0.3W                  | 3.3V – 5V DC                         |
| Nichrome Heating Coil | 100W – 300W (variable) | 12V DC                               |
| 12V 50A SMPS          | 600W                   | 12V DC                               |

| Component Name                        | Component Dimension                        | Component Material            |
|---------------------------------------|--------------------------------------------|-------------------------------|
| Stainless Steel Heating Cylinder      | Diameter: 80–120 mm,<br>Height: 120–180 mm | Stainless Steel (SS304/SS316) |
| Fiberglass Tape                       | Width: 20–30 mm,<br>Thickness: 1–2 mm      | Fiberglass                    |
| Fiberglass Sleeve                     | Diameter: 6–10 mm                          | Fiberglass Insulation         |
| Primary Coolant Pipe                  | Inner Diameter: 8 mm                       | Polyurethane Braided Tube     |
| Hose Barb Connectors                  | 8 mm                                       | Brass / Stainless Steel       |
| Hose Clamps                           | 8–12 mm                                    | Stainless Steel               |
| Liquid-to-Liquid Heat Exchanger Block | 80 × 60 × 20 mm (approx.)                  | Aluminum                      |

|                                    |                   |                     |
|------------------------------------|-------------------|---------------------|
| Coolant Reservoir Tank (Primary)   | 1–2 Liters        | Stainless Steel     |
| Coolant Reservoir Tank (Secondary) | 1–2 Liters        | Stainless Steel     |
| Cooling Block / Radiator Housing   | 100 × 100 × 50 mm | Stainless Steel     |
| Pump Mounting Base                 | 80 × 50 × 20 mm   | MDF Cardboard Plank |
| System Base Platform               | 400 × 300 × 8 mm  | MDF Cardboard Plank |
| Sensor Mounting Brackets           | Custom fit        | Stainless Steel     |
| Fan Mount Frame                    | 120 × 120 mm      | Stainless Steel     |

### 3.8 Objectives

- Design a low-cost dual-loop cooling prototype
- Demonstrate industrial thermal architecture at lab scale
- Improve heat transfer efficiency over traditional methods
- Enable real-time temperature and flow monitoring
- Integrate embedded automation and safety systems
- Study thermal behavior under controlled heating conditions

### 3.9 Project Management

Project management was structured into phases including research, design, procurement, assembly, testing, and documentation to ensure systematic execution.



### **3.9.1 Identifying Milestones**

#### **1. November 2025 : Literature Survey and Problem Identification**

Conducted extensive research on thermal management systems, liquid cooling technologies, and industrial cooling architectures. Identified limitations of traditional air and single-loop cooling systems.

#### **2. December 2025: Prior Art Search and Product Benchmarking**

Analyzed patents, industrial products, and existing thermal systems. Established the novelty and feasibility of a low-cost dual-loop educational cooling prototype.

#### **3. January 2026 :System Architecture Design and Design Thinking Implementation**

Developed the complete dual-loop system architecture, customer journey map, empathy map, and finalized problem statement, objectives, and design specifications.

#### **4. February 2026: Component Selection and Procurement**

Finalized electronic and mechanical components including ESP32-S3, pumps, sensors, relays, MOSFET, heat exchanger, and heating chamber materials. Completed procurement process.

#### **5. March 2026: Thermal Chamber Fabrication and Heater Integration**

Fabricated the stainless-steel heating cylinder, developed the nichrome heating jacket, implemented fiberglass insulation layers, and assembled the primary thermal unit.

#### **6. April 2026: Cooling Loop Assembly and Electronics Integration**

Built primary and secondary coolant loops, integrated pumps, heat exchanger, fan, sensors, relays, LCD, and RTC modules with the ESP32 controller.

#### **7. May 2026: Embedded Programming, Testing, and Safety Validation**

Developed the complete embedded logic for heating, monitoring, cooling control, and fault detection. Conducted temperature and flow testing under various operating conditions.

### 8. June 2026: Performance Analysis, Documentation, and Final Prototype Completion

Collected thermal performance data, analyzed system efficiency, prepared graphs, completed literature documentation, feasibility studies, and finalized the working prototype.

### 3.9.2 Project Execution Plan

The execution plan as mentioned in Table 5 begins with conceptual design and thermal architecture planning. This is followed by component selection and system fabrication. After mechanical assembly, the electronics and sensors are integrated with the ESP32-S3 controller. The heater system, pumps, and cooling fan are programmed for automated operation. Testing is then performed to validate heat transfer, coolant flow, and safety conditions. Finally, performance results are documented, analysed, and prepared for future improvements and scalability.

Table 3.4: Gantt Chart explaining Project Milestones

| Milestone ID | Milestone                                        | NOV 2025                                                                            | DEC 2025                                                                            | JAN 2026                                                                            | FEB 2026                                                                             | MAR 2026                                                                              | APR 2026                                                                              | MAY 2026                                                                              | JUN 2026                                                                              |
|--------------|--------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| A1           | Literature Survey & Problem Identification       |  |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| A2           | Prior Art Search & Product Benchmarking          |                                                                                     |  |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| A3           | System Architecture Design & Design Thinking     |                                                                                     |                                                                                     |  |                                                                                      |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| A4           | Component Selection & Procurement                |                                                                                     |                                                                                     |                                                                                     |  |                                                                                       |                                                                                       |                                                                                       |                                                                                       |
| A5           | Thermal Chamber Fabrication & Heater Integration |                                                                                     |                                                                                     |                                                                                     |                                                                                      |  |                                                                                       |                                                                                       |                                                                                       |
| A6           | Cooling Loop Assembly & Electronics Integration  |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |  |                                                                                       |                                                                                       |
| A7           | Embedded Programming & Testing                   |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |  |                                                                                       |
| A8           | Performance Analysis & Documentation             |                                                                                     |                                                                                     |                                                                                     |                                                                                      |                                                                                       |                                                                                       |                                                                                       |  |

## Summary

The reviewed research papers collectively highlight the growing importance of advanced thermal management systems in modern engineering applications, especially in AI data centers, high-performance computing, and industrial systems. The studies emphasize that traditional air cooling and single-loop liquid cooling are becoming insufficient for handling increasing thermal loads. Key findings show that liquid cooling systems, particularly dual-loop architectures, provide superior heat transfer efficiency, better temperature stability, improved fault isolation, and reduced energy consumption. Several papers focused on optimizing heat exchangers, coolant flow dynamics, and embedded sensor integration for real-time monitoring and intelligent thermal control. The literature also explored sustainable cooling methods, nanofluid-based thermal enhancement, and adaptive cooling strategies. Overall, the research strongly supports the development of modular dual-loop cooling systems as a practical and efficient solution for future thermal challenges, validating the project's objective of creating a low-cost, scalable, and sensor-integrated thermal management prototype.

# CHAPTER 4

## FEASIBILITY ANALYSIS

## CHAPTER 4

### FEASIBILITY ANALYSIS

#### 4.1. Executive summary

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|                                      |                                                                                                                                                                                                                                                                                                                          |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product concept</b>               | Intelligent Dual-Loop Controlled Cooling System with Real-Time Monitoring and Automated Thermal Management                                                                                                                                                                                                               |
| <b>Problem it solves</b>             | Existing small-scale thermal systems lack real-time monitoring, automated temperature control, efficient heat transfer, and safety mechanisms. The system demonstrates controlled heat generation, heat transfer, and cooling using a dual-loop architecture with sensor-based monitoring.                               |
| <b>Target user</b>                   | Engineering students, researchers, thermal system designers, educational institutions, and prototype developers working on cooling systems, heat exchangers, and reactor-inspired thermal management systems.                                                                                                            |
| <b>Overall feasibility statement</b> | <b>Conditionally Feasible</b> – The proposed prototype can be successfully developed using commercially available components and standard manufacturing methods. Performance depends on proper thermal insulation, coolant circulation, leak-proof plumbing, and accurate control algorithms.                            |
| <b>Key strengths</b>                 | <ol style="list-style-type: none"><li>1. Real-time temperature and flow monitoring using ESP32 and sensors.</li><li>2. Modular dual-loop architecture allowing independent heat generation and cooling circuits.</li><li>3. Low-cost implementation using readily available components and recycled materials.</li></ol> |
| <b>Key risks</b>                     | <ol style="list-style-type: none"><li>1. Coolant leakage due to multiple hose connections and fittings.</li><li>2. Inefficient heat transfer between primary and secondary loops if heat exchanger contact is poor.</li><li>3. Overheating of heater assembly or pump failure leading to thermal instability.</li></ol>  |

|                             |                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Recommendation</b>       | <b>Proceed with Conditions</b> – Continue development after validating heater operation, coolant circulation, leak testing, and temperature control functionality under safe operating conditions.                                                                                                                                                       |
| <b>Immediate next steps</b> | <ol style="list-style-type: none"> <li>1. Complete fabrication of the heated sphere chamber and primary coolant loop.</li> <li>2. Assemble and test the dual-loop heat exchanger and cooling subsystem.</li> <li>3. Integrate ESP32 monitoring, flow sensing, relay control, and safety shutdown mechanisms, followed by full system testing.</li> </ol> |

## 4.2. Project overview

### 4.2.1 Product description

The Controlled Cooling System is a smart thermal management prototype designed to demonstrate heat generation, heat transfer, and cooling using a dual-loop circulation architecture. The system uses an electrically heated aluminum thermal core, temperature sensors, flow sensors, and an ESP32 microcontroller to monitor and control thermal behavior in real time. Heat generated in the primary loop is transferred through a compact liquid-to-liquid heat exchanger and dissipated through a secondary cooling loop. The modular design enables users to study thermal dynamics, heat exchanger performance, coolant circulation, and automated cooling control in a safe and low-cost environment. The system serves as an educational and research platform for understanding advanced thermal management principles used in industrial cooling and energy systems.

### 4.2.2 Problem statement

Many educational institutions, engineering students, and researchers lack access to affordable and practical thermal management platforms for studying heat generation, heat transfer, coolant circulation, and cooling system performance. Existing industrial thermal systems are expensive, complex, and difficult to safely replicate in laboratory environments. As a result, learners often struggle to gain hands-on experience with real-time thermal monitoring, heat exchanger operation, and automated cooling control. There is a need for a low-cost, modular, and sensor-

integrated thermal management system that enables practical experimentation and performance analysis without requiring specialized industrial infrastructure.

### 4.2.3 Proposed solution

The proposed solution is a dual-loop controlled cooling system that combines heat generation, coolant circulation, heat exchange, and automated monitoring into a single integrated platform. A controlled heating element generates thermal energy within an aluminum thermal core, while a primary coolant loop absorbs and transports the heat. The thermal energy is then transferred through a compact aluminum block heat exchanger to a secondary cooling loop, where it is dissipated using a dedicated cooling chamber and forced-air cooling. An ESP32 microcontroller continuously monitors temperatures, coolant flow rates, and system status through connected sensors and provides real-time feedback via a display interface. The modular architecture allows users to evaluate heat transfer efficiency, cooling performance, and thermal control strategies under various operating conditions.

### 4.2.4 Project objectives

1. Validate the technical feasibility of a dual-loop thermal management and controlled cooling system using readily available components.
2. Evaluate the effectiveness of heat generation, heat transfer, and heat dissipation mechanisms under controlled operating conditions.
3. Assess the performance of real-time monitoring and control using ESP32, temperature sensors, and flow sensors.
4. Estimate the overall prototype cost, scalability, and practical implementation requirements.
5. Determine the viability of further development into an educational, research, or industrial thermal management platform.

### 4.2.5 Scope and assumptions

#### Scope

This feasibility study focuses on the design, fabrication, and evaluation of a prototype dual-loop controlled cooling system. The study includes heat generation, coolant circulation, heat transfer through a liquid-to-liquid heat exchanger, temperature monitoring, flow monitoring, and automated control using a microcontroller-based architecture. The objective is to validate thermal performance, operational stability, and overall system feasibility at prototype scale.

## Out of Scope

The following activities are outside the scope of this study:

- Industrial-scale deployment
- Commercial manufacturing planning
- Regulatory certification and compliance
- Long-term reliability testing
- Advanced thermal optimization
- Production-ready enclosure design
- Large-scale safety validation.

Key assumptions:

- The prototype will operate at laboratory scale using low-voltage electrical systems.
- Commercially available components such as ESP32 controllers, sensors, pumps, relays, and heat exchangers will be used.
- The primary and secondary coolant loops will remain physically isolated during operation.
- The prototype will be operated within safe temperature and pressure limits.
- The project team has access to standard fabrication tools, electronic testing equipment, and basic workshop facilities.
- The cooling system will be evaluated primarily for educational, demonstration, and research purposes rather than industrial use.
- Prototype performance data will be sufficient to determine technical feasibility and identify future design improvements.

## 4.3. Technical feasibility

---

Technical feasibility assesses whether the product can be designed, built, and tested using available or accessible technology, materials, skills, and tools.



### 4.3.1 Core technology assessment

| Technology Component                                                                               | Maturity Level                                                                                                    | In-House Expertise                                                                                         | Gap / Action Required                                                                                              |
|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <b>Core sensing / actuation mechanism</b> (MAX6675, Thermocouples, Flow Sensors, Relays, MOSFETs)  | <input checked="" type="checkbox"/> Proven<br><input type="checkbox"/> Emerging<br><input type="checkbox"/> Novel | <input type="checkbox"/> Yes<br><input checked="" type="checkbox"/> Partial<br><input type="checkbox"/> No | Integrate sensors, calibrate temperature and flow measurements, validate control logic under operating conditions. |
| <b>Power supply / energy system</b> (12V SMPS, LM2596, Fuse Protection, Heater System)             | <input checked="" type="checkbox"/> Proven<br><input type="checkbox"/> Emerging<br><input type="checkbox"/> Novel | <input checked="" type="checkbox"/> Yes<br><input type="checkbox"/> Partial<br><input type="checkbox"/> No | Verify electrical safety, current handling capability, fuse coordination, and thermal loading of heater circuit.   |
| <b>Enclosure and materials</b> (Aluminum thermal core, stainless-steel chambers, tubing, fittings) | <input checked="" type="checkbox"/> Proven<br><input type="checkbox"/> Emerging<br><input type="checkbox"/> Novel | <input type="checkbox"/> Yes<br><input checked="" type="checkbox"/> Partial<br><input type="checkbox"/> No | Fabricate leak-proof chambers, validate material compatibility with coolant and operating temperatures.            |
| <b>Electronics / PCB design</b> (ESP32-S3, wiring, interfaces, control circuitry)                  | <input checked="" type="checkbox"/> Proven<br><input type="checkbox"/> Emerging<br><input type="checkbox"/> Novel | <input type="checkbox"/> Yes<br><input checked="" type="checkbox"/> Partial<br><input type="checkbox"/> No | Complete schematic validation, improve wiring reliability, and develop PCB for future versions.                    |
| <b>Firmware / embedded software</b> (ESP32 monitoring and control system)                          | <input checked="" type="checkbox"/> Proven<br><input type="checkbox"/> Emerging<br><input type="checkbox"/> Novel | <input type="checkbox"/> Yes<br><input checked="" type="checkbox"/> Partial<br><input type="checkbox"/> No | Develop sensor acquisition, display interface, relay control, alarm system, and safety shutdown logic.             |
| <b>Assembly and testing process</b>                                                                | <input checked="" type="checkbox"/> Proven<br><input type="checkbox"/> Emerging<br><input type="checkbox"/> Novel | <input type="checkbox"/> Yes<br><input checked="" type="checkbox"/> Partial<br><input type="checkbox"/> No | Perform subsystem testing, leak testing, thermal characterization, and end-to-end system validation.               |

### 4.3.2 Key technical risks

#### **Risk 1: Coolant Leakage and Fluid Containment Failure\**

**Description:**

Multiple hose connections, barbed fittings, pumps, reservoirs, and heat exchanger interfaces increase the probability of leaks during operation.

**Impact:**

- Reduced cooling performance
- Potential damage to electronics
- Unsafe operating conditions

**Mitigation:**

- Use hose clamps on all tubing connections
- Perform leak testing with water before introducing coolant
- Apply RTV silicone to all container penetrations
- Conduct 24-hour static leak testing prior to operation

#### **Risk 2: Inefficient Heat Transfer Between Primary and Secondary Loops**

**Description:**

Poor thermal contact between the aluminum heat exchanger blocks may reduce heat transfer efficiency.

**Impact:**

- Elevated coolant temperatures
- Reduced cooling effectiveness
- Inaccurate thermal performance evaluation

**Mitigation:**

- Apply a thin, uniform layer of thermal paste
- Use bolted clamping arrangement to maximize contact pressure
- Monitor inlet and outlet temperatures to quantify performance
- Optimize block mounting during testing

### Risk 3: Heater Overheating or Thermal Runaway

#### Description:

Failure of the coolant circulation system or inadequate heat removal may cause excessive temperatures in the aluminum thermal core.

#### Impact:

- Heater damage
- Component failure
- Potential safety hazard

#### Mitigation:

- Implement software temperature limits using MAX6675 sensors
- Use relay/MOSFET shutdown when temperature exceeds threshold
- Add audible buzzer alarms for over-temperature conditions
- Validate cooling system operation before full-power testing

## 4.4. Financial feasibility

Financial feasibility determines whether the product can be developed and commercialised within available resources, and whether it will generate sufficient returns to justify the investment.

### 4.4.1 Estimated development costs

At the idea stage, use rough order-of-magnitude estimates. These will be refined as the project progresses.

| Cost category                       | Low estimate (Rs) | High estimate (Rs) | Confidence                                                         |
|-------------------------------------|-------------------|--------------------|--------------------------------------------------------------------|
| Research and concept validation     | Rs 500            | Rs 2000            | [ ] High<br>[ <input checked="" type="checkbox"/> ] Med<br>[ ] Low |
| Prototype development (hardware)    | Rs 4000           | Rs 8000            | [ <input checked="" type="checkbox"/> ] High<br>[ ] Med<br>[ ] Low |
| Component and materials procurement | Rs 5000           | Rs 8000            | [ ] High<br>[ ] Med                                                |

|                                       |                 |                 |                                                                    |
|---------------------------------------|-----------------|-----------------|--------------------------------------------------------------------|
|                                       |                 |                 | [ <input checked="" type="checkbox"/> ] Low                        |
| Software / firmware development       | Rs 0            | Rs 5000         | [ ] High<br>[ <input checked="" type="checkbox"/> ] Med<br>[ ] Low |
| Testing and certification             | Rs 1000         | Rs 3500         | [ ] High<br>[ <input checked="" type="checkbox"/> ] Med<br>[ ] Low |
| Tooling and manufacturing setup       | Rs 100          | Rs 2000         | [ <input checked="" type="checkbox"/> ] High<br>[ ] Med<br>[ ] Low |
| Marketing and launch                  | Rs 0            | Rs 6000         | [ ] High<br>[ ] Med<br>[ <input checked="" type="checkbox"/> ] Low |
| Working capital (first 6 months)      | Rs 0            | Rs 5000         | [ <input checked="" type="checkbox"/> ] High<br>[ ] Med<br>[ ] Low |
| Contingency (15–20%)                  | Rs 1500         | Rs 4500         | [ ] High<br>[ <input checked="" type="checkbox"/> ] Med<br>[ ] Low |
| <b>TOTAL ESTIMATED<br/>INVESTMENT</b> | <b>Rs 12100</b> | <b>Rs 44000</b> |                                                                    |

#### 4.4.2 Revenue model

Describe how the product will generate revenue. For a hardware product at concept stage, consider the following questions:

- What is the proposed unit selling price? What is the basis for this price?
- What is the estimated bill of materials (BOM) cost per unit?
- What is the target gross margin? (Typically, 40–60% for hardware products)
- What is the minimum viable sales volume to break even?

| Parameter              | Value                                                                                                                        |
|------------------------|------------------------------------------------------------------------------------------------------------------------------|
| Target Unit Price (Rs) | ₹2,50,000                                                                                                                    |
| Basis                  | Value-based pricing benchmarked against educational thermal systems, laboratory demonstrators, and engineering training kits |

|                                |                                                                                                     |
|--------------------------------|-----------------------------------------------------------------------------------------------------|
| Estimated Unit BOM Cost        | ₹1,00,000                                                                                           |
| Source                         | Prototype component cost estimate, supplier quotations, and market benchmarks                       |
| Gross Margin Target            | 150%                                                                                                |
| Break-even Volume (Units/Year) | 25 units                                                                                            |
| Estimated Time to Break-even   | 12–18 months from launch                                                                            |
| Funding Requirement            | ₹1,00,000–₹2,00,000                                                                                 |
| Source                         | Own funds, academic grants, innovation grants, incubation support, or prototype development funding |

4.5. Market feasibility

Market feasibility examines whether a sufficiently large, reachable, and willing-to-pay market exists for the product, and whether competitive conditions allow a viable position.

4.5.1 Target market definition

|                                |                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Primary target segment         | Engineering colleges, universities, technical institutions, and research laboratories focusing on thermal systems, heat transfer, fluid mechanics, embedded systems, and energy engineering. Primary users include undergraduate and postgraduate engineering students, faculty, and lab instructors. |
| Secondary target segment       | Industrial training centres, R&D organizations, startup incubators, and prototype development labs working on cooling systems, thermal management, and process control.                                                                                                                               |
| Total addressable market (TAM) | Rs 250 Cr   Source: [research / estimate]                                                                                                                                                                                                                                                             |

|                                             |                                                                                                                                                          |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Serviceable addressable market (SAM)</b> | Rs 75 Cr   Rationale: [what portion can you realistically reach?]                                                                                        |
| <b>Initial target market share</b>          | 2 % within 3 years                                                                                                                                       |
| <b>Key buying trigger</b>                   | Need for affordable hands-on thermal system demonstrators, heat exchanger training kits, research experiments, and embedded control-based lab equipment. |
| <b>Key purchase barrier</b>                 | Budget limitations, preference for established lab equipment vendors, lack of awareness, and concerns regarding long-term durability and support.        |

### 4.5.2 Customer need validation

At the idea stage, evidence of customer need may be qualitative. Document what you know and what remains to be validated.

| <b>Validation activity</b>         | <b>Status</b>                                                                                                     | <b>Key finding</b>                                                                                                                                            |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer interviews / surveys      | <input type="checkbox"/> Done<br><input checked="" type="checkbox"/> Planned<br><input type="checkbox"/> Not done | Initial discussions with engineering students and faculty indicate demand for affordable thermal management demonstration systems for practical lab learning. |
| Observation of current behaviour   | <input checked="" type="checkbox"/> Done<br><input type="checkbox"/> Planned<br><input type="checkbox"/> Not done | Existing thermal lab systems are expensive, bulky, and offer limited flexibility for real-time experimentation and embedded integration.                      |
| Review of existing market research | <input checked="" type="checkbox"/> Done<br><input type="checkbox"/> Planned<br><input type="checkbox"/> Not done | Educational thermal system markets show increasing demand for modular and low-cost hands-on training systems in engineering institutions.                     |

|                                           |                                                                                                                   |                                                                                                                                                             |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Analysis of competitor reviews / feedback | <input checked="" type="checkbox"/> Done<br><input type="checkbox"/> Planned<br><input type="checkbox"/> Not done | Existing competitors focus on industrial-grade or expensive academic setups, leaving a gap for compact, affordable, student-friendly thermal demonstrators. |
| Expert / industry consultation            | <input type="checkbox"/> Done<br><input checked="" type="checkbox"/> Planned<br><input type="checkbox"/> Not done | Consultation with thermal engineering faculty and embedded systems experts is planned to refine design and improve heat transfer efficiency.                |
| Willingness-to-pay testing                | <input type="checkbox"/> Done<br><input checked="" type="checkbox"/> Planned<br><input type="checkbox"/> Not done | Pricing validation will be conducted with engineering colleges and labs to determine acceptable cost range and procurement feasibility.                     |

### 4.5.3 Competitive landscape

Identify the key alternatives your target customer currently uses and how your product compares.

| Competitor / Alternative                                 | Type                                       | Price (Rs)          | Key Strength                                                                | Key Weakness                                                                |
|----------------------------------------------------------|--------------------------------------------|---------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Industrial Thermal Training Kits (Lab Equipment Vendors) | <input checked="" type="checkbox"/> Direct | ₹50,000 – ₹2,00,000 | High precision, industrial-grade reliability, established academic adoption | Expensive, bulky, less customizable, limited real-time embedded integration |
| Commercial Heat Exchanger Demonstration Units            | <input checked="" type="checkbox"/> Direct | ₹30,000 – ₹1,50,000 | Proven performance, standardized experiments, structured training modules   | High cost, fixed architecture, difficult to modify for research             |

|                                                    |                                                 |                                          |                                                                  |                                                                                  |
|----------------------------------------------------|-------------------------------------------------|------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Simulation Software (ANSYS, MATLAB Thermal Models) | <input checked="" type="checkbox"/><br>Indirect | ₹10,000 – ₹1,00,000+ (license dependent) | High analytical accuracy, no physical risk, scalable simulations | No hands-on practical learning, lacks physical fluid/thermal behavior validation |
| DIY / Manual Alternative                           | <input checked="" type="checkbox"/><br>Indirect | ₹5,000 – ₹15,000                         | Low cost, flexible customization, educational build process      | Inconsistent quality, high trial-and-error, poor documentation, safety concerns  |
| Status Quo (Do Nothing)                            | <input checked="" type="checkbox"/><br>Indirect | ₹0                                       | No investment required, no operational complexity                | No practical thermal learning, no experimentation, no research capability        |

#### 4.5.4 Competitive differentiation

Describe your product's unique value proposition — what it does better, cheaper, faster, or differently than existing alternatives.

The proposed Controlled Cooling System differentiates itself by combining affordability, modularity, and real-time embedded control into a single educational and research-oriented thermal management platform.

Unlike expensive industrial thermal training kits and rigid commercial heat exchanger systems, this product provides a low-cost, customizable, and hands-on learning environment for studying heat generation, transfer, and cooling.

Its dual-loop architecture, integrated sensor network, and ESP32-based automation enable real-time thermal monitoring, flow analysis, and active control, making it both a practical learning tool and an experimental research platform. The system bridges the gap between low-cost DIY setups and high-cost industrial equipment by delivering flexibility, accessibility, and strong technical depth at a significantly lower price point.

##### Unique Value Proposition:

Affordable + Modular + Smart + Research-Friendly



## 4.6. Operational feasibility

Operational feasibility assesses whether the organisation has — or can acquire — the people, processes, infrastructure, and partnerships needed to develop, manufacture, and deliver the product.

### 4.6.1 Team and skills assessment

| Required Capability                   | Available? | Gap and Plan to Address                                                                                                                                                            |
|---------------------------------------|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hardware / Mechanical Engineering     | ☑ Partial  | Basic fabrication and assembly skills are available. Advanced thermal optimization and structural refinement will be improved through iterative prototyping and faculty guidance.  |
| Electronics and PCB Design            | ☑ Yes      | Core circuit integration, wiring, sensor interfacing, and power management are available. Future PCB optimization can be developed for compactness and reliability.                |
| Embedded Software / Firmware          | ☑ Yes      | ESP32 programming, sensor acquisition, relay control, and automation logic are available. Further refinement for safety logic and data logging will be implemented during testing. |
| Industrial Design (Form Factor, UX)   | ☑ Partial  | Functional prototype design is available, but product-level enclosure aesthetics and user-friendly packaging need future improvement.                                              |
| Manufacturing / Production Management | ☑ Partial  | Prototype-level fabrication is possible, but large-scale production planning and assembly optimization need future development.                                                    |
| Supply Chain and Procurement          | ☑ Yes      | Component sourcing and procurement are manageable using local and online electronics suppliers. Supply chain is stable for prototype scale.                                        |
| Quality Assurance and Testing         | ☑ Partial  | Basic testing capability exists for electrical, thermal, and flow validation. More structured testing procedures will be developed during prototype validation.                    |

|                                     |                                             |                                                                                                                                                                                  |
|-------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Regulatory and Compliance Knowledge | <input checked="" type="checkbox"/> No      | Regulatory certification is outside current prototype scope. Compliance requirements will be studied if the product enters commercialization.                                    |
| Sales, Marketing and Distribution   | <input checked="" type="checkbox"/> No      | No current sales or distribution framework exists. Initial outreach will be through academic networks, faculty contacts, and technical exhibitions.                              |
| Finance and Project Management      | <input checked="" type="checkbox"/> Partial | Budgeting, component planning, and development tracking are manageable at prototype scale. Larger financial planning and commercialization strategy will require future support. |

#### 4.6.2 Manufacturing and supply chain

| Category                                   | Answer                                                                                                                                                                |
|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Proposed Manufacturing Approach</b>     | <input checked="" type="checkbox"/> <b>In-house</b> <input type="checkbox"/> Contract Manufacturer <input type="checkbox"/> Hybrid                                    |
| <b>Key Component Suppliers Identified?</b> | <input checked="" type="checkbox"/> <b>Yes, confirmed</b>                                                                                                             |
| <b>Lead Time for Critical Components</b>   | <b>1–2 weeks</b>   Risk: <input checked="" type="checkbox"/> <b>Low</b>                                                                                               |
| <b>Minimum Order Quantities (MOQ)</b>      | Component 1: <b>1 unit (ESP32-S3)</b>   Component 2: <b>1 unit (MAX6675)</b>   Component 3: <b>1 unit (Pump)</b>                                                      |
| <b>Quality Control Plan</b>                | <input checked="" type="checkbox"/> <b>Partially defined</b>                                                                                                          |
| <b>Target Production Location</b>          | <b>Bangalore, India</b> – selected due to easy access to electronics markets, fabrication workshops, industrial hardware suppliers, and educational ecosystem support |

### 4.6.3 Regulatory and compliance requirements (if applicable)

Hardware products are subject to mandatory certifications before they can be sold. Identify applicable requirements early to avoid costly delays.

| <b>Regulation / Certification</b>        | <b>Applicable Market</b>  | <b>Status</b>                                                           | <b>Estimated Cost and Timeline</b>  |
|------------------------------------------|---------------------------|-------------------------------------------------------------------------|-------------------------------------|
| <b>BIS Certification (India)</b>         | India                     | <input checked="" type="checkbox"/> N/A (Prototype Stage)               | ₹25,000 – ₹75,000 / 3–6 months      |
| <b>CE Marking (Europe)</b>               | Europe                    | <input checked="" type="checkbox"/> N/A                                 | ₹50,000 – ₹2,00,000 / 4–8 months    |
| <b>FCC Certification (USA)</b>           | USA                       | <input checked="" type="checkbox"/> N/A                                 | ₹1,00,000 – ₹3,00,000 / 3–6 months  |
| <b>RoHS Compliance</b>                   | Global Electronics Market | <input checked="" type="checkbox"/> Required (Future Commercialization) | ₹10,000 – ₹30,000 / 1–2 months      |
| <b>IP Rating (if applicable)</b>         | Industrial / Outdoor Use  | <input checked="" type="checkbox"/> N/A                                 | ₹20,000 – ₹50,000 / 2–4 months      |
| <b>Safety / UL Certification</b>         | Global Industrial Market  | <input checked="" type="checkbox"/> Required (Future Commercialization) | ₹1,50,000 – ₹5,00,000 / 6–12 months |
| <b>Other (Thermal Safety Validation)</b> | Research / Education      | <input checked="" type="checkbox"/> Required (Prototype Validation)     | ₹5,000 – ₹15,000 / 1–2 weeks        |

### 4.6.4 Go-to-market and distribution

| <b>Category</b>              | <b>Answer</b>                           |
|------------------------------|-----------------------------------------|
| <b>Primary Sales Channel</b> | <input checked="" type="checkbox"/> B2B |

|                                              |                                                                                                                                                        |
|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Key Partners (if known)</b>               | Engineering colleges, technical institutions, university laboratories, educational equipment vendors, academic exhibitions, and engineering symposiums |
| <b>After-Sales Service Model</b>             | <input checked="" type="checkbox"/> <b>Warranty / Repair</b> <input checked="" type="checkbox"/> <b>Replacement</b>                                    |
| <b>Launch Geography</b>                      | <b>Bangalore, Karnataka, India</b> (initial), followed by expansion across engineering institutions in India                                           |
| <b>Time to First Shipment (from concept)</b> | <b>3–6 months</b>                                                                                                                                      |

#### 4.7. Risk register

Identify and rate the key risks across all feasibility dimensions. Use this register to guide mitigation planning and stage-gate decision-making.

| <b>Risk</b>                                                                          | <b>Likelihood</b> | <b>Impact</b> | <b>Rating</b> | <b>Mitigation Strategy</b>                                                                                                                   |
|--------------------------------------------------------------------------------------|-------------------|---------------|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Core thermal system fails to achieve target temperature or cooling efficiency</b> | Medium            | High          | <b>High</b>   | Perform early-stage proof-of-concept testing, validate heater resistance, optimize coolant flow rate, and improve heat exchanger efficiency. |
| <b>Coolant leakage from chambers, barbs, or tubing connections</b>                   | High              | High          | <b>High</b>   | Conduct leak testing before operation, use RTV sealing, proper hose clamps, and pressure validation under working conditions.                |
| <b>Heater overheating or thermal runaway due to pump failure</b>                     | Medium            | High          | <b>High</b>   | Implement temperature cutoff logic using MAX6675 sensors, relay safety cutoff, buzzer alarms, and flow-based shutdown conditions.            |

|                                                                |        |        |               |                                                                                                                   |
|----------------------------------------------------------------|--------|--------|---------------|-------------------------------------------------------------------------------------------------------------------|
| <b>Component procurement delays or material unavailability</b> | Low    | Medium | <b>Medium</b> | Identify backup suppliers, use alternative local sources, and maintain spare stock for critical components.       |
| <b>Electrical overload, short circuit, or fuse failure</b>     | Medium | High   | <b>High</b>   | Validate wiring gauge, fuse sizing, MOSFET switching, and proper isolation between heater, pumps, and controller. |
| <b>Sensor inaccuracies or unstable readings</b>                | Medium | Medium | <b>Medium</b> | Calibrate sensors before testing, verify thermocouple placement, and use filtered software averaging.             |
| <b>Pump failure or insufficient coolant flow</b>               | Medium | High   | <b>High</b>   | Monitor flow using ZJ-S201, use pump priming procedures, and maintain spare pumps for replacement.                |
| <b>Heat exchanger efficiency lower than expected</b>           | Medium | Medium | <b>Medium</b> | Improve thermal contact using thermal paste, optimize clamping pressure, and test different flow rates.           |
| <b>Prototype cost exceeds budget</b>                           | Low    | Medium | <b>Medium</b> | Track expenses, reuse existing materials, and simplify non-critical subsystems.                                   |
| <b>System fails continuous operation stability test</b>        | Medium | High   | <b>High</b>   | Conduct short-duration test cycles first, progressively increase runtime, and monitor thermal behavior closely.   |
| <b>User safety hazard due to exposed hot surfaces</b>          | Medium | High   | <b>High</b>   | Add insulation layers, warning indicators, physical shielding, and emergency stop logic.                          |
| <b>Market adoption lower than expected</b>                     | Low    | Medium | <b>Low</b>    | Conduct live demonstrations, collect academic feedback, and refine educational value proposition.                 |

## 4.8. Recommended next steps

Based on the feasibility assessment, the following actions are recommended before the project proceeds to the next stage gate (prototype development).

| # | Action                                                                                   | Owner        | Deadline | Success Criterion                                                                            |
|---|------------------------------------------------------------------------------------------|--------------|----------|----------------------------------------------------------------------------------------------|
| 1 | Complete final mechanical assembly of thermal chamber, coolant tanks, and heat exchanger | Project Team | 2 days   | All physical components assembled and secured properly                                       |
| 2 | Complete electrical wiring of ESP32, relays, MOSFET, pumps, sensors, and heater          | Project Team | 2 days   | Full wiring verified without short circuits or loose connections                             |
| 3 | Perform heater resistance testing and validate safe current draw                         | Project Team | 1 day    | Heater resistance within target range ( $2\Omega$ – $4\Omega$ ) and current below fuse limit |
| 4 | Conduct coolant loop leak testing for both primary and secondary loops                   | Project Team | 1 day    | Zero leakage under continuous circulation test                                               |
| 5 | Integrate and calibrate MAX6675 temperature sensors and flow sensors                     | Project Team | 1 day    | Stable and accurate sensor readings obtained                                                 |
| 6 | Develop and upload embedded firmware for system monitoring and automated control         | Project Team | 2 days   | ESP32 successfully controls pumps, fan, heater, and safety logic                             |
| 7 | Conduct full thermal performance testing under controlled conditions                     | Project Team | 2 days   | Stable heat generation, transfer, and cooling achieved without overheating                   |
| 8 | Document prototype performance, collect observations, and identify optimization areas    | Project Team | 1 day    | Complete test report and improvement plan prepared                                           |

# CHAPTER 5

## METHODOLOGY AND IMPLEMENTATION

## CHAPTER 5

### METHODOLOGY AND IMPLEMENTATION

#### 5.1 Introduction

The proposed prototype is a thermal energy generation and transfer system inspired by reactor cooling principles. The system is designed to generate heat using an electrically heated aluminum spherical core and transfer the generated thermal energy through a multi-stage cooling architecture. The primary objective is to demonstrate controlled heat generation, temperature monitoring, fluid circulation, and heat exchange using readily available materials and electronic control systems. The prototype consists of a heat generation unit, a primary coolant loop, a heat exchanger, and a secondary cooling loop. An ESP32 microcontroller is used to monitor temperature, coolant flow rate, and system status through various sensors such as thermocouples and flow sensors. The collected data is displayed on an LCD interface and can be used for real-time system analysis. The project demonstrates the principles of thermal energy transfer, coolant circulation, heat exchange, and automated monitoring in a compact and cost-effective manner.

#### 5.2 System Architecture

As shown in Figure 5.1, The proposed system architecture consists of four major subsystems: Heat Generation Unit, Primary Coolant Loop, Heat Exchanger Unit, and Secondary Cooling Loop. All subsystems are monitored and controlled using an ESP32-S3 microcontroller-based control system. The Heat Generation Unit consists of an aluminum spherical chamber fabricated from two hemispherical aluminum vessels. Nichrome wire wrapped around the sphere acts as the heating element and is controlled using an IRFZ44N MOSFET. K-type thermocouples interfaced through MAX6675 modules continuously monitor the temperature of the sphere. The Primary Coolant Loop contains a coolant reservoir, DC circulation pump, flow sensor, and a stainless-steel chamber housing the heated sphere. Coolant absorbs thermal energy from the sphere and transports it to the heat exchanger. The coolant flow rate is measured using a ZJ-S201 flow sensor to ensure proper circulation. The Heat Exchanger Unit consists of two aluminum cooling blocks placed in direct contact with each other using thermal paste.



One block is connected to the primary coolant loop while the other is connected to the secondary coolant loop. This arrangement enables thermal energy transfer between the two loops through conduction.

The Secondary Cooling Loop comprises a coolant reservoir, DC circulation pump, cooling chamber, and a 12V DC fan. Heat received from the heat exchanger is dissipated into the surrounding environment, thereby cooling the circulating fluid before it returns to the reservoir. The Control and Monitoring Unit is centered around the ESP32-S3 microcontroller. It receives inputs from thermocouples, flow sensors, and the DS3231 real-time clock module. System parameters are displayed on an I2C LCD display, while relays and MOSFETs control pumps, cooling fans, and the heating elements. Power is supplied through a 12V SMPS with dedicated fused branches for the heater, pumps, and electronics, ensuring safe and reliable operation. The overall architecture demonstrates the complete cycle of heat generation, heat absorption, heat exchange, heat rejection, and real-time monitoring, providing a compact representation of a multi-loop thermal energy transfer system.

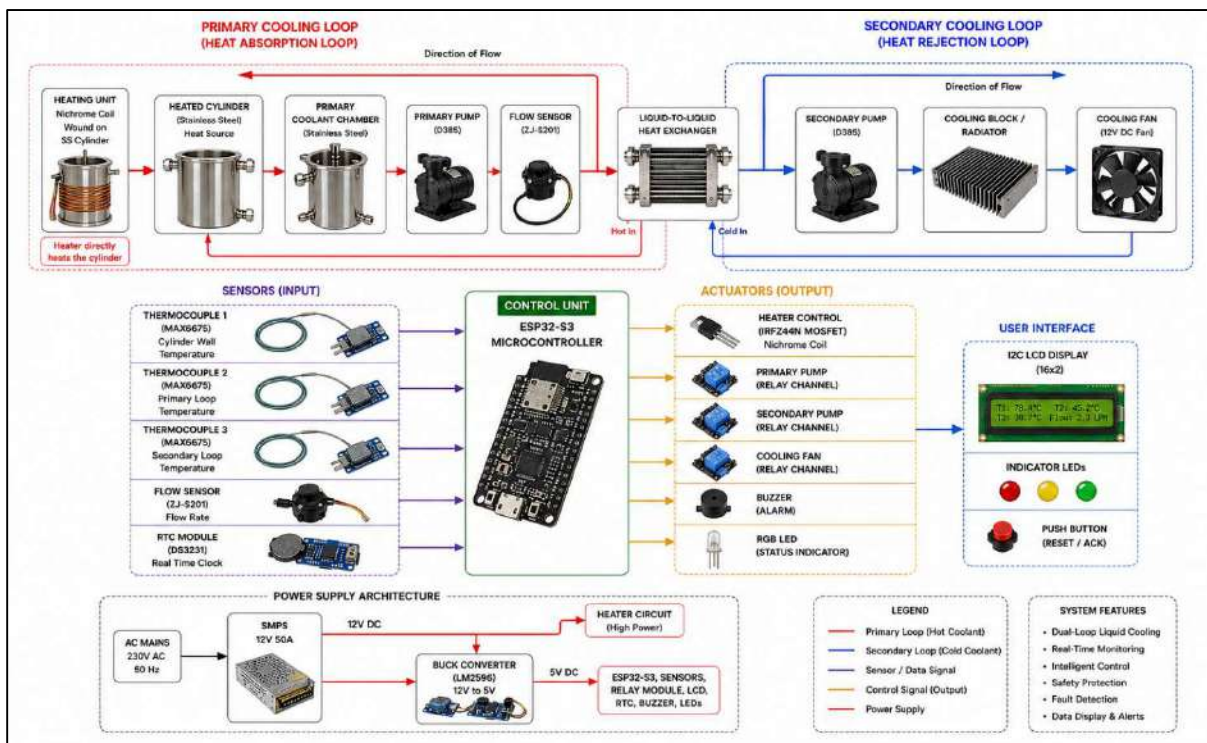


Figure 5.1: System Architecture of Dual Loop Cooling System

### 5.3 Flow Diagram / Algorithm

1. Start the system and initialize the ESP32 microcontroller.
2. Initialize all peripherals including MAX6675 temperature sensors, flow sensors, LCD display, RTC module, relays, pumps, and cooling fan.
3. Read the initial temperature and flow sensor values.
4. Activate the heating element (nichrome wire) through the MOSFET control circuit.
5. Heat is transferred from the nichrome wire to the aluminum spherical core.
6. Continuously monitor the sphere temperature using the thermocouple sensor.
7. Start the primary coolant pump and circulate coolant through the heated chamber.
8. The coolant absorbs heat from the aluminum sphere and carries it to the heat exchanger.
9. The heat exchanger transfers thermal energy from the primary coolant loop to the secondary coolant loop through the aluminum block interface.
10. Activate the secondary coolant pump to circulate coolant through the cooling loop.
11. Operate the cooling fan to enhance heat dissipation from the secondary loop.
12. Continuously monitor:
  - Sphere temperature
  - Primary loop temperature
  - Secondary loop temperature
  - Coolant flow rate
13. Display all measured parameters on the LCD screen.
14. Compare sensor values with predefined safety limits.
15. If abnormal temperature or flow conditions are detected:
  - Trigger buzzer alarm.
  - Indicate warning through RGB LED.
  - Shut down the heater if necessary.
16. Repeat the monitoring and control cycle continuously until the system is turned off.

# Coolant System Control – Process Flowchart

ESP32-S3 Microcontroller | Temperature, Flow & Safety Monitoring

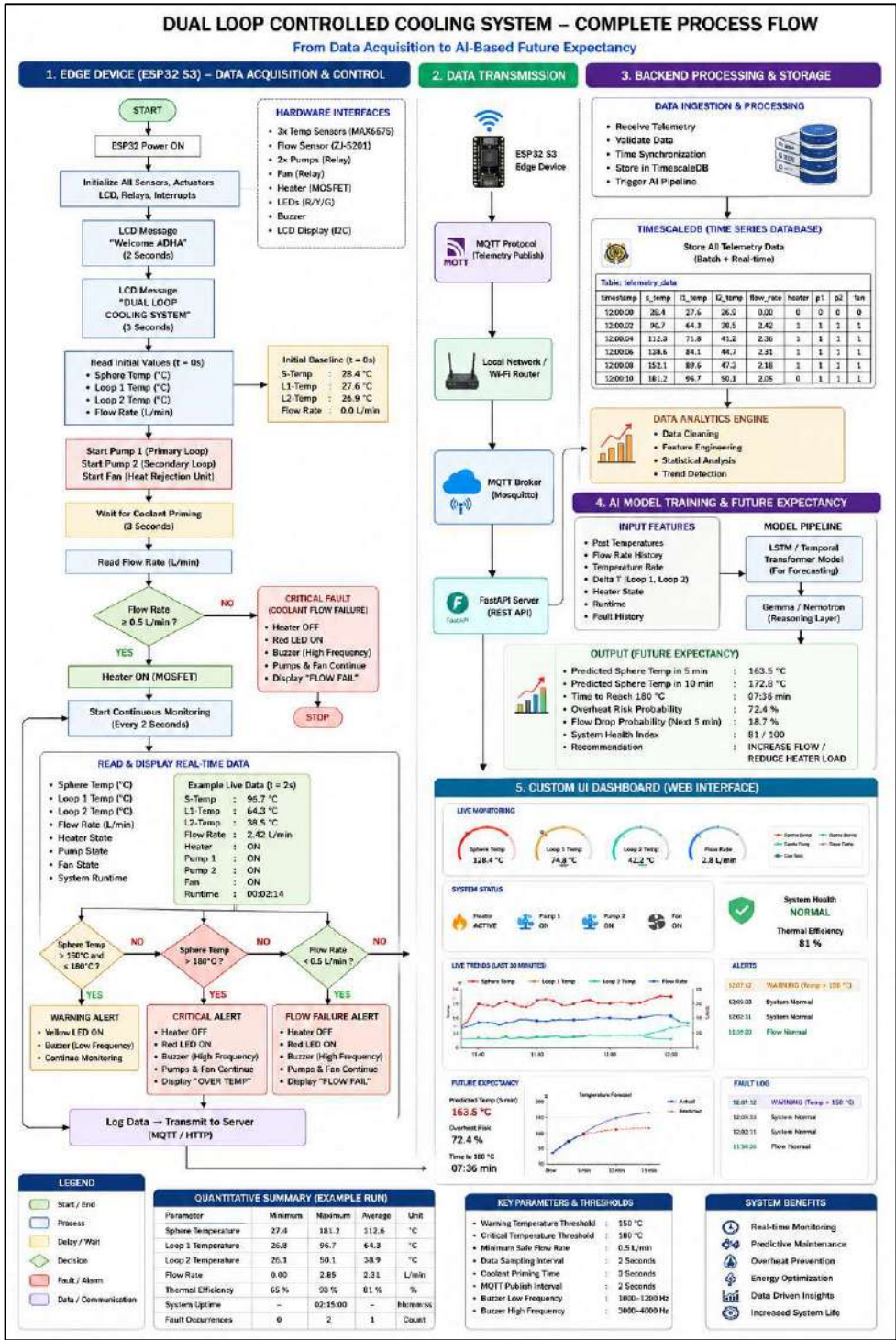


Figure 5.2: Workflow Flowchart for solving problem

## 5.4 Performance Parameters (with Mathematical Equations)

The performance of the thermal energy transfer system is evaluated using several thermal and fluid flow parameters. These parameters help determine the efficiency of heat generation, heat transfer, coolant circulation, and cooling performance.

### 1. Heat Power Generated by Heater

The electrical power supplied to the heater is calculated as:

$$P = VI$$

Where:

$P$  = Heater Power (W)

$V$  = Supply Voltage (V)

$I$  = Current (A)

### 2. Heat Energy Produced

The total thermal energy generated over a period of time is:

$$Q = Pt$$

Where:

$Q$  = Heat Energy (J)

$P$  = Heater Power (W)

$t$  = Time (s)

### 3. Mass Flow Rate of Coolant

The mass flow rate of the coolant is given by:

$$m' = \rho Qf$$

Where:

$m$  = Mass Flow Rate (kg/s)

$\rho$  = Coolant Density (kg/m<sup>3</sup>)

$Qf$  = Volumetric Flow Rate (m<sup>3</sup>/s)

### 4. Heat Absorbed by Coolant

The heat transferred to the coolant is:

$$Q' = m' C_p (T_o - T_i)$$

Where:

$Q'$  = Heat Transfer Rate (W)

$m'$  = Mass Flow Rate (kg/s)

$C_p$  = Specific Heat Capacity (J/kg·K)

$T_i$  = Inlet Temperature (°C)

$T_o$  = Outlet Temperature (°C)

### 5. Heat Exchanger Effectiveness

The effectiveness of the heat exchanger is:

$$\varepsilon = \frac{Q'_{actual}}{Q'_{max}}$$

Where:

$\varepsilon$  = Heat Exchanger Effectiveness

$Q'_{actual}$  = Actual Heat Transfer Rate (W)

$Q'_{max}$  = Maximum Possible Heat Transfer Rate (W)

## 6. Thermal Efficiency

The thermal efficiency of the system is:

$$\eta = \frac{Q'_{\text{useful}}}{P_{\text{input}}}$$

$$\eta = \frac{Q'_{\text{useful}}}{P_{\text{input}}} \times 100$$

Where:

$\eta$  = Thermal Efficiency (%)

$Q'_{\text{useful}}$  = Useful Heat Transfer Rate (W)

$P_{\text{input}}$  = Electrical Power Input (W)

## 7. Temperature Difference Across Heat Exchanger

The temperature difference driving heat transfer is:

$$\Delta T = T_h - T_c$$

$$\Delta T = T_h - T_c$$

Where:

$T_h$  = Hot Side Temperature (°C)

$T_c$  = Cold Side Temperature (°C)

## 8. Coolant Velocity

The velocity of coolant flowing through the pipe is:

$$v = \frac{Q}{A}$$

Where:

$v$  = Fluid Velocity (m/s)

$Qf$  = Volumetric Flow Rate (m<sup>3</sup>/s)

$A$  = Cross-Sectional Area of Pipe (m<sup>2</sup>)

For a circular pipe:

$$A = \pi D^2 / 4$$

where  $D$  is the pipe diameter (m).

## 9. Reynolds Number

The Reynolds number determines the flow regime:

$$Re = \frac{\rho v D}{\mu}$$

Where:

$Re$  = Reynolds Number

$\rho$  = Fluid Density (kg/m<sup>3</sup>)

$v$  = Fluid Velocity (m/s)

$D$  = Pipe Diameter (m)

$\mu$  = Dynamic Viscosity (Pa · s)

Flow Classification

$Re < 2300 \rightarrow$  Laminar Flow

$2300 \leq Re \leq 4000 \rightarrow$  Transitional Flow

$Re > 4000 \rightarrow$  Turbulent Flow

## 10. Temperature Measurement Error

The temperature measurement error is:

$$\text{Error} = [T_{\text{measured}} - T_{\text{actual}}]$$

Where:

$T_{\text{measured}}$  = Sensor Reading (°C)

$T_{\text{actual}}$  = Reference Temperature (°C)

## 11. System Response Time

The response time of the system is:

$$tr = tf - ti$$

Where:

$tr$  = Response Time (s)

$ti$  = Initial Time (s)

$tf$  = Final Time (s)

## 12. Flow Sensor Pulse–Frequency Relation

For the ZJ-S201 water flow sensor:

$$F = 7.5Q$$

Where:

$F$  = Pulse Frequency (Hz)

$Q$  = Flow Rate (L/min)

Therefore,

$$Q = 7.5F$$

This relation is used by the ESP32 microcontroller to calculate coolant flow rate in real time.



These performance parameters provide a complete evaluation of the thermal generation, heat transfer, coolant circulation, and cooling effectiveness of the proposed system.

## 5.5 Hardware Requirements

### 1. ESP32-S3 Development Board

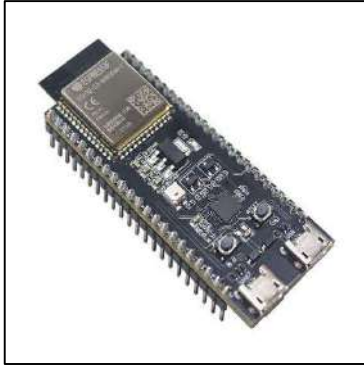


Figure 5.3.1: ESP32-S3.

The ESP32-S3 is the central control unit of the project. It is a high-performance microcontroller with Wi-Fi and Bluetooth capabilities, designed for real-time sensor interfacing, automation, and embedded control. It processes temperature and flow data, controls relays and MOSFETs, and executes the safety logic for the thermal system.

#### Function in this Project:

- Reads temperature from MAX6675 modules.
- Monitors coolant flow using the flow sensor.
- Controls pumps, fan, and heater switching.
- Executes thermal safety and emergency shutdown logic.

### 2. Nichrome Heating Coil



Figure 5.3.2: Nichrome Coil

Nichrome wire is used as the primary heat generation element due to its high electrical resistance and excellent thermal stability. When electric current passes through it, it converts electrical energy into heat, making it ideal for controlled heating applications.

**Function in this Project:**

- Generates the primary thermal load.
- Simulates heat-producing systems like CPUs/GPUs.
- Transfers heat into the primary coolant chamber.
- Provides controllable heating for experimentation.

**3. IRFZ44N MOSFET**

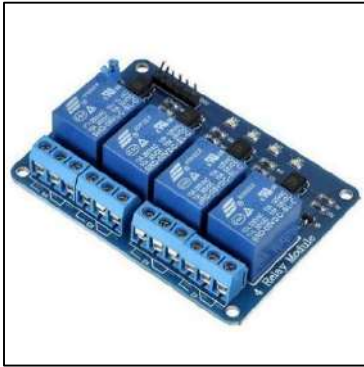
The IRFZ44N MOSFET acts as an electronic switch for controlling high-current loads efficiently. It allows the ESP32 to regulate the nichrome heater safely without directly carrying heavy current.

Figure 5.3.3: MOSFET

**Function in this Project:**

- Controls heater ON/OFF switching.
- Enables PWM-based heat regulation.
- Protects the ESP32 from high-current load.
- Improves switching efficiency.

#### 4. 4-Channel Relay Module



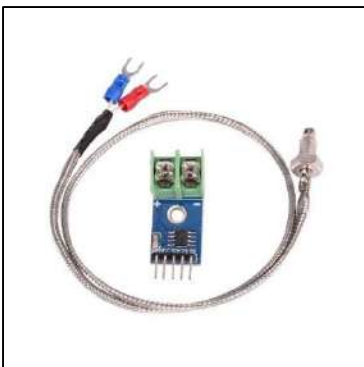
The relay module provides electrical isolation between low-voltage control signals and high-power devices. It is used to switch pumps, fan, and heater master supply safely.

Figure 5.3.4: Relay

#### Function in this Project:

- Controls coolant pumps.
- Controls cooling fan.
- Provides heater master cutoff.
- Implements hardware-level isolation.

#### 5. MAX6675 Temperature Modules



The MAX6675 is a thermocouple-to-digital converter used for accurate temperature measurement at high temperatures. It allows the system to monitor thermal behavior in real time.

Figure 5.3.5: MAX6675

**Function in this Project:**

- Measures heater wall temperature.
- Monitors primary loop temperature.
- Monitors secondary loop temperature.
- Enables thermal safety feedback

**6. K-Type Thermocouples**

K-type thermocouples are high-temperature sensors used to measure thermal variations in the heating and cooling system. They are durable and suitable for industrial thermal applications.

Figure 5.3.6: Thermocouples

**Function in this Project:**

- Detects real-time chamber temperature.
- Measures coolant heat absorption.
- Measures heat exchanger output temperature.
- Provides sensor data to MAX6675.

## 7. D385 DC Pumps



The D385 pump circulates coolant through both cooling loops. It ensures continuous fluid flow for heat transfer and system stability.

Figure 5.3.7: DC Pumps

### Function in this Project:

- Drives primary coolant loop.
- Drives secondary coolant loop.
- Maintains coolant circulation.
- Supports thermal transfer across loops.

## 8. ZJ-S201 Flow Sensor



The flow sensor measures coolant flow rate inside the system. It ensures the pumps are operating correctly and prevents heater operation under no-flow conditions.

Figure 5.3.8: ZJ-S201 Flow Sensor

**Function in this Project:**

- Measures coolant flow rate.
- Detects pump failure.
- Enables flow-based safety cutoff.
- Supports system diagnostics.

**9. Stainless Steel Thermal Chamber**

The stainless-steel chamber acts as the primary thermal containment structure where heat is generated and transferred into the coolant. It provides durability and thermal conductivity.

Figure 5.3.9: Steel Thermal Chamber

**Function in this Project:**

- Holds primary coolant.
- Transfers heat from heater to fluid.
- Acts as thermal chamber.
- Supports heater jacket mounting.

## 10. Liquid-to-Liquid Heat Exchanger Block



The heat exchanger transfers heat from the primary coolant loop to the secondary coolant loop without mixing the fluids. It forms the core of the dual-loop architecture.

Figure 5.3.10: Aluminum Heat Exchanger Block

### Function in this Project:

- Transfers heat between loops.
- Separates coolant circuits.
- Improves cooling efficiency.
- Enables modular thermal dissipation.

## 11. Cooling Fan



The fan improves heat rejection in the secondary cooling stage by increasing airflow around the cooling block or radiator.

Figure 5.3.11: 12V DC Fan

**Function in this Project:**

- Enhances heat dissipation.
- Improves secondary loop cooling.
- Maintains stable outlet temperature.
- Supports forced convection.

**12. LM2596 Buck Converter**

The LM2596 reduces the 12V supply to a stable 5V output required for powering control electronics safely.

Figure 5.3.12: LM2596

**Function in this Project:**

- Step down voltage for ESP32.
- Powers relay module.
- Powers sensors.
- Stabilizes low-voltage electronics.



### 13. 12V 50A SMPS



The SMPS serves as the primary power source for the entire thermal system, supplying sufficient current for pumps, heater, and control electronics.

Figure 5.3.13: 12V 50A SMPS

#### Function in this Project:

- Powers heater circuit.
- Powers pumps.
- Powers cooling fan.
- Supplies LM2596 input power.

### 14. I2C LCD Display

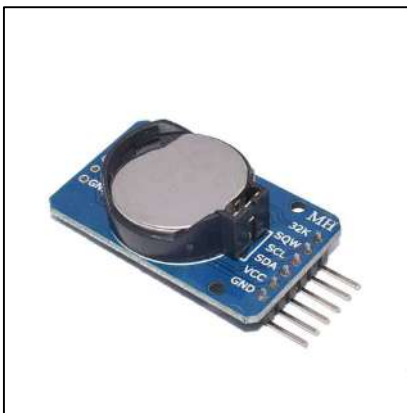


The LCD display provides real-time visualization of temperature, flow, and operational data, allowing easier monitoring of system performance.

Figure 5.3.14: I2C Display

**Function in this Project:**

- Displays thermal readings.
- Displays flow values.
- Shows system status.
- Shows safety alerts.

**15. DS3231 RTC Module**

The DS3231 is a highly accurate real-time clock used for time-based logging and monitoring system events.

Figure 5.3.15: RTC Module Function in this Project:

**Function in this Project:**

- Tracks system runtime.
- Timestamps readings.
- Supports thermal logging.
- Enables performance analysis.

## 16. Fiberglass Tape



Fiberglass tape acts as a high-temperature insulation layer between the metal chamber and nichrome heater, preventing electrical shorts and improving thermal containment.

Figure 5.3.16: Fiberglass Sleeves

### Function in this Project:

- Electrically insulates heaters.
- Supports nichrome winding.
- Improves thermal retention.
- Prevents direct metal contact.

## 17. Coolant Fluid (Distilled Water / Machine Coolant)



Figure 5.3.17: Distilled Water



Figure 5.3.18: CNC Cooling Oil

The coolant absorbs, transfers, and dissipates thermal energy through both loops. It acts as the heat transport medium.

**Function in this Project:**

- Absorbs generate heat.
- Transfers heat to exchanger.
- Remove heat in secondary loop.
- Stabilizes thermal behavior.

## 5.6 Software Requirements

### 5.6.1 Arduino IDE

Purpose:

Primary development environment used for writing, compiling, and uploading firmware to the ESP32-S3 microcontroller.

Used for:

- Writing embedded C/C++ code
- Sensor interfacing
- Relay control
- LCD integration
- Safety logic implementation

### 5.6.2 ESP32 Board Support Package (BSP)

Purpose:

Required package inside Arduino IDE to enable ESP32-S3 compilation and hardware compatibility.

Used for:

- Flashing firmware
- GPIO mapping
- WiFi/Bluetooth libraries
- Peripheral support

### **5.6.3 CH340 USB Drivers**

Purpose:

USB-UART communication drivers for uploading code and serial monitoring.

Used for:

- Establishing PC ↔ ESP32 communication
- Serial debugging
- Uploading firmware

### **5.6.4 Proteus / Wokwi Simulator**

Purpose:

Used for virtual simulation of sensor and controller logic before physical implementation.

Used for:

- Circuit validation
- Logic testing
- Pin mapping verification
- Relay testing

### 5.6.4 Altium Designer

Purpose:

Used for schematic design and PCB planning if migrating from breadboard to custom PCB.

Used for:

- Circuit diagram generation
- PCB routing
- BOM generation
- Hardware optimization

### 5.6.5 Google Sheets

Purpose:

Used for data logging, tabulation, and graph analysis.

Used for:

- Performance tables
- Experimental logs
- Result plotting
- Comparative analysis

### Embedded Libraries Used

1. Wire.h → I2C communication
2. LiquidCrystal\_I2C.h → LCD display
3. max6675.h → Thermocouple interface
4. RTCLib.h → RTC handling
5. EEPROM.h → Data storage
6. Ticker.h → Timed tasks

## 5.7 Hardware Implementation (Circuit Diagram)

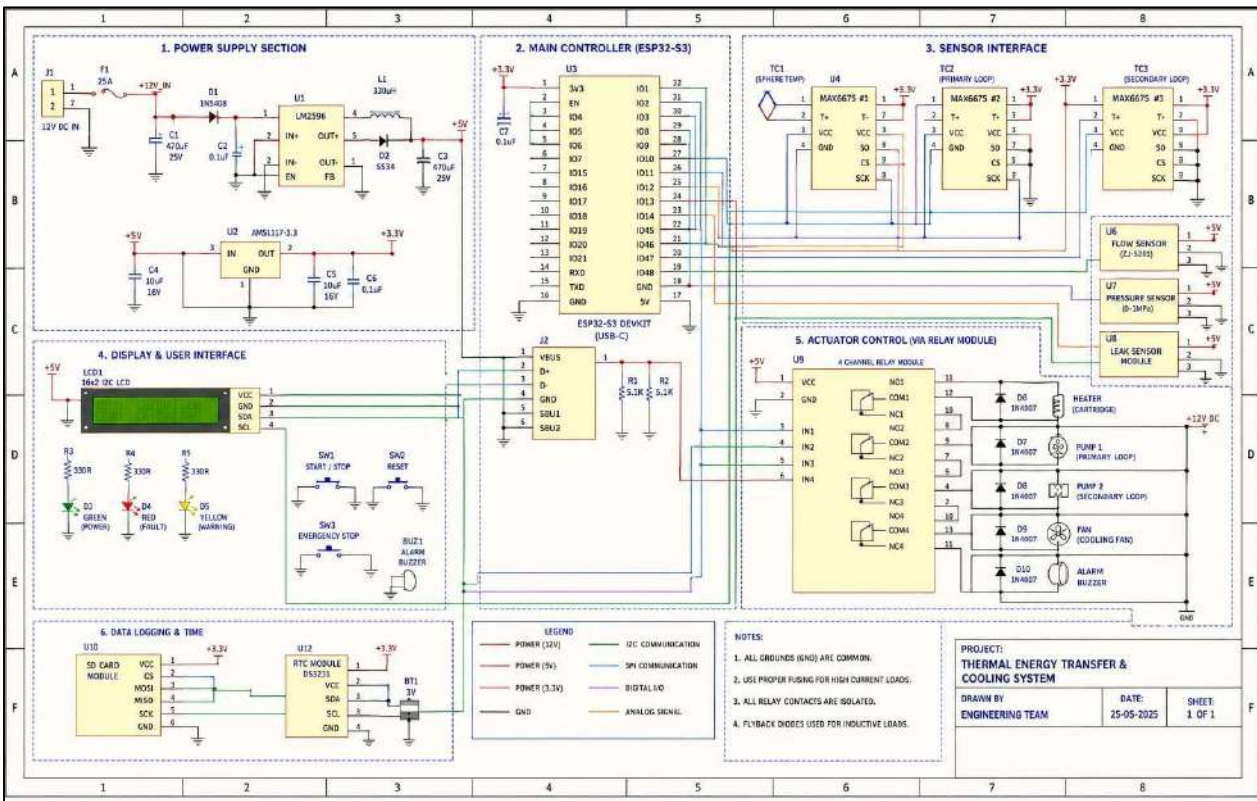


Figure 5.4: Circuit Diagram of Connection Chart

As shown in Figure 5.4, The hardware implementation of the proposed Thermal Energy Transfer and Cooling System was carried out by integrating thermal, mechanical, and electronic subsystems into a single operational prototype. The complete system consists of a power supply section, ESP32-S3 microcontroller unit, temperature sensing modules, flow monitoring system, relay control unit, heating section, coolant circulation system, display interface, and data logging modules.

The power supply section receives a 12V DC input from the SMPS unit. The input power is protected using a fuse and diode arrangement to prevent overcurrent and reverse polarity conditions. The LM2596 buck converter is used to generate a regulated 5V supply for sensors, display modules, relay circuits, and peripheral devices. An AMS1117-3.3V voltage regulator further converts the 5V supply into 3.3V required by the ESP32-S3 controller and communication modules.

The ESP32-S3 microcontroller acts as the central processing unit of the system. It continuously receives input data from the temperature sensors, flow sensor, leak sensor, and other monitoring

devices. The controller processes the collected information and performs decision-making functions related to heating control, pump activation, cooling fan operation, and alarm generation.

Temperature measurement is achieved using three K-Type thermocouples interfaced through MAX6675 modules. The first thermocouple monitors the temperature of the heated aluminum sphere, the second measures the primary coolant loop temperature, and the third measures the secondary coolant loop temperature. These sensors provide real-time temperature measurements with high accuracy and thermal stability.

The flow monitoring section consists of a ZJ-S201 flow sensor installed in the coolant circulation line. The sensor generates pulse signals proportional to the coolant flow rate, enabling continuous monitoring of fluid circulation. Additional safety monitoring is provided through leak detection and pressure sensing modules to ensure reliable system operation.

A 16×2 I2C LCD display is used as the primary user interface. The display continuously shows important operating parameters such as thermal core temperature, coolant temperatures, flow rate, and system status. Visual indicators including RGB LEDs provide quick system status information such as normal operation, heating active, and fault conditions. A buzzer is included to generate audible alarms during abnormal operating conditions.

The actuator control section consists of a four-channel relay module interfaced with the ESP32 controller. The relay outputs control the cartridge heater, primary coolant pump, secondary coolant pump, and cooling fan. Flyback protection diodes are connected across inductive loads to prevent voltage spikes generated during switching operations.

The heat generation section consists of an aluminum thermal core heated using a nichrome heating element. Electrical energy supplied to the heater is converted into thermal energy, which is absorbed by the primary coolant loop. The heated coolant then transports thermal energy to the heat exchanger assembly. The heat exchanger is constructed using two aluminum cooling blocks placed in direct thermal contact with thermal paste between them. One block carries the hot primary coolant while the second block carries the cooler secondary coolant. Heat transfer occurs through conduction across the aluminium blocks and thermal interface material.



The secondary cooling loop removes heat from the heat exchanger and transports it to the cooling chamber. A cooling fan assists in heat rejection by increasing airflow across the cooling section. This arrangement ensures continuous heat removal and stable thermal operation.

The hardware components were mounted on a common base platform and interconnected using appropriate wire gauges, tubing, hose fittings, clamps, and connectors. The completed hardware assembly successfully integrates heat generation, heat transfer, heat exchange, coolant circulation, sensing, monitoring, and control functions into a compact prototype. The implemented system demonstrates the practical principles of thermal energy management and serves as an effective educational platform for studying heat transfer and cooling technologies.

### **Summary**

This phase of the project focuses on the technical foundation and implementation of the Dual-Loop Controlled Cooling System. It begins with an introduction to the system and its architecture, explaining the separation of heat absorption and heat rejection into two independent coolant loops for improved thermal efficiency. The flow diagram and algorithm define the operational sequence, including initialization, heating, cooling, monitoring, and safety decisions. Key performance parameters such as temperature, flow rate, heat transfer rate, and thermal efficiency are analyzed using relevant mathematical equations. The required hardware components include ESP32-S3, sensors, pumps, relays, MOSFETs, and heat exchangers, while the software requirements include Arduino IDE and embedded control logic. Finally, the hardware implementation and circuit design integrate all components into a functional prototype for real-time thermal monitoring, control, and fault detection.

# CHAPTER 6

## RESULTS AND DISCUSSION

## CHAPTER 6

# RESULTS AND DISCUSSION

### 6.1 INTRODUCTION

The results and discussion chapter presents the performance evaluation of the proposed thermal energy generation and cooling system. The prototype was designed to demonstrate heat generation, heat transfer, coolant circulation, temperature monitoring, and heat dissipation using a dual-loop cooling architecture. The system integrates a nichrome-heated aluminum thermal core, coolant circulation loops, heat exchanger blocks, temperature sensors, flow sensors, pumps, and an ESP32-based monitoring unit.

The primary objective of testing was to verify the ability of the system to generate controlled heat, transfer thermal energy through the coolant loop, and remove heat effectively through the secondary cooling system. Various parameters such as temperature, flow rate, coolant performance, and heat transfer characteristics were monitored throughout the experimentation process. The obtained results confirm the successful operation of the proposed prototype and demonstrate its capability as an educational thermal management platform.

### 6.2 Soft Prototype

Before fabricating the physical prototype, a soft prototype was developed to verify the overall system architecture and analyze the interaction between different components. The soft prototype played an important role in minimizing design errors and improving system reliability before hardware implementation.

The soft prototype included the complete electrical and thermal architecture of the system. The ESP32 microcontroller was selected as the central control unit due to its processing capability, multiple communication interfaces, and compatibility with various sensors. Temperature monitoring was implemented using three MAX6675 modules connected to K-type thermocouples. These sensors were positioned to monitor the thermal core temperature, primary coolant temperature, and secondary coolant temperature respectively.

The coolant circulation system was represented using dual-loop architecture. The first loop was designed to absorb heat from the thermal core and transport it to the heat exchanger. The second loop was designed to remove heat from the heat exchanger and dissipate it to the surroundings. Flow monitoring was incorporated using the ZJ-S201 flow sensor to measure coolant movement and verify circulation efficiency.

The soft prototype also included relay-controlled pump operation, fan control, alarm indication through a buzzer, and visual status indication using RGB LEDs. The circuit schematic was prepared using electronic design software, and component connectivity was verified before hardware assembly.

Simulation and architecture verification confirmed that the proposed design could successfully perform temperature monitoring, coolant circulation, and heat transfer functions. The soft prototype significantly reduced implementation challenges and provided confidence for physical development.

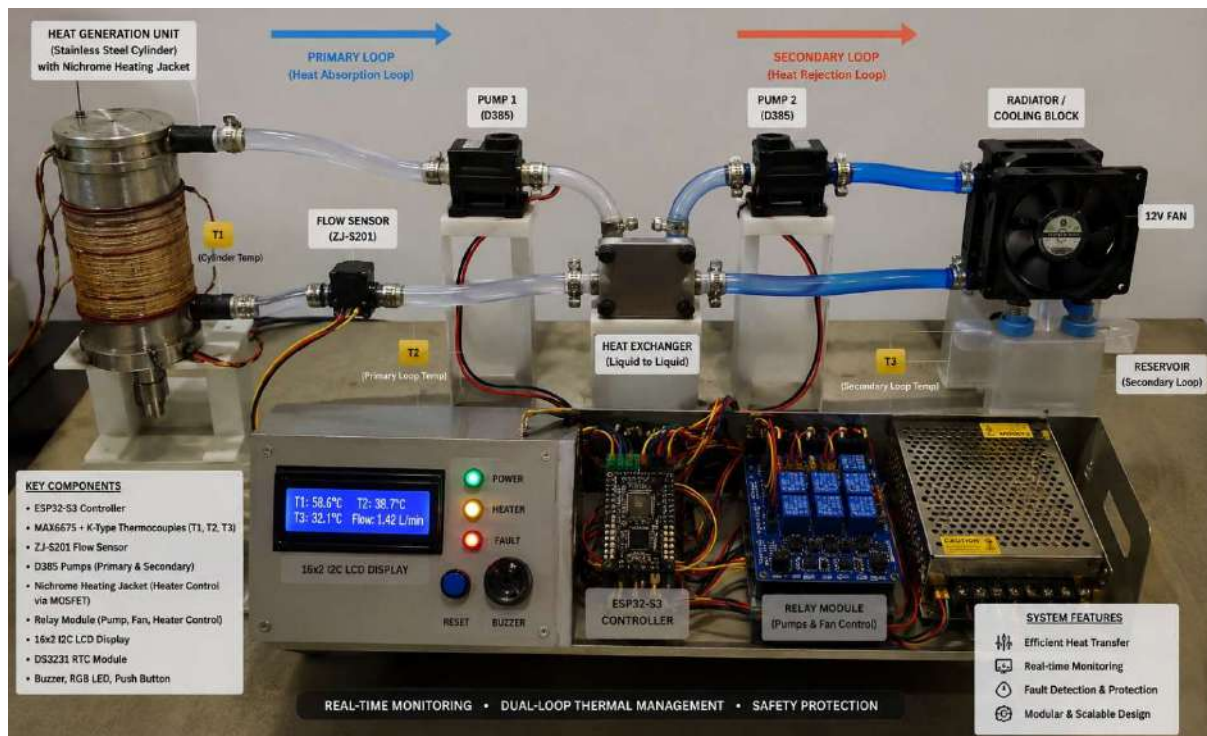


Figure 6.1: Soft Prototype (Ideation Version)

### 6.3 Validation of the Solution

Validation of the proposed solution was performed by testing each subsystem individually and subsequently evaluating the integrated system. The validation process aimed to confirm that the prototype satisfied all functional requirements identified during the design stage.

The heating section was first tested independently. When electrical power was supplied to the nichrome heating element, the aluminum thermal core exhibited a gradual increase in temperature. The thermocouples accurately detected temperature variations and transmitted data to the ESP32 controller through the MAX6675 modules. The displayed values matched the expected thermal behavior of the system.

The primary coolant loop was then activated. The circulation pump successfully transported coolant around the heated thermal core. Temperature measurements indicated that the coolant absorbed heat from the thermal core, resulting in a measurable increase in coolant temperature. This confirmed effective heat absorption by the primary loop.

The heat exchanger assembly was subsequently evaluated. Thermal energy carried by the primary coolant was transferred through the aluminum block heat exchanger to the secondary coolant loop. The temperature difference observed between the inlet and outlet sections demonstrated successful heat exchange. The thermal paste applied between the aluminum blocks improved thermal conductivity and enhanced heat transfer performance.

The secondary cooling loop effectively removed heat from the heat exchanger and transported it to the cooling section. The cooling fan assisted in heat dissipation and contributed to temperature tabilization. Flow sensor readings confirmed continuous coolant circulation throughout the testing period.

The integrated system demonstrated stable operation and achieved all intended objectives. Heat generation, thermal transport, heat exchange, and cooling functions operated as expected. The results validate the effectiveness of the proposed design and confirm the feasibility of using the system for educational demonstrations and thermal engineering studies.

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## 6.4 Performance Parameters (Tabulation and Graphs)

The performance of the system was evaluated using several key parameters including temperature distribution, coolant flow rate, heat transfer effectiveness, and system stability.

### Thermal Core Temperature

The temperature of the aluminum thermal core increased progressively after activation of the heating element. The observed temperature rise confirmed effective conversion of electrical energy into thermal energy. The thermal core served as the primary heat source for the entire system.

### Primary Coolant Temperature

The primary coolant absorbed thermal energy from the heated core and transported it toward the heat exchanger. The increase in coolant temperature verified successful heat extraction from the thermal core.

### Secondary Coolant Temperature

The secondary coolant received thermal energy from the heat exchanger and carried it to the cooling section. Temperature measurements demonstrated the effectiveness of heat transfer between the two loops.

### Flow Rate Performance

Continuous circulation of coolant is essential for effective thermal management. The ZJ-S201 flow sensor monitored coolant movement and provided real-time flow measurements. Stable flow rate values indicated reliable pump operation and consistent heat transfer performance.

The obtained data indicates that the coolant loops successfully removed heat from the thermal core and maintained stable thermal transport throughout operation.

## 6.5 Final Prototype

The final prototype represents the successful integration of thermal, mechanical, and electronic subsystems into a single operational platform. The completed system consists of a heated aluminum thermal core, coolant circulation chambers, heat exchanger assembly, pumps, sensors, ESP32 control system, display interface, and cooling arrangement.

The aluminum thermal core was fabricated using two hemispherical aluminum containers joined together to form a spherical structure. Nichrome wire wrapped around the sphere served as the heat generation element. Ceramic insulation was used to reduce heat loss and improve thermal efficiency. The primary coolant loop transported thermal energy from the heated core to the heat exchanger. The

heat exchanger consisted of two aluminum cooling blocks clamped together with thermal paste applied between their contact surfaces. This arrangement enabled thermal energy transfer from the primary loop to the secondary loop.

The secondary coolant loop removed heat from the heat exchanger and circulated it through the cooling chamber. A fan-assisted cooling arrangement enhanced heat dissipation and improved system stability.

The ESP32 continuously monitored temperature and flow data and displayed the information through the LCD interface. Relay modules-controlled pumps and cooling devices based on operating conditions. The buzzer and RGB LED provided visual and audible status indications.

The final prototype successfully demonstrated heat generation, heat absorption, heat transfer, heat exchange, coolant circulation, and heat rejection processes. The system operated reliably throughout testing and achieved the objectives established during the design phase. The results indicate that the proposed prototype can serve as an effective educational tool for studying thermal engineering concepts and cooling system operation.



Figure 6.2 : Prototype demonstration





Fig 6.3 : Result Graphs

Summary

This section presents the **results and discussion** of the Dual-Loop Controlled Cooling System, evaluating the practical implementation and performance of the proposed thermal architecture. It begins with an introduction to the testing phase, followed by the **soft prototype**, which visually represents the system design and operational flow before physical fabrication. The **validation of the solution** confirms the effectiveness of the dual-loop architecture in improving heat transfer, maintaining temperature stability, and ensuring system safety. Key **performance parameters** such as cylinder temperature, coolant flow rate, primary and secondary loop temperatures, and thermal dissipation efficiency are recorded, tabulated, and graphically analyzed for better understanding. The **final prototype** includes the actual hardware assembly, integrated circuit connections, and live sample outputs displayed through sensors and LCD monitoring. The results demonstrate improved thermal management, real-time monitoring, and fault detection, proving the system’s effectiveness as a low cost and scalable thermal management solution for data centres and AI training centres.



## CHAPTER 7

# CONCLUSION AND FUTURE SCOPE

## CHAPTER 7

### CONCLETION AND FUTURE SCOPE

#### 7.1 CONCLUSION

The Thermal Energy Transfer and Cooling System was successfully designed, developed, and implemented to demonstrate the principles of heat generation, heat transfer, coolant circulation, heat exchange, and real-time monitoring. The project achieved its primary objective of creating a compact thermal management system capable of generating heat using a nichrome-based heating arrangement and transferring the generated thermal energy through a multi-stage cooling architecture.

The aluminum spherical core effectively absorbed and stored thermal energy generated by the heating element. The primary coolant loop successfully extracted heat from the heated chamber and transported it to the heat exchanger. The custom-designed aluminum block heat exchanger enabled efficient thermal energy transfer between the primary and secondary coolant loops without direct fluid mixing. The secondary cooling loop, assisted by forced-air cooling, effectively dissipated heat to the surrounding environment and maintained continuous circulation of the coolant.

The integration of the ESP32-S3 microcontroller with temperature sensors, flow sensors, relay modules, and display units enabled accurate real-time monitoring and control of the system. Temperature measurements obtained from multiple locations confirmed the effectiveness of the heat transfer process, while the flow sensor verified continuous coolant circulation. The LCD display provided a user-friendly interface for observing system parameters, and the implemented safety mechanisms ensured reliable operation under varying conditions.

The experimental results demonstrated that the proposed system successfully performed heat generation, thermal energy transport, heat exchange, and heat dissipation in a controlled and stable manner. The results obtained validated the effectiveness of the design and confirmed the feasibility of using low-cost, readily available components for thermal management applications.

Furthermore, the project provided valuable practical experience in thermal engineering, fluid circulation systems, embedded systems, sensor integration, power electronics, and system automation. The developed prototype serves as an educational platform for understanding thermal energy management and can be further enhanced through the incorporation of advanced heat exchangers,

improved insulation techniques, automated data logging, remote monitoring capabilities, and higher-capacity cooling systems.

In conclusion, the project successfully achieved all the intended objectives and demonstrated an effective and reliable thermal energy transfer system capable of illustrating the fundamental concepts of heat generation, heat transfer, cooling, and intelligent monitoring.

The prototype represents a practical and scalable approach to thermal management and provides a strong foundation for future research and development in thermal energy systems.

## 7.2 Future scope

The developed Thermal Energy Transfer and Cooling System successfully demonstrates the principles of heat generation, heat transfer, coolant circulation, and thermal management. However, several improvements and enhancement

s can be incorporated in future versions to increase system efficiency, reliability, and practical applicability. One potential improvement is the implementation of a high-efficiency heat exchanger. Replacing the current aluminum block heat exchanger with a plate-type or finned heat exchanger can significantly improve the rate of thermal energy transfer and reduce heat losses. Advanced thermal insulation materials can also be incorporated around the heating chamber to minimize energy loss to the surroundings and improve overall thermal efficiency.

The current prototype uses basic temperature and flow monitoring. Future versions can include additional sensors such as pressure sensors, liquid level sensors, and ambient temperature sensors to provide more comprehensive system monitoring and performance analysis. The collected data can be stored using SD card modules or cloud-based databases for long-term performance evaluation and research purposes.

Wireless monitoring and control can be introduced using IoT technologies. The ESP32 controller can transmit real-time system data to a web dashboard or mobile application through Wi-Fi connectivity. This would allow remote monitoring of temperatures, flow rates, pump status, and system alarms from any location.

The cooling subsystem can be upgraded by incorporating dedicated radiators, larger cooling fans, or liquid-to-air heat exchangers to improve heat rejection capability. Variable-speed pumps and

intelligent flow control mechanisms can also be implemented to optimize coolant circulation according to thermal load conditions.

Future development may include automatic temperature regulation using advanced control algorithms such as PID control. This would enable precise temperature maintenance, reduce power consumption, and improve overall system stability.

The system can also be scaled to handle higher thermal loads by increasing heater capacity, coolant flow rates, and heat exchanger size. Such modifications would make the design suitable for larger experimental setups, industrial thermal management studies, and advanced educational demonstrations.

Furthermore, artificial intelligence and machine learning techniques can be integrated for predictive maintenance, fault detection, and thermal performance optimization. These technologies would enhance system reliability and support intelligent decision-making during operation. Overall, the proposed system provides a strong foundation for future research and development in thermal energy management, smart cooling systems, and automated heat transfer applications. The flexibility and modularity of the design allow it to be expanded and adapted for various engineering, educational, and research-oriented applications.

### 7.3 Summary

The future scope of the **Dual-Loop Controlled Cooling System** lies in expanding its application from an educational prototype to an advanced industrial thermal management platform. One major improvement would be integrating **IoT and cloud connectivity**, enabling remote monitoring, predictive maintenance, and real-time thermal analytics. Advanced machine learning algorithms can also be incorporated to optimize coolant flow, predict overheating conditions, and improve overall energy efficiency.

The system can be further enhanced by experimenting with **high-performance coolants** such as nanofluids or phase-change materials to improve heat transfer efficiency. The current liquid-to-liquid heat exchanger can be upgraded to industrial-grade compact heat exchangers for better thermal performance. Future versions may also include **closed-loop PID temperature control**, improving precision and stability.

# CHAPTER 8

## OUR JOURNEY EXPERIENCE AND TAKEAWAY



Name: **Dhanashree**

USN: **1DT25EC039**

Branch: ECE

Throughout this project, I gained valuable practical experience in designing and developing a reactor-inspired thermal energy transfer system. I was involved in understanding heat generation, coolant circulation, heat exchange mechanisms, and sensor integration. While working on the project, I learned how theoretical concepts studied in engineering can be applied to solve real-world problems. Apart from technical knowledge, the IDT (Innovation and Design Thinking) subject played a major role in improving my interpersonal skills. Through interactions with people from different backgrounds, conducting discussions, gathering feedback, and working collaboratively within a team, I developed better communication, presentation, and teamwork skills. This experience taught me the importance of understanding user requirements, effective collaboration, and continuous learning in engineering projects.



Name: **Harshitha L M**

USN: **1DT25EC051**

Branch: ECE

Working on this project was a challenging yet rewarding experience. I learned about advanced cooling systems, coolant properties, heat transfer principles, and system design. The process of selecting components, troubleshooting issues, and optimizing the design enhanced my problem-solving and analytical abilities. Through the IDT subject, I had the opportunity to interact with various individuals, understand different perspectives, and participate in team discussions. These activities helped me improve my communication skills, confidence, leadership qualities, and ability to express technical ideas clearly. The project taught me that successful engineering solutions require both technical expertise and effective human interaction and also deep understanding on critical thinking and hard code implementation.



Name: **Anurag Kumar Singh**

USN: **1DT25EC012**

Branch: ECE

This project provided me with hands-on exposure to electronics integration, sensor interfacing, thermal system design, and prototype development. I learned the importance of careful planning, testing, and coordination while working on a multidisciplinary project. One of the most valuable aspects of this journey was the learning experience gained through the IDT course. By engaging with people, understanding their viewpoints, and working together to solve problems, I developed stronger communication and collaboration skills. The subject helped me appreciate the role of empathy, teamwork, and creative thinking in engineering design. Overall, the project enhanced both my technical knowledge and personal development.



Name: **Ayushman Patro**

USN: **1DT25EC017**

Branch: ECE

My journey through this project has been a continuous process of redesigning, experimenting, and solving real-world engineering challenges. Starting from the idea of aluminum spheres and evolving into heater jacket systems, every stage demanded critical thinking and adaptability. Decisions such as shifting from silicone oil to alternative coolants, redesigning heat transfer mechanisms, and optimizing component integration taught me the practical side of engineering beyond theory. I gained hands-on experience in thermal system design, fluid dynamics, electronics interfacing, embedded control, and safety planning. Most importantly, I learned how simplifying complex concepts leads to practical, buildable, and effective solutions.

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- [37] EKWB, **Industrial Liquid Cooling Solutions User Guide**, 2024.
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- [41] UL Standards, **Electrical Safety Certification for Industrial Equipment**, Underwriters Laboratories.
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- [43] NVIDIA Corporation, **AI Data Center Thermal Design Guide**, 2024.
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ANNEXURE I

DATA SHEET OF SENSORS

# Thermocouple type K

Order nr.: 803910 1211

Thermocouples have a simple structure. The two thermocouple wires are welded at the tip and remain without a protection sleeve. Therefore, thermocouples are characterized by particularly fast response times and a wide range of applications. The thermocouples can be used in a temperature range from -40 °C to +350 °C. You will find thermocouples of type J (Fe-CuNi) and type K (NiCr-Ni), we will be pleased to offer you others on request.



| General Information                           |                                                               |
|-----------------------------------------------|---------------------------------------------------------------|
| Measuring range                               | -40 °C to +350 °C                                             |
| Perm. °C range cable                          | -50 °C to +350 °C                                             |
| Accuracy                                      | -40 °C to +375 °C: ±1,5 °C according to DIN IEC 60584 Class 1 |
| Response time                                 | t63 / t99: information is available on request                |
| Pull-out force                                | ≥ 30 N                                                        |
| Supply and output                             |                                                               |
| Measuring element                             | Thermocouple Type K                                           |
| Measuring point                               | Welding point not insulated                                   |
| Measurement signal                            | Thermovoltage                                                 |
| Ambient conditions                            |                                                               |
| Protection class                              | IP20 according DIN 60529                                      |
| Humidity and moisture condensation resistance | according to application-specific qualification               |
| Certificates and Standards                    |                                                               |
| Standards                                     | DIN EN 61326-1:2013   DIN EN IEC 63000:2019-05                |
| Directive                                     | RoHS 2011/65/EU   2014/30/EU                                  |
| Certificates                                  | Certificate of suitability (on request)                       |

**Customizable options**  
E - Material connection cable  
F - Length connection cable  
G - Connector

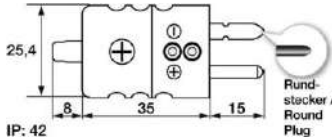
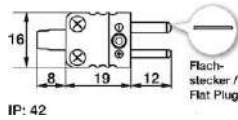


| E - Cable material and configuration connection cable                               |       |                    |                      |      |                         |                       |                  |             |                           |                      |             |                     |
|-------------------------------------------------------------------------------------|-------|--------------------|----------------------|------|-------------------------|-----------------------|------------------|-------------|---------------------------|----------------------|-------------|---------------------|
|  | Code  | Type               | Color                | IP   | From (°C) <sup>1)</sup> | To (°C) <sup>1)</sup> | Outside material | Isol. Leads | Sheath (mm) <sup>2)</sup> | Q (mm <sup>2</sup> ) | Color Leads | Ω / m <sup>4)</sup> |
|                                                                                     | E8321 | Thermocouple cable | Type K <sup>3)</sup> | IP20 | -50                     | +350                  | Glass fibre      | Glass fibre | 2,0 x 1,2                 | 0,2                  | gn, wt      | 2,00                |

Insulation resistance: ≥ 100 MOhm at min. 100 VDC | <sup>1)</sup>Perm. range °C | <sup>2)</sup>Tolerance ± 0,2 mm | <sup>3)</sup>Color according to IEC 584 | <sup>4)</sup>per thermocouple

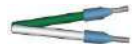
| F - Length |      |      |      |      |      |      |      |      |
|------------|------|------|------|------|------|------|------|------|
| Code       | F010 | F020 | F030 | F040 | F050 | F100 | F150 | F200 |
| m          | 1    | 2    | 3    | 4    | 5    | 10   | 15   | 20   |

## Technical drawing - Connector



| Your order code |                           |                         |           |
|-----------------|---------------------------|-------------------------|-----------|
| Order nr.       | Material connection cable | Length connection cable | Connector |
| 803910 1211     | E_____                    | F_____                  | G_____    |

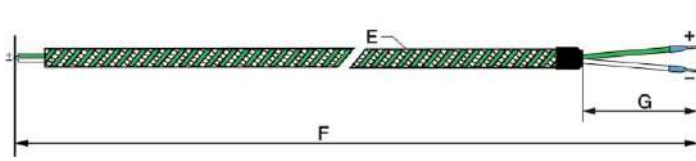
RL / KS / 17.03.2022

| G - Connector                                                                     |                   |                                |                                                                                   |       |                             |
|-----------------------------------------------------------------------------------|-------------------|--------------------------------|-----------------------------------------------------------------------------------|-------|-----------------------------|
| Picture                                                                           | Code              | Feature                        | Bild                                                                              | Code  | Merkmal                     |
|  | G01 <sup>1)</sup> | Insulated end ferrules (50 mm) |                                                                                   |       |                             |
|  | G12               | Mini-TE-Stecker Typ K gn       |  | G19   | Mini-TC connector Type K ye |
|  | G321}             | TE-Stecker Typ K gn            |  | G391} | TC connector Type K ye      |

1)auf Anfrage | Weitere Stecker auf Anfrage erhältlich

# Technical drawing

Version with insulated end ferrules



Konfigurierbare Optionen

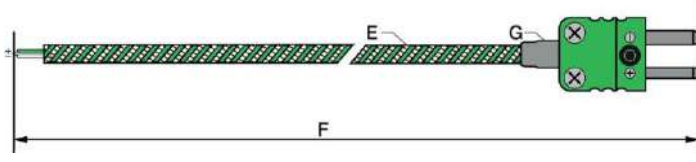
E - Material Anschlussleitung

F - Länge Anschlussleitung

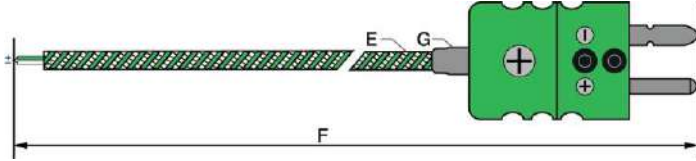
G - Stecker

All dimensions in mm

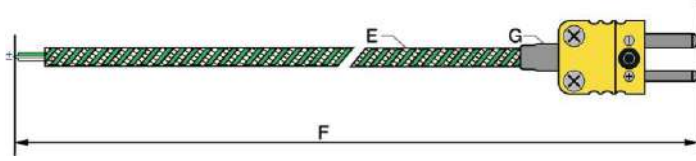
Variante mit Mini-TE-Stecker



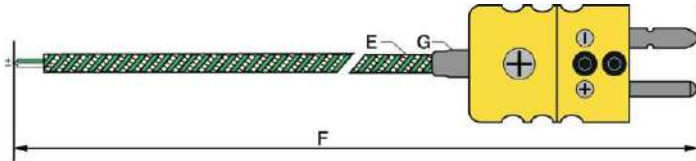
Variante mit TE-Stecker



Version with mini TE connector



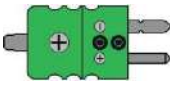
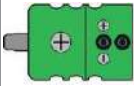
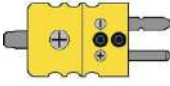
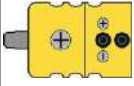
Version with TE connector



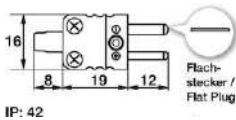
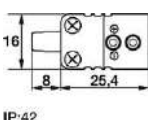
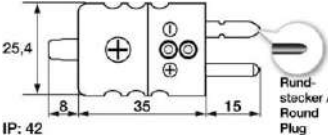
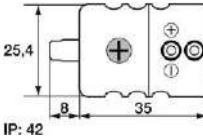
RL / KS / 17.03.2022

# Matching accessories: Stecker

Details of accessories can be found on our website.

| Connector                                                                         |             |                             |                                                                                   |             |                            |
|-----------------------------------------------------------------------------------|-------------|-----------------------------|-----------------------------------------------------------------------------------|-------------|----------------------------|
| Picture                                                                           | Code        | Feature                     | Picture                                                                           | Code        | Feature                    |
|  | 809140 2000 | Mini-TC connector Type K gn |  | 809100 2000 | Mini-TC coupling Type K gn |
|  | 809150 2000 | TC connector Type K gn      |  | 809110 2000 | TC coupling Type K gn      |
|  | 809140 2001 | Mini-TC connector Type K ye |  | 809100 2001 | Mini-TC coupling Type K ye |
|  | 809150 2001 | TC connector Type K ye      |  | 809110 2001 | TC coupling Type K ye      |

1)on request | Other connectors available on request

| Technische Zeichnung - Stecker                                                                                                 |                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
|  <p>Flachstecker / Flat Plug<br/>IP: 42</p> |  <p>IP: 42</p> |
|  <p>Rundstecker / Round Plug<br/>IP: 42</p> |  <p>IP: 42</p> |

Click [here](#) to ask about the production status of specific part numbers.

## MAX6675

## Cold-Junction-Compensated K-Thermocouple-to-Digital Converter (0°C to +1024°C)

### General Description

The MAX6675 performs cold-junction compensation and digitizes the signal from a type-K thermocouple. The data is output in a 12-bit resolution, SPI-compatible, read-only format.

This converter resolves temperatures to 0.25°C, allows readings as high as +1024°C, and exhibits thermocouple accuracy of 8 LSBs for temperatures ranging from 0°C to +700°C.

The MAX6675 is available in a small, 8-pin SO package.

### Applications

- Industrial
- Appliances
- HVAC

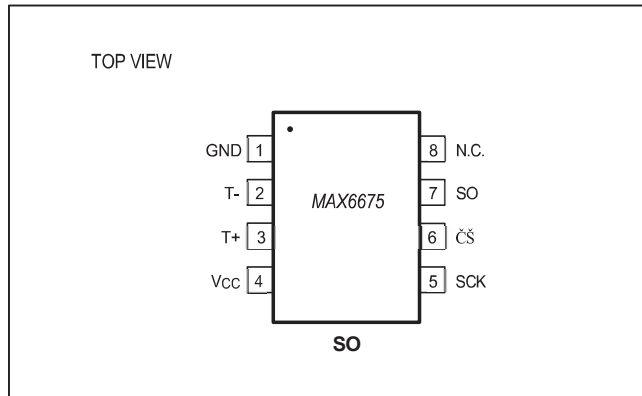
### Features

- Direct Digital Conversion of Type -K Thermocouple Output
- Cold-Junction Compensation
- Simple SPI-Compatible Serial Interface
- 12-Bit, 0.25°C Resolution
- Open Thermocouple Detection

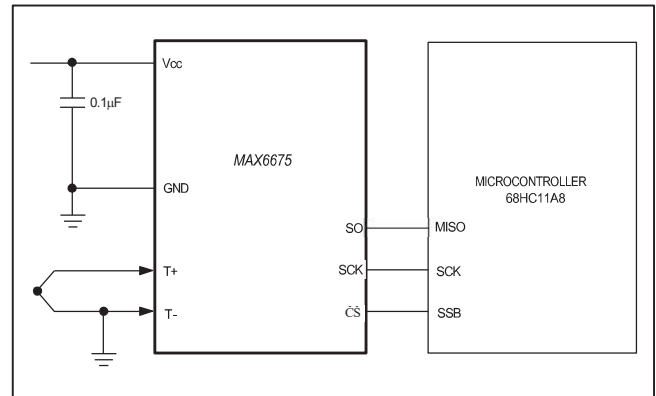
### Ordering Information

| PART       | TEMP RANGE     | PIN-PACKAGE |
|------------|----------------|-------------|
| MAX6675ISA | -20°C to +85°C | 8 SO        |

### Pin Configuration



### Typical Application Circuit



**Absolute Maximum Ratings**

Supply Voltage ( $V_{CC}$  to GND)..... V to +6V  
 SO, SCK, CS, T-, T+ to GND..... V to  $V_{CC} + 0.3V$   
 SO Current ..... 50mA  
 ESD Protection (Human Body Model).....  $\pm 2000V$   
 Continuous Power Dissipation ( $T_A = +70^\circ C$ )  
 8-Pin SO (derate 5.88mW/ $^\circ C$  above  $+70^\circ C$ )..... 471mW  
 Operating Temperature Range.....  $-20^\circ C$  to  $+85^\circ C$

Storage Temperature Range .....  $-65^\circ C$  to  $+150^\circ C$   
 Junction Temperature.....  $+150^\circ C$   
 SO Package  
 Vapor Phase (60s).....  $+215^\circ C$   
 Infrared (15s).....  $+220^\circ C$   
 Lead Temperature (soldering, 10s).....  $+300^\circ C$

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

**Electrical Characteristics**

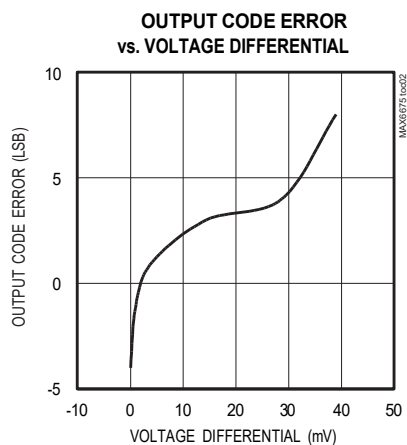
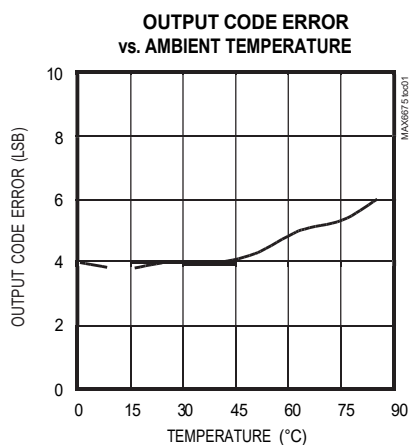
( $V_{CC} = +3.0V$  to  $+5.5V$ ,  $T_A = -20^\circ C$  to  $+85^\circ C$ , unless otherwise noted. Typical values specified at  $+25^\circ C$ .) (Note 1)

| PARAMETER                        | SYMBOL     | CONDITIONS                                                                             | MIN                 | TYP   | MAX                 | UNITS       |
|----------------------------------|------------|----------------------------------------------------------------------------------------|---------------------|-------|---------------------|-------------|
| Temperature Error                |            | $T_{THERMOCOUPLE} = +700^\circ C$ ,<br>$T_A = +25^\circ C$ (Note 2)                    | $V_{CC} = +3.3V$    | -5    | +5                  | LSB         |
|                                  |            |                                                                                        | $V_{CC} = +5V$      | -6    | +6                  |             |
|                                  |            | $T_{THERMOCOUPLE} = 0^\circ C$ to<br>$+700^\circ C$ , $T_A = +25^\circ C$ (Note 2)     | $V_{CC} = +3.3V$    | -8    | +8                  |             |
|                                  |            |                                                                                        | $V_{CC} = +5V$      | -9    | +9                  |             |
|                                  |            | $T_{THERMOCOUPLE} = +700^\circ C$<br>to $+1000^\circ C$ , $T_A = +25^\circ C$ (Note 2) | $V_{CC} = +3.3V$    | -17   | +17                 |             |
|                                  |            |                                                                                        | $V_{CC} = +5V$      | -19   | +19                 |             |
| Thermocouple Conversion Constant |            |                                                                                        |                     | 10.25 |                     | $\mu V/LSB$ |
| Cold-Junction Compensation Error |            | $T_A = -20^\circ C$ to<br>$+85^\circ C$ (Note 2)                                       | $V_{CC} = +3.3V$    | -3.0  | +3.0                | $^\circ C$  |
|                                  |            |                                                                                        | $V_{CC} = +5V$      | -3.0  | +3.0                |             |
| Resolution                       |            |                                                                                        |                     | 0.25  |                     | $^\circ C$  |
| Thermocouple Input Impedance     |            |                                                                                        |                     | 60    |                     | $k\Omega$   |
| Supply Voltage                   | $V_{CC}$   |                                                                                        | 3.0                 |       | 5.5                 | V           |
| Supply Current                   | $I_{CC}$   |                                                                                        |                     | 0.7   | 1.5                 | mA          |
| Power-On Reset Threshold         |            | $V_{CC}$ rising                                                                        | 1                   | 2     | 2.5                 | V           |
| Power-On Reset Hysteresis        |            |                                                                                        |                     | 50    |                     | mV          |
| Conversion Time                  |            | (Note 2)                                                                               |                     | 0.17  | 0.22                | s           |
| <b>SERIAL INTERFACE</b>          |            |                                                                                        |                     |       |                     |             |
| Input Low Voltage                | $V_{IL}$   |                                                                                        |                     |       | $0.3 \times V_{CC}$ | V           |
| Input High Voltage               | $V_{IH}$   |                                                                                        | $0.7 \times V_{CC}$ |       |                     | V           |
| Input Leakage Current            | $I_{LEAK}$ | $V_{IN} = GND$ or $V_{CC}$                                                             |                     |       | $\pm 5$             | $\mu A$     |
| Input Capacitance                | $C_{IN}$   |                                                                                        |                     | 5     |                     | pF          |



**Electrical Characteristics (continued)**(V<sub>CC</sub> = +3.0V to +5.5V, T<sub>A</sub> = -20°C to +85°C, unless otherwise noted. Typical values specified at +25°C.) (Note 1)

| PARAMETER                     | SYMBOL           | CONDITIONS                  | MIN                   | TYP | MAX | UNITS |
|-------------------------------|------------------|-----------------------------|-----------------------|-----|-----|-------|
| Output High Voltage           | V <sub>OH</sub>  | I <sub>SOURCE</sub> = 1.6mA | V <sub>CC</sub> - 0.4 |     |     | V     |
| Output Low Voltage            | V <sub>OL</sub>  | I <sub>SINK</sub> = 1.6mA   |                       |     | 0.4 | V     |
| <b>TIMING</b>                 |                  |                             |                       |     |     |       |
| Serial Clock Frequency        | f <sub>SCL</sub> |                             |                       |     | 4.3 | MHz   |
| SCK Pulse High Width          | t <sub>CH</sub>  |                             | 100                   |     |     | ns    |
| SCK Pulse Low Width           | t <sub>CL</sub>  |                             | 100                   |     |     | ns    |
| CSB Fall to SCK Rise          | t <sub>CSS</sub> | C <sub>L</sub> = 10pF       | 100                   |     |     | ns    |
| CSB Fall to Output Enable     | t <sub>DV</sub>  | C <sub>L</sub> = 10pF       |                       |     | 100 | ns    |
| CSB Rise to Output Disable    | t <sub>TR</sub>  | C <sub>L</sub> = 10pF       |                       |     | 100 | ns    |
| SCK Fall to Output Data Valid | t <sub>DO</sub>  | C <sub>L</sub> = 10pF       |                       |     | 100 | ns    |

**Note 1:** All specifications are 100% tested at T<sub>A</sub> = +25°C. Specification limits over temperature (T<sub>A</sub> = T<sub>MIN</sub> to T<sub>MAX</sub>) are guaranteed by design and characterization, not production tested.**Note 2:** Guaranteed by design. Not production tested.**Typical Operating Characteristics**(V<sub>CC</sub> = +3.3V, T<sub>A</sub> = +25°C, unless otherwise noted.)

Pin Description

| PIN | NAME | FUNCTION                                                                      |
|-----|------|-------------------------------------------------------------------------------|
| 1   | GND  | Ground                                                                        |
| 2   | T-   | Alumel Lead of Type-K Thermocouple. Should be connected to ground externally. |
| 3   | T+   | Chromel Lead of Type-K Thermocouple                                           |
| 4   | VCC  | Positive Supply. Bypass with a 0.1μF capacitor to GND.                        |
| 5   | SCK  | Serial Clock Input                                                            |
| 6   | CS   | Chip Select. Set CS low to enable the serial interface.                       |
| 7   | SO   | Serial Data Output                                                            |
| 8   | N.C. | No Connection                                                                 |

Detailed Description

The MAX6675 is a sophisticated thermocouple-to-digital converter with a built-in 12-bit analog-to-digital converter (ADC). The MAX6675 also contains cold-junction compensation sensing and correction, a digital controller, an SPI-compatible interface, and associated control logic.

The MAX6675 is designed to work in conjunction with an external microcontroller (μC) or other intelligence in thermostatic, process-control, or monitoring applications.

Temperature Conversion

The MAX6675 includes signal-conditioning hardware to convert the thermocouple’s signal into a voltage compatible with the input channels of the ADC. The T+and T- inputs connect to internal circuitry that reduces the introduction of noise errors from the thermocouple wires.

Before converting the thermoelectric voltages into equivalent temperature values, it is necessary to compensate for the difference between the thermocouple cold-junction side (MAX6675 ambient temperature) and a 0°C virtual reference. For a type-K thermocouple, the voltage changes by 41μV/°C, which approximates the thermocouple characteristic with the following linear equation:

$$V_{OUT} = (41\mu V / ^\circ C) \times (T_R - T_{AMB})$$

Where:

V<sub>OUT</sub> is the thermocouple output voltage (μV).

T<sub>R</sub> is the temperature of the remote thermocouple junction (°C).

T<sub>AMB</sub> is the ambient temperature (°C).

Cold-Junction Compensation

The function of the thermocouple is to sense a difference in temperature between two ends of the thermocouple wires. The thermocouple’s hot junction can be read from 0°C to +1023.75°C. The cold end (ambient temperature of the board on which the MAX6675 is mounted) can only range from -20°C to +85°C. While the temperature at the cold end fluctuates, the MAX6675 continues to accurately sense the temperature difference at the opposite end.

The MAX6675 senses and corrects for the changes in the ambient temperature with cold-junction compensation. The device converts the ambient temperature reading into a voltage using a temperature-sensing diode. To make the actual thermocouple temperature measurement, the MAX6675 measures the voltage from the thermocouple’s output and from the sensing diode. The device’s internal circuitry passes the diode’s voltage (sensing ambient temperature) and thermocouple voltage (sensing remote temperature minus ambient temperature) to the conversion function stored in the ADC to calculate the thermocouple’s hot-junction temperature.

Optimal performance from the MAX6675 is achieved when the thermocouple cold junction and the MAX6675 are at the same temperature. Avoid placing heat-generating devices or components near the MAX6675 because this may produce cold-junction-related errors.

Digitization

The ADC adds the cold-junction diode measurement with the amplified thermocouple voltage and reads out the 12-bit result onto the SO pin. A sequence of all zeros means the thermocouple reading is 0°C. A sequence of all ones means the thermocouple reading is +1023.75°C.

## Applications Information

### Serial Interface

The *Typical Application Circuit* shows the MAX6675 interfaced with a microcontroller. In this example, the MAX6675 processes the reading from the thermocouple and transmits the data through a serial interface. Force CS low and apply a clock signal at SCK to read the results at SO. Forcing CS low immediately stops any conversion process. Initiate a new conversion process by forcing CS high.

Force CS low to output the first bit on the SO pin. A complete serial interface read requires 16 clock cycles. Read the 16 output bits on the falling edge of the clock. The first bit, D15, is a dummy sign bit and is always zero. Bits D14–D3 contain the converted temperature in the order of MSB to LSB. Bit D2 is normally low and goes high when the thermocouple input is open. D1 is low to provide a device ID for the MAX6675 and bit D0 is three-state.

[Figure 1a](#) is the serial interface protocol and [Figure 1b](#) shows the serial interface timing. [Figure 2](#) is the SO output.

### Open Thermocouple

Bit D2 is normally low and goes high if the thermocouple input is open. In order to allow the operation of the open thermocouple detector, T<sub>-</sub> must be grounded. Make the ground connection as close to the GND pin as possible.

### Noise Considerations

The accuracy of the MAX6675 is susceptible to power-supply coupled noise. The effects of power-supply noise can be minimized by placing a 0.1µF ceramic bypass capacitor close to the supply pin of the device.

### Thermal Considerations

Self-heating degrades the temperature measurement accuracy of the MAX6675 in some applications. The magnitude of the temperature errors depends on the thermal conductivity of the MAX6675 package, the mounting technique, and the effects of airflow. Use a large ground plane to improve the temperature measurement accuracy of the MAX6675.

The accuracy of a thermocouple system can also be improved by following these precautions:

- Use the largest wire possible that does not shunt heat away from the measurement area.
- If small wire is required, use it only in the region of the measurement and use extension wire for the region with no temperature gradient.
- Avoid mechanical stress and vibration, which could strain the wires.
- When using long thermocouple wires, use a twisted-pair extension wire.
- Avoid steep temperature gradients.
- Try to use the thermocouple wire well within its temperature rating.
- Use the proper sheathing material in hostile environments to protect the thermocouple wire.
- Use extension wire only at low temperatures and only in regions of small gradients.
- Keep an event log and a continuous record of thermocouple resistance.

### Reducing Effects of Pick-Up Noise

The input amplifier (A1) is a low-noise amplifier designed to enable high-precision input sensing. Keep the thermocouple and connecting wires away from electrical noise sources.

MAX6675

Cold-Junction-Compensated K-Thermocouple-to-Digital Converter (0°C to +1024°C)

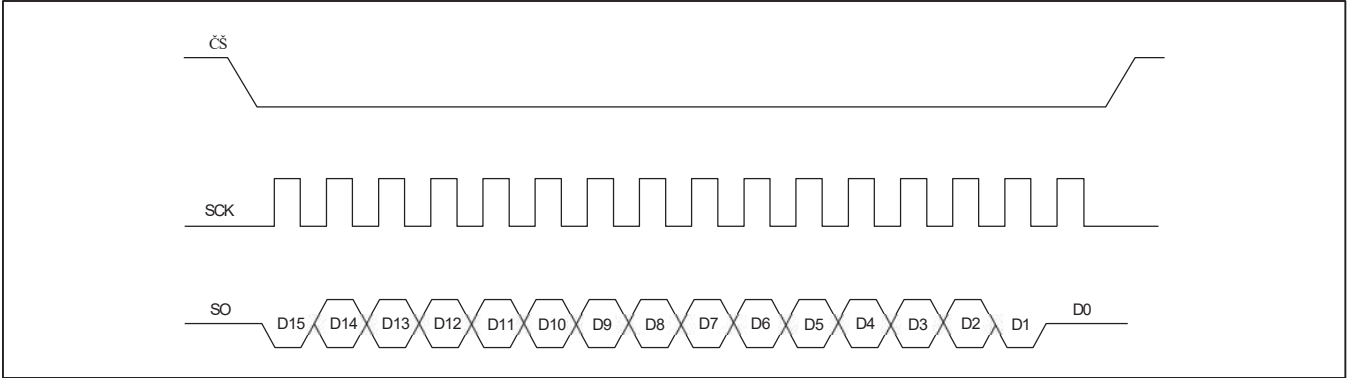


Figure 1a. Serial Interface Protocol

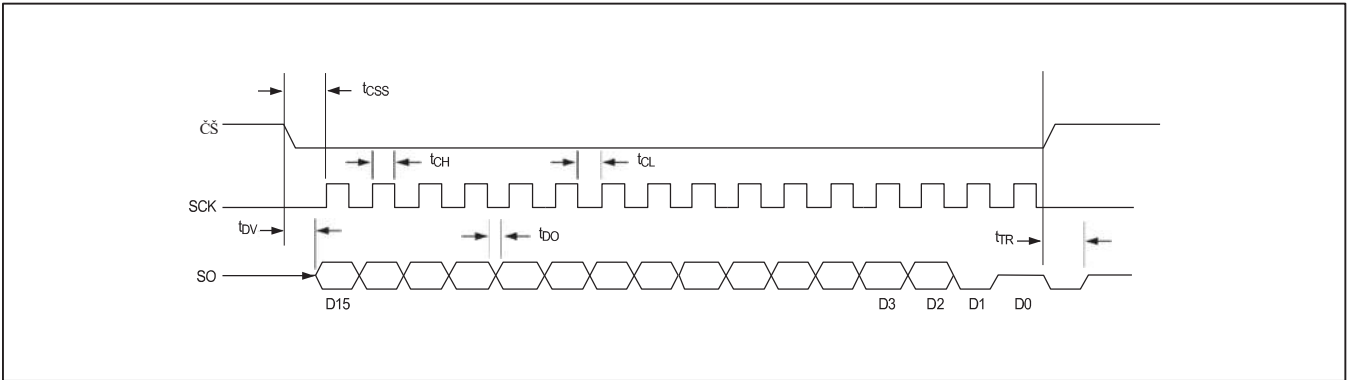


Figure 1b. Serial Interface Timing

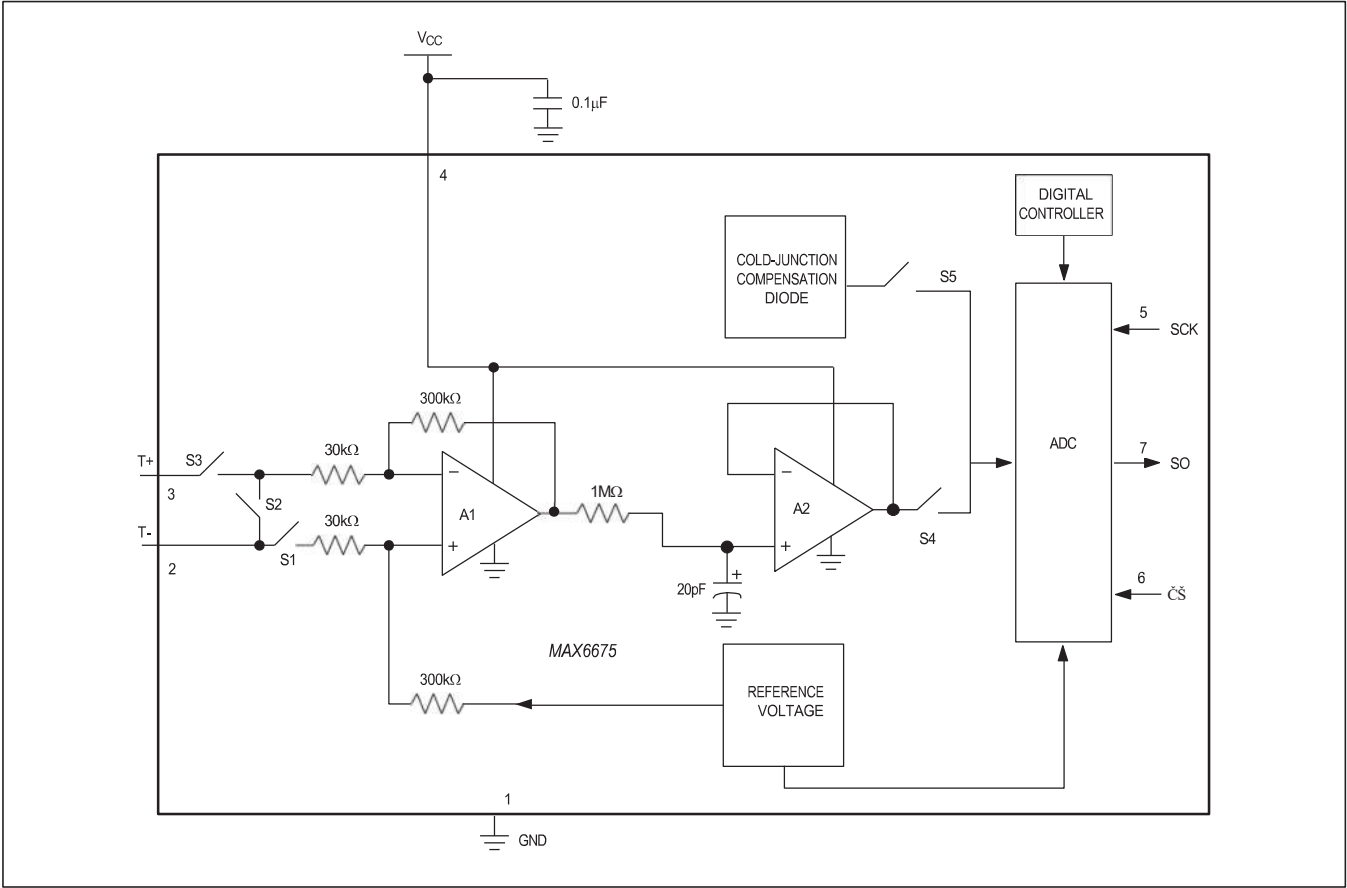
| BIT | DUMMY<br>SIGN BIT | 12-BIT<br>TEMPERATURE READING |    |    |    |    |   |   |   |   |   |   |     | THERMOCOUPLE<br>INPUT | DEVICE<br>ID | STATE           |
|-----|-------------------|-------------------------------|----|----|----|----|---|---|---|---|---|---|-----|-----------------------|--------------|-----------------|
| Bit | 15                | 14                            | 13 | 12 | 11 | 10 | 9 | 8 | 7 | 6 | 5 | 4 | 3   | 2                     | 1            | 0               |
|     | 0                 | MSB                           |    |    |    |    |   |   |   |   |   |   | LSB |                       | 0            | Three-<br>state |

Figure 2. SO Output

MAX6675

Cold-Junction-Compensated K-Thermocouple-to-Digital Converter (0°C to +1024°C)

Block Diagram



Chip Information

TRANSISTOR COUNT: 6720  
PROCESS: BiCMOS

Package Information

For the latest package outline information and land patterns (footprints), go to [www.maximintegrated.com/packages](http://www.maximintegrated.com/packages). Note that a "+", "#", or "-" in the package code indicates RoHS status only. Package drawings may show a different suffix character, but the drawing pertains to the package regardless of RoHS status.

| PACKAGE TYPE | PACKAGE CODE | OUTLINE NO. | LAND PATTERN NO. |
|--------------|--------------|-------------|------------------|
| 8 SO         | S8+2         | 21-0041     | 90-0096          |

**Revision History**

| REVISION<br>NUMBER | REVISION<br>DATE | DESCRIPTION                                                  | PAGES<br>CHANGED |
|--------------------|------------------|--------------------------------------------------------------|------------------|
| 2                  | 4/14             | Removed automotive reference                                 | 1                |
| 3                  | 6/21             | Updated equation in <i>Temperature Compensation</i> section. | 4                |

For pricing, delivery, and ordering information, please visit Maxim Integrated's online storefront at <https://www.maximintegrated.com/en/storefront/storefront.html>.

*Maxim Integrated cannot assume responsibility for use of any circuitry other than circuitry entirely embodied in a Maxim Integrated product. No circuit patent licenses are implied. Maxim Integrated reserves the right to change the circuitry and specifications without notice at any time. The parametric values (min and max limits) shown in the Electrical Characteristics table are guaranteed. Other parametric values quoted in this data sheet are provided for guidance.*

MODEL: YF-S201

Description:

Water flow sensor consists of a plastic valve body, a water rotor, and a hall-effect sensor. When water flows through the rotor, rotor rolls. Its speed changes with different rate of flow. The hall-effect sensor outputs the corresponding pulse signal. This one is suitable to detect flow in water dispenser or coffee machine. We have a comprehensive line of water flow sensors in different diameters. Check them out to find the one that meets your need most.

Features:

- Compact, Easy to Install
- High Sealing Performance
- High Quality Hall Effect Sensor
- RoHS Compliant

Specifications:

- Working Voltage: DC 4.5V~24V
- Normal Voltage: DC 5V~18V
- Max. Working Current: 15mA (DC 5V)
- Load capacity: ≤ 10 mA (DC 5V)
- Flow Rate Range: 1~30L/min
- Load Capacity: ≤10mA (DC 5V)
- Operating Temperature: ≤80°C
- Liquid Temperature: ≤120°C
- Operating Humidity: 35% ~ 90%RH
- Allowing Pressure: ≤1.75MPa
- Storage Temperature: -25 ~ + 80°C
- Storage Humidity: 25% ~ 95%RH
- Electric strength 1250V/min
- Insulation resistance ≥ 100MΩ
- External threads: 1/2"
- Outer diameter: 20mm
- Intake diameter: 9mm
- Outlet diameter: 12mm



Application:

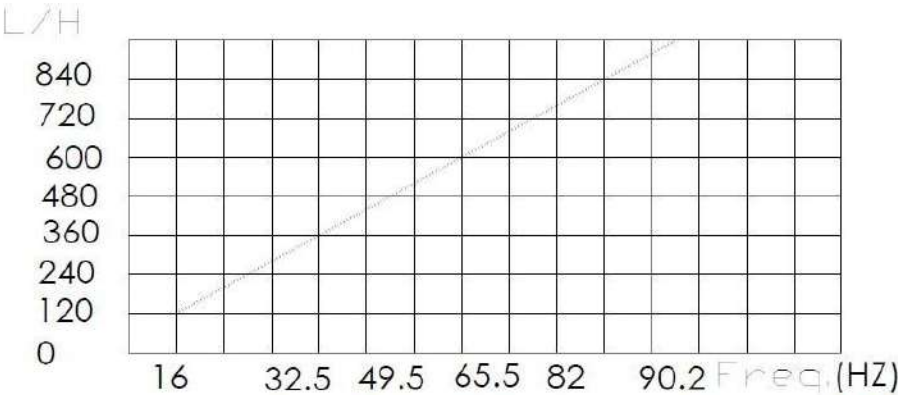
Water heaters, credit card machines, water vending machine, flow measurement device!

Cercuit:

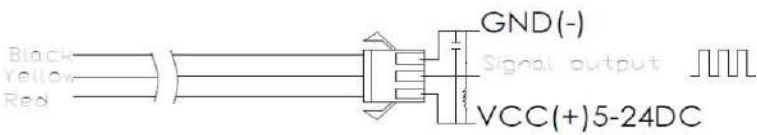
- Red: Positive
- Black: GND
- Yellow: Output signal

Flow Range: 100L/H-/1800H-L/H

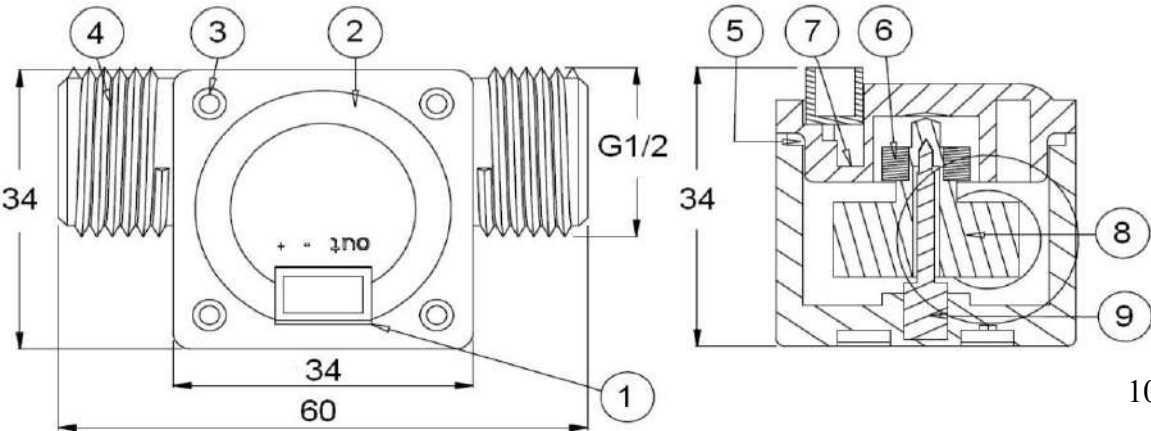
| Flow (L/H) | Frezq.(Hz) | Erro range |
|------------|------------|------------|
| 120        | 16         | ±10<br>5%  |
| 240        | 32.5       |            |
| 360        | 49.3       |            |
| 480        | 65.5       |            |
| 600        | 82         |            |
| 720        | 90.2       |            |



Connection method:



| N° | Item         | Material    |
|----|--------------|-------------|
| 1  | Wire         | PVC         |
| 2  | Bonnet       | PA          |
| 3  | Screw        | Zinc Plated |
| 4  | Valve Body   | PA          |
| 5  | Press Valve  |             |
| 6  | Magnet       |             |
| 7  | Hall         |             |
| 8  | Impeller     | POM         |
| 9  | Steel Sharft | SUS304      |



## DS3231

## Extremely Accurate I2C-Integrated RTC/TCXO/Crystal

### General Description

The DS3231 is a low-cost, extremely accurate I<sup>2</sup>C real-time clock (RTC) with an integrated temperature-compensated crystal oscillator (TCXO) and crystal. The device incorporates a battery input, and maintains accurate timekeeping when main power to the device is interrupted. The integration of the crystal resonator enhances the long-term accuracy of the device as well as reduces the piece-part count in a manufacturing line. The DS3231 is available in commercial and industrial temperature ranges, and is offered in a 16-pin, 300-mil SO package.

The RTC maintains seconds, minutes, hours, day, date, month, and year information. The date at the end of the month is automatically adjusted for months with fewer than 31 days, including corrections for leap year. The clock operates in either the 24-hour or 12-hour format with an AM/PM indicator. Two programmable time-of-day alarms and a programmable square-wave output are provided. Address and data are transferred serially through an I<sup>2</sup>C bidirectional bus.

A precision temperature-compensated voltage reference and comparator circuit monitors the status of V<sub>CC</sub> to detect power failures, to provide a reset output, and to automatically switch to the backup supply when necessary. Additionally, the RST pin is monitored as a pushbutton input for generating a  $\mu$ P reset.

### Benefits and Features

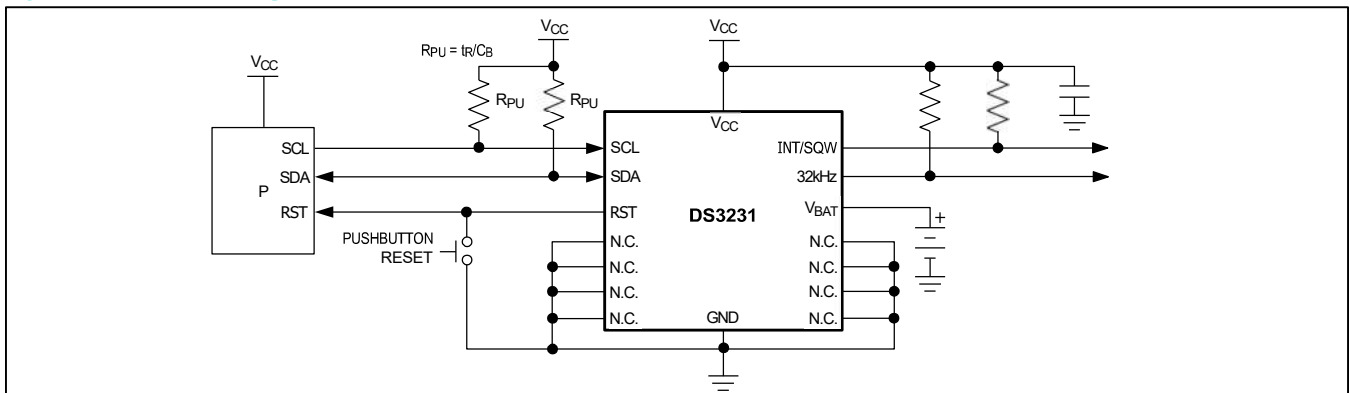
- Highly Accurate RTC Completely Manages All Timekeeping Functions
  - Real-Time Clock Counts Seconds, Minutes, Hours, Date of the Month, Month, Day of the Week, and Year, with Leap-Year Compensation Valid Up to 2100
  - Accuracy  $\pm 2$ ppm from 0°C to +40°C
  - Accuracy  $\pm 3.5$ ppm from -40°C to +85°C
  - Digital Temp Sensor Output:  $\pm 3^\circ\text{C}$  Accuracy
  - Register for Aging Trim
  - RST Output/Pushbutton Reset Debounce Input
  - Two Time-of-Day Alarms
  - Programmable Square-Wave Output Signal
- Simple Serial Interface Connects to Most Microcontrollers
  - Fast (400kHz) I<sup>2</sup>C Interface
- Battery-Backup Input for Continuous Timekeeping
  - Low Power Operation Extends Battery-Backup Run Time
  - 3.3V Operation
- Operating Temperature Ranges: Commercial (0°C to +70°C) and Industrial (-40°C to +85°C)
- Underwriters Laboratories® (UL) Recognized

### Applications

- Servers
- Telematics
- Utility Power Meters
- GPS

*Ordering Information and Pin Configuration appear at end of data sheet.*

### Typical Operating Circuit



*Underwriters Laboratories is a registered certification mark of Underwriters Laboratories Inc.*



## Absolute Maximum Ratings

Voltage Range on Any Pin Relative to Ground ..... V to +6.0V  
 Junction-to-Ambient Thermal Resistance ( $\theta_{JA}$ ) (Note 1) 73°C/W  
 Junction-to-Case Thermal Resistance ( $\theta_{JC}$ ) (Note 1) ..... 23°C/W  
 Operating Temperature Range  
   DS3231S ..... 0°C to +70°C  
   DS3231SN ..... -40°C to +85°C

Junction Temperature ..... +125°C  
 Storage Temperature Range ..... -40°C to +85°C  
 Lead Temperature (soldering, 10s) ..... +260°C  
 Soldering Temperature (reflow, 2 times max) ..... +260°C  
 (see the *Handling, PCB Layout, and Assembly* section)

**Note 1:** Package thermal resistances were obtained using the method described in JEDEC specification JESD51-7, using a four-layer board. For detailed information on package thermal considerations, refer to [www.maximintegrated.com/thermal-tutorial](http://www.maximintegrated.com/thermal-tutorial).

*Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.*

## Recommended Operating Conditions

( $T_A = T_{MIN}$  to  $T_{MAX}$ , unless otherwise noted.) (Notes 2, 3)

| PARAMETER              | SYMBOL    | CONDITIONS | MIN            | TYP | MAX            | UNITS |
|------------------------|-----------|------------|----------------|-----|----------------|-------|
| Supply Voltage         | $V_{CC}$  |            | 2.3            | 3.3 | 5.5            | V     |
|                        | $V_{BAT}$ |            | 2.3            | 3.0 | 5.5            | V     |
| Logic 1 Input SDA, SCL | $V_{IH}$  |            | 0.7 x $V_{CC}$ |     | $V_{CC} + 0.3$ | V     |
| Logic 0 Input SDA, SCL | $V_{IL}$  |            | -0.3           |     | 0.3 x $V_{CC}$ | V     |

## Electrical Characteristics

( $V_{CC} = 2.3V$  to  $5.5V$ ,  $V_{CC}$  = Active Supply (see Table 1),  $T_A = T_{MIN}$  to  $T_{MAX}$ , unless otherwise noted.) (Typical values are at  $V_{CC} = 3.3V$ ,  $V_{BAT} = 3.0V$ , and  $T_A = +25^\circ C$ , unless otherwise noted.) (Notes 2, 3)

| PARAMETER                                    | SYMBOL        | CONDITIONS                                                              | MIN              | TYP   | MAX  | UNITS   |
|----------------------------------------------|---------------|-------------------------------------------------------------------------|------------------|-------|------|---------|
| Active Supply Current                        | $I_{CCA}$     | (Notes 4, 5)                                                            | $V_{CC} = 3.63V$ |       | 200  | $\mu A$ |
|                                              |               |                                                                         | $V_{CC} = 5.5V$  |       | 300  |         |
| Standby Supply Current                       | $I_{CCS}$     | I <sup>2</sup> C bus inactive, 32kHz output on, SQW output off (Note 5) | $V_{CC} = 3.63V$ |       | 110  | $\mu A$ |
|                                              |               |                                                                         | $V_{CC} = 5.5V$  |       | 170  |         |
| Temperature Conversion Current               | $I_{CCSCONV}$ | I <sup>2</sup> C bus inactive, 32kHz output on, SQW output off          | $V_{CC} = 3.63V$ |       | 575  | $\mu A$ |
|                                              |               |                                                                         | $V_{CC} = 5.5V$  |       | 650  |         |
| Power-Fail Voltage                           | $V_{PF}$      |                                                                         | 2.45             | 2.575 | 2.70 | V       |
| Logic 0 Output, 32kHz, INT/SQW, SDA          | $V_{OL}$      | $I_{OL} = 3mA$                                                          |                  |       | 0.4  | V       |
| Logic 0 Output, RST                          | $V_{OL}$      | $I_{OL} = 1mA$                                                          |                  |       | 0.4  | V       |
| Output Leakage Current 32kHz, INT/SQW, SDA   | $I_{LO}$      | Output high impedance                                                   | -1               | 0     | +1   | $\mu A$ |
| Input Leakage SCL                            | $I_{LI}$      |                                                                         | -1               |       | +1   | $\mu A$ |
| RST Pin I/O Leakage                          | $I_{OL}$      | RST high impedance (Note 6)                                             | -200             |       | +10  | $\mu A$ |
| $V_{BAT}$ Leakage Current ( $V_{CC}$ Active) | $I_{BATLKG}$  |                                                                         |                  | 25    | 100  | nA      |

## Electrical Characteristics (continued)

( $V_{CC} = 2.3V$  to  $5.5V$ ,  $V_{CC}$  = Active Supply (see Table 1),  $T_A = T_{MIN}$  to  $T_{MAX}$ , unless otherwise noted.) (Typical values are at  $V_{CC} = 3.3V$ ,  $V_{BAT} = 3.0V$ , and  $T_A = +25^\circ C$ , unless otherwise noted.) (Notes 2, 3)

| PARAMETER                                        | SYMBOL             | CONDITIONS                                               | MIN            | TYP    | MAX  | UNITS |
|--------------------------------------------------|--------------------|----------------------------------------------------------|----------------|--------|------|-------|
| Output Frequency                                 | $f_{OUT}$          | $V_{CC} = 3.3V$ or $V_{BAT} = 3.3V$                      |                | 32.768 |      | kHz   |
| Frequency Stability vs. Temperature (Commercial) | $\Delta f/f_{OUT}$ | $V_{CC} = 3.3V$ or $V_{BAT} = 3.3V$ , aging offset = 00h | 0°C to +40°C   |        | ±2   | ppm   |
|                                                  |                    |                                                          | >40°C to +70°C |        | ±3.5 |       |
| Frequency Stability vs. Temperature (Industrial) | $\Delta f/f_{OUT}$ | $V_{CC} = 3.3V$ or $V_{BAT} = 3.3V$ , aging offset = 00h | -40°C to <0°C  |        | ±3.5 | ppm   |
|                                                  |                    |                                                          | 0°C to +40°C   |        | ±2   |       |
|                                                  |                    |                                                          | >40°C to +85°C |        | ±3.5 |       |
| Frequency Stability vs. Voltage                  | $\Delta f/V$       |                                                          |                | 1      |      | ppm/V |
| Trim Register Frequency Sensitivity per LSB      | $\Delta f/LSB$     | Specified at:                                            | -40°C          | 0.7    |      | ppm   |
|                                                  |                    |                                                          | +25°C          | 0.1    |      |       |
|                                                  |                    |                                                          | +70°C          | 0.4    |      |       |
|                                                  |                    |                                                          | +85°C          | 0.8    |      |       |
| Temperature Accuracy                             | Temp               | $V_{CC} = 3.3V$ or $V_{BAT} = 3.3V$                      | -3             |        | +3   | °C    |
| Crystal Aging                                    | $\Delta f/f_O$     | After reflow, not production tested                      | First year     | ±1.0   |      | ppm   |
|                                                  |                    |                                                          | 0–10 years     | ±5.0   |      |       |

## Electrical Characteristics

( $V_{CC} = 0V$ ,  $V_{BAT} = 2.3V$  to  $5.5V$ ,  $T_A = T_{MIN}$  to  $T_{MAX}$ , unless otherwise noted.) (Note 2)

| PARAMETER                      | SYMBOL      | CONDITIONS                                                                         | MIN               | TYP  | MAX | UNITS |
|--------------------------------|-------------|------------------------------------------------------------------------------------|-------------------|------|-----|-------|
| Active Battery Current         | $I_{BATA}$  | EOSC = 0, BBSQW = 0, SCL = 400kHz (Note 5)                                         | $V_{BAT} = 3.63V$ |      | 70  | μA    |
|                                |             |                                                                                    | $V_{BAT} = 5.5V$  |      | 150 |       |
| Timekeeping Battery Current    | $I_{BATT}$  | EOSC = 0, BBSQW = 0, EN32kHz = 1, SCL = SDA = 0V or SCL = SDA = $V_{BAT}$ (Note 5) | $V_{BAT} = 3.63V$ | 0.84 | 3.0 | μA    |
|                                |             |                                                                                    | $V_{BAT} = 5.5V$  | 1.0  | 3.5 |       |
| Temperature Conversion Current | $I_{BATTC}$ | EOSC = 0, BBSQW = 0, SCL = SDA = 0V or SCL = SDA = $V_{BAT}$                       | $V_{BAT} = 3.63V$ |      | 575 | μA    |
|                                |             |                                                                                    | $V_{BAT} = 5.5V$  |      | 650 |       |
| Data-Retention Current         | $I_{BATDR}$ | EOSC = 1, SCL = SDA = 0V, +25°C                                                    |                   |      | 100 | nA    |

## AC Electrical Characteristics

(V<sub>CC</sub> = V<sub>CC(MIN)</sub> to V<sub>CC(MAX)</sub> or V<sub>BAT</sub> = V<sub>BAT(MIN)</sub> to V<sub>BAT(MAX)</sub>, V<sub>BAT</sub> > V<sub>CC</sub>, T<sub>A</sub> = T<sub>MIN</sub> to T<sub>MAX</sub>, unless otherwise noted.) (Note 2)

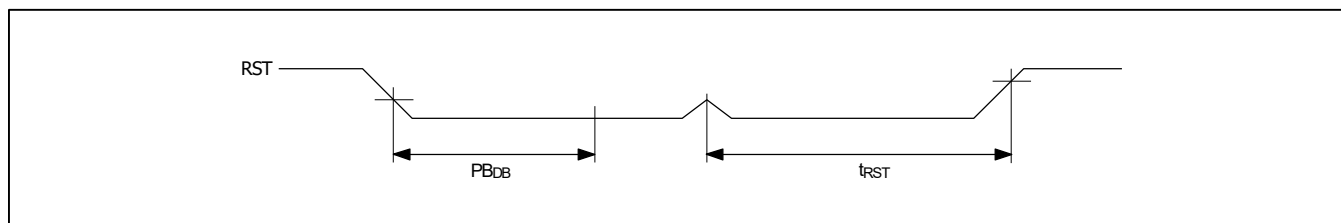
| PARAMETER                                                         | SYMBOL              | CONDITIONS    | MIN               | TYP | MAX  | UNITS |
|-------------------------------------------------------------------|---------------------|---------------|-------------------|-----|------|-------|
| SCL Clock Frequency                                               | f <sub>SCL</sub>    | Fast mode     | 100               |     | 400  | kHz   |
|                                                                   |                     | Standard mode | 0                 |     | 100  |       |
| Bus Free Time Between STOP and START Conditions                   | t <sub>BUF</sub>    | Fast mode     | 1.3               |     |      | μs    |
|                                                                   |                     | Standard mode | 4.7               |     |      |       |
| Hold Time (Repeated) START Condition (Note 7)                     | t <sub>HD:STA</sub> | Fast mode     | 0.6               |     |      | μs    |
|                                                                   |                     | Standard mode | 4.0               |     |      |       |
| Low Period of SCL Clock                                           | t <sub>LOW</sub>    | Fast mode     | 1.3               |     |      | μs    |
|                                                                   |                     | Standard mode | 4.7               |     |      |       |
| High Period of SCL Clock                                          | t <sub>HIGH</sub>   | Fast mode     | 0.6               |     |      | μs    |
|                                                                   |                     | Standard mode | 4.0               |     |      |       |
| Data Hold Time (Notes 8, 9)                                       | t <sub>HD:DAT</sub> | Fast mode     | 0                 |     | 0.9  | μs    |
|                                                                   |                     | Standard mode | 0                 |     | 0.9  |       |
| Data Setup Time (Note 10)                                         | t <sub>SU:DAT</sub> | Fast mode     | 100               |     |      | ns    |
|                                                                   |                     | Standard mode | 250               |     |      |       |
| START Setup Time                                                  | t <sub>SU:STA</sub> | Fast mode     | 0.6               |     |      | μs    |
|                                                                   |                     | Standard mode | 4.7               |     |      |       |
| Rise Time of Both SDA and SCL Signals (Note 11)                   | t <sub>R</sub>      | Fast mode     | 20 + _____        |     | 300  | ns    |
|                                                                   |                     | Standard mode | 0.1C <sub>B</sub> |     | 1000 |       |
| Fall Time of Both SDA and SCL Signals (Note 11)                   | t <sub>F</sub>      | Fast mode     | 20 + _____        |     | 300  | ns    |
|                                                                   |                     | Standard mode | 0.1C <sub>B</sub> |     | 300  |       |
| Setup Time for STOP Condition                                     | t <sub>SU:STO</sub> | Fast mode     | 0.6               |     |      | μs    |
|                                                                   |                     | Standard mode | 4.7               |     |      |       |
| Capacitive Load for Each Bus Line                                 | C <sub>B</sub>      | (Note 11)     |                   |     | 400  | pF    |
| Capacitance for SDA, SCL                                          | C <sub>I/O</sub>    |               |                   | 10  |      | pF    |
| Pulse Width of Spikes That Must Be Suppressed by the Input Filter | t <sub>SP</sub>     |               |                   | 30  |      | ns    |
| Pushbutton Debounce                                               | PB <sub>DB</sub>    |               |                   | 250 |      | ms    |
| Reset Active Time                                                 | t <sub>RST</sub>    |               |                   | 250 |      | ms    |
| Oscillator Stop Flag (OSF) Delay                                  | t <sub>OSF</sub>    | (Note 12)     |                   | 100 |      | ms    |
| Temperature Conversion Time                                       | t <sub>CONV</sub>   |               |                   | 125 | 200  | ms    |

## Power-Switch Characteristics

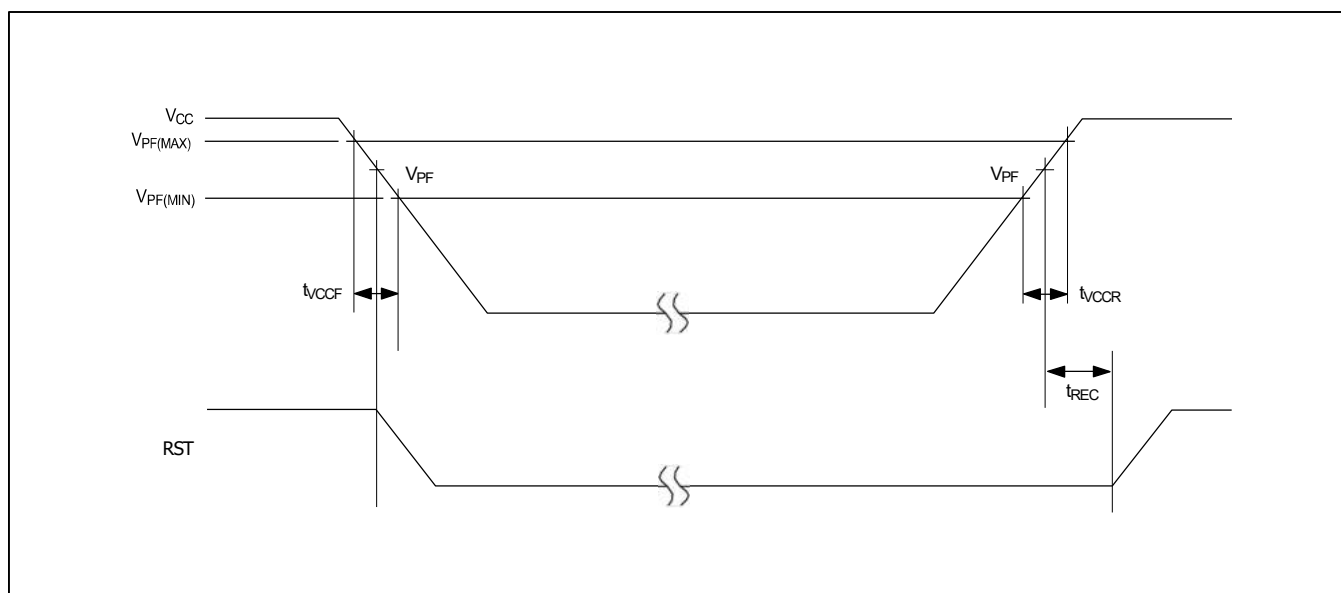
(T<sub>A</sub> = T<sub>MIN</sub> to T<sub>MAX</sub>)

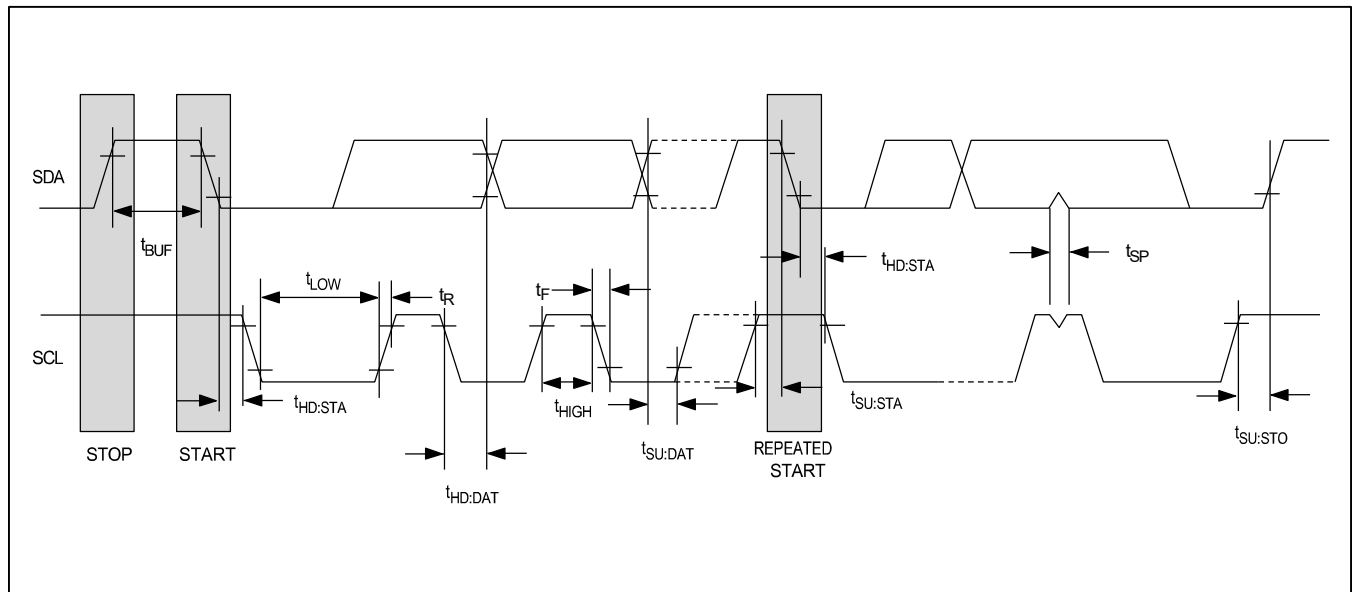
| PARAMETER                                                               | SYMBOL            | CONDITIONS | MIN | TYP | MAX | UNITS |
|-------------------------------------------------------------------------|-------------------|------------|-----|-----|-----|-------|
| V <sub>CC</sub> Fall Time; V <sub>PF(MAX)</sub> to V <sub>PF(MIN)</sub> | t <sub>VCCF</sub> |            | 300 |     |     | μs    |
| V <sub>CC</sub> Rise Time; V <sub>PF(MIN)</sub> to V <sub>PF(MAX)</sub> | t <sub>VCCR</sub> |            | 0   |     |     | μs    |
| Recovery at Power-Up                                                    | t <sub>REC</sub>  | (Note 13)  |     | 250 | 300 | ms    |

## Pushbutton Reset Timing



## Power-Switch Timing



Data Transfer on I<sup>2</sup>C Serial Bus

**WARNING:** Negative undershoots below -0.3V while the part is in battery-backed mode may cause loss of data.

**Note 2:** Limits at -40°C are guaranteed by design and not production tested.

**Note 3:** All voltages are referenced to ground.

**Note 4:** I<sub>CCA</sub>—SCL clocking at max frequency = 400kHz.

**Note 5:** Current is the averaged input current, which includes the temperature conversion current.

**Note 6:** The RST pin has an internal 50kΩ (nominal) pullup resistor to V<sub>CC</sub>.

**Note 7:** After this period, the first clock pulse is generated.

**Note 8:** A device must internally provide a hold time of at least 300ns for the SDA signal (referred to the V<sub>IH(MIN)</sub> of the SCL signal) to bridge the undefined region of the falling edge of SCL.

**Note 9:** The maximum t<sub>HD:DAT</sub> needs only to be met if the device does not stretch the low period (t<sub>LOW</sub>) of the SCL signal.

**Note 10:** A fast-mode device can be used in a standard-mode system, but the requirement t<sub>SU:DAT</sub> ≥ 250ns must then be met. This is automatically the case if the device does not stretch the low period of the SCL signal. If such a device does stretch the low period of the SCL signal, it must output the next data bit to the SDA line t<sub>R(MAX)</sub> + t<sub>SU:DAT</sub> = 1000 + 250 = 1250ns before the SCL line is released.

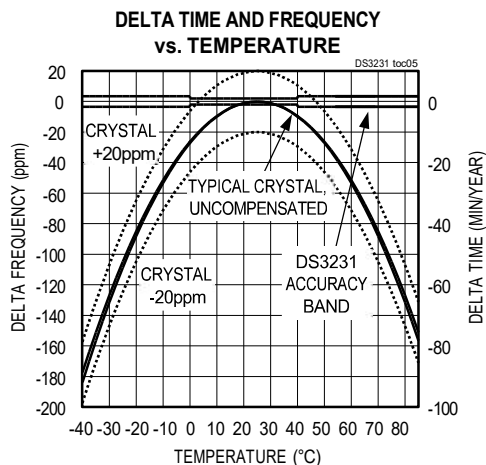
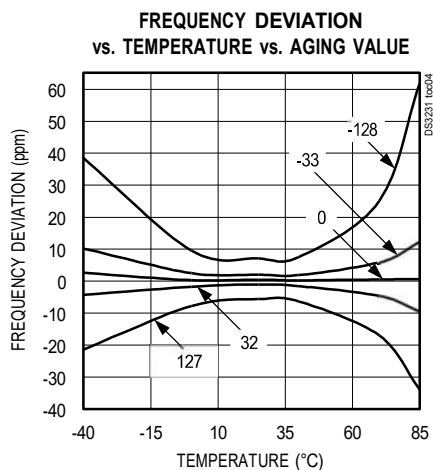
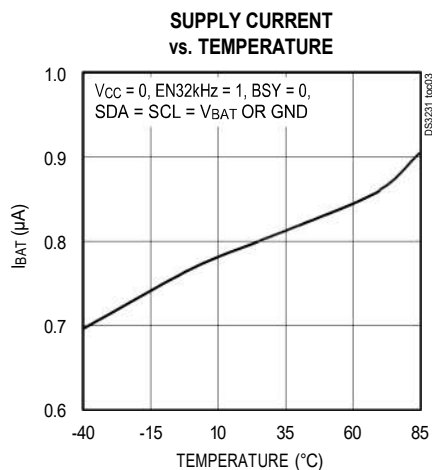
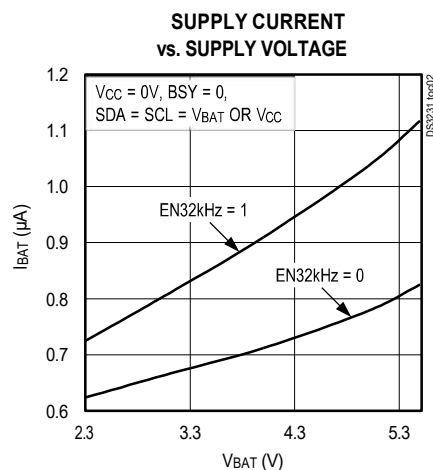
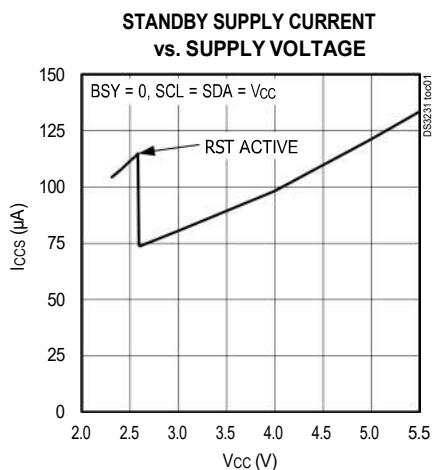
**Note 11:** C<sub>B</sub>—total capacitance of one bus line in pF.

**Note 12:** The parameter t<sub>OSF</sub> is the period of time the oscillator must be stopped for the OSF flag to be set over the voltage range of 0.0V ≤ V<sub>CC</sub> ≤ V<sub>CC(MAX)</sub> and 2.3V ≤ V<sub>BAT</sub> ≤ 3.4V.

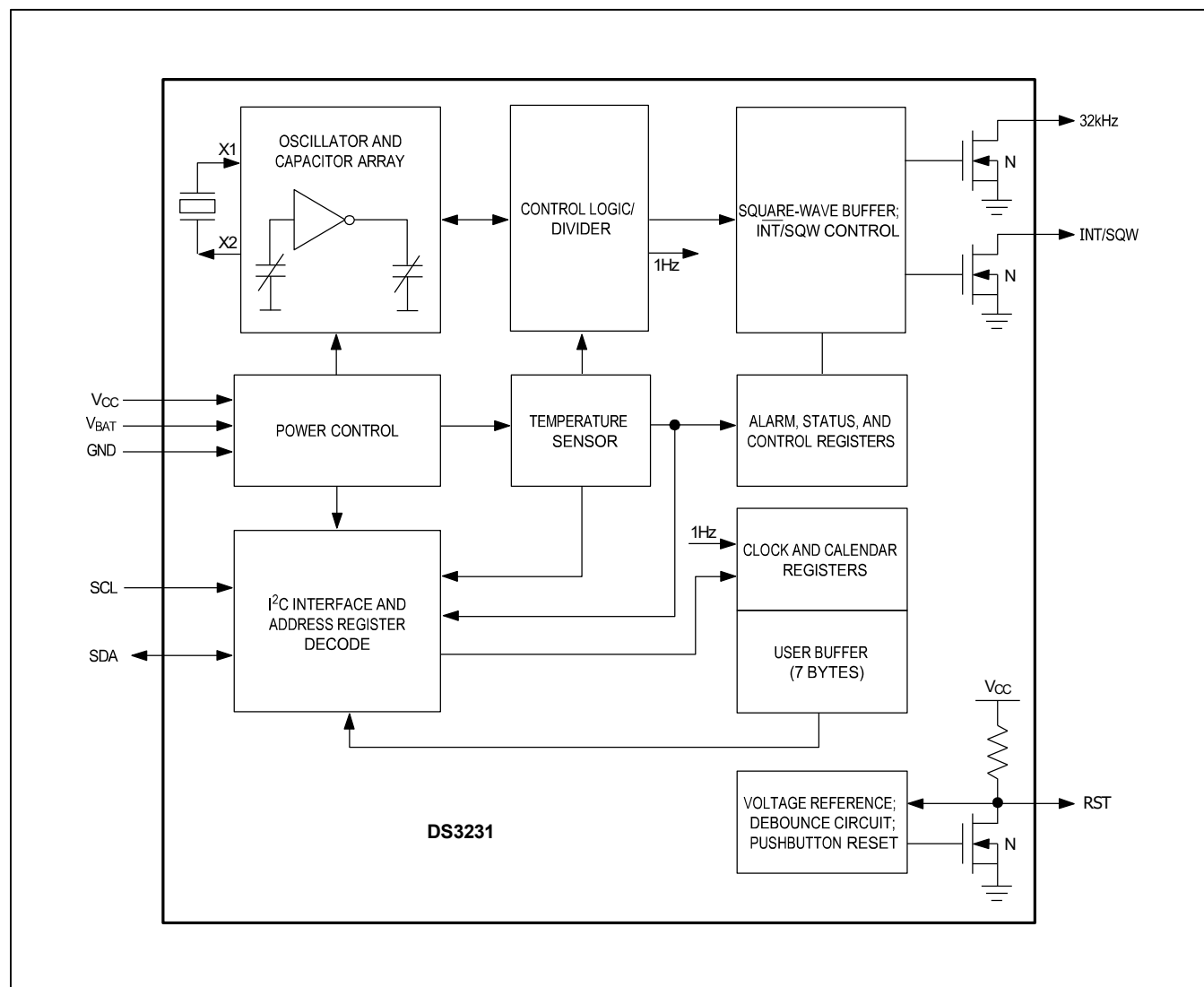
**Note 13:** This delay applies only if the oscillator is enabled and running. If the EOSC bit is a 1, t<sub>REC</sub> is bypassed and RST immediately goes high. The state of RST does not affect the I<sup>2</sup>C interface, RTC, or TCXO.

## Typical Operating Characteristics

( $V_{CC} = +3.3V$ ,  $T_A = +25^\circ C$ , unless otherwise noted.)



## Block Diagram



## Pin Description

| PIN  | NAME             | FUNCTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | 32kHz            | 32kHz Output. This open-drain pin requires an external pullup resistor. When enabled, the output operates on either power supply. It may be left open if not used.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 2    | V <sub>CC</sub>  | DC Power Pin for Primary Power Supply. This pin should be decoupled using a 0.1μF to 1.0μF capacitor. If not used, connect to ground.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 3    | INT/SQW          | Active-Low Interrupt or Square-Wave Output. This open-drain pin requires an external pullup resistor connected to a supply at 5.5V or less. This multifunction pin is determined by the state of the INTCN bit in the Control Register (0Eh). When INTCN is set to logic 0, this pin outputs a square wave and its frequency is determined by RS2 and RS1 bits. When INTCN is set to logic 1, then a match between the timekeeping registers and either of the alarm registers activates the INT/SQW pin (if the alarm is enabled). Because the INTCN bit is set to logic 1 when power is first applied, the pin defaults to an interrupt output with alarms disabled. The pullup voltage can be up to 5.5V, regardless of the voltage on V <sub>CC</sub> . If not used, this pin can be left unconnected. |
| 4    | RST              | Active-Low Reset. This pin is an open-drain input/output. It indicates the status of V <sub>CC</sub> relative to the V <sub>PF</sub> specification. As V <sub>CC</sub> falls below V <sub>PF</sub> , the RST pin is driven low. When V <sub>CC</sub> exceeds V <sub>PF</sub> , for t <sub>RST</sub> , the RST pin is pulled high by the internal pullup resistor. The active-low, open-drain output is combined with a debounced pushbutton input function. This pin can be activated by a pushbutton reset request. It has an internal 50kΩ nominal value pullup resistor to V <sub>CC</sub> . No external pullup resistors should be connected. If the oscillator is disabled, t <sub>REC</sub> is bypassed and RST immediately goes high.                                                               |
| 5–12 | N.C.             | No Connection. Must be connected to ground.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 13   | GND              | Ground                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 14   | V <sub>BAT</sub> | Backup Power-Supply Input. When using the device with the V <sub>BAT</sub> input as the primary power source, this pin should be decoupled using a 0.1μF to 1.0μF low-leakage capacitor. When using the device with the V <sub>BAT</sub> input as the backup power source, the capacitor is not required. If V <sub>BAT</sub> is not used, connect to ground. The device is UL recognized to ensure against reverse charging when used with a primary lithium battery.<br>Go to <a href="http://www.maximintegrated.com/qa/info/ul">www.maximintegrated.com/qa/info/ul</a> .                                                                                                                                                                                                                               |
| 15   | SDA              | Serial Data Input/Output. This pin is the data input/output for the I <sup>2</sup> C serial interface. This open-drain pin requires an external pullup resistor. The pullup voltage can be up to 5.5V, regardless of the voltage on V <sub>CC</sub> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 16   | SCL              | Serial Clock Input. This pin is the clock input for the I <sup>2</sup> C serial interface and is used to synchronize data movement on the serial interface. Up to 5.5V can be used for this pin, regardless of the voltage on V <sub>CC</sub> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

## Detailed Description

The DS3231 is a serial RTC driven by a temperature-compensated 32kHz crystal oscillator. The TCXO provides a stable and accurate reference clock, and maintains the RTC to within ±2 minutes per year accuracy from -40°C to +85°C. The TCXO frequency output is available at the 32kHz pin. The RTC is a low-power clock/calendar with two programmable time-of-day alarms and a programmable square-wave output. The INT/SQW provides either an interrupt signal due to alarm conditions or a square-wave output. The clock/calendar provides seconds, minutes, hours, day, date, month, and year information. The date at the end of the month is automatically adjusted for months with fewer than 31 days, including corrections for leap

year. The clock operates in either the 24-hour or 12-hour format with an AM/PM indicator. The internal registers are accessible through an I<sup>2</sup>C bus interface.

A temperature-compensated voltage reference and comparator circuit monitors the level of V<sub>CC</sub> to detect power failures and to automatically switch to the backup supply when necessary. The RST pin provides an external pushbutton function and acts as an indicator of a power-fail event.

## Operation

The block diagram shows the main elements of the DS3231. The eight blocks can be grouped into four functional groups: TCXO, power control, pushbutton function, and RTC. Their operations are described separately in the following sections.



### 32kHz TCXO

The temperature sensor, oscillator, and control logic form the TCXO. The controller reads the output of the on-chip temperature sensor and uses a lookup table to determine the capacitance required, adds the aging correction in AGE register, and then sets the capacitance selection registers. New values, including changes to the AGE register, are loaded only when a change in the temperature value occurs, or when a user-initiated temperature conversion is completed. Temperature conversion occurs on initial application of  $V_{CC}$  and once every 64 seconds afterwards.

### Power Control

This function is provided by a temperature-compensated voltage reference and a comparator circuit that monitors the  $V_{CC}$  level. When  $V_{CC}$  is greater than  $V_{PF}$ , the part is powered by  $V_{CC}$ . When  $V_{CC}$  is less than  $V_{PF}$  but greater than  $V_{BAT}$ , the DS3231 is powered by  $V_{CC}$ . If  $V_{CC}$  is less than  $V_{PF}$  and is less than  $V_{BAT}$ , the device is powered by  $V_{BAT}$ . See Table 1.

**Table 1. Power Control**

| SUPPLY CONDITION                       | ACTIVE SUPPLY |
|----------------------------------------|---------------|
| $V_{CC} < V_{PF}$ , $V_{CC} < V_{BAT}$ | $V_{BAT}$     |
| $V_{CC} < V_{PF}$ , $V_{CC} > V_{BAT}$ | $V_{CC}$      |
| $V_{CC} > V_{PF}$ , $V_{CC} < V_{BAT}$ | $V_{CC}$      |
| $V_{CC} > V_{PF}$ , $V_{CC} > V_{BAT}$ | $V_{CC}$      |

To preserve the battery, the first time  $V_{BAT}$  is applied to the device, the oscillator will not start up until  $V_{CC}$  exceeds  $V_{PF}$ , or until a valid I2C address is written to the part. Typical oscillator startup time is less than one second. Approximately 2 seconds after  $V_{CC}$  is applied, or a valid I2C address is written, the device makes a temperature measurement and applies the calculated correction to the oscillator. Once the oscillator is running, it continues to run as long as a valid power source is available ( $V_{CC}$  or  $V_{BAT}$ ), and the device continues to measure the temperature and correct the oscillator frequency every 64 seconds.

On the first application of power ( $V_{CC}$ ) or when a valid I2C address is written to the part ( $V_{BAT}$ ), the time and date registers are reset to 01/01/00 01 00:00:00 (DD/MM/YY DOW HH:MM:SS).

### $V_{BAT}$ Operation

There are several modes of operation that affect the amount of  $V_{BAT}$  current that is drawn. While the device

is powered by  $V_{BAT}$  and the serial interface is active, active battery current,  $I_{BATA}$ , is drawn. When the serial interface is inactive, timekeeping current ( $I_{BATT}$ ), which includes the averaged temperature conversion current,  $I_{BATTC}$ , is used (refer to Application Note 3644: *Power Considerations for Accurate Real-Time Clocks* for details). Temperature conversion current,  $I_{BATTC}$ , is specified since the system must be able to support the periodic higher current pulse and still maintain a valid voltage level. Data retention current,  $I_{BATTD}$ , is the current drawn by the part when the oscillator is stopped ( $EOSC = 1$ ). This mode can be used to minimize battery requirements for times when maintaining time and date information is not necessary, e.g., while the end system is waiting to be shipped to a customer.

### Pushbutton Reset Function

The DS3231 provides for a pushbutton switch to be connected to the RST output pin. When the DS3231 is not in a reset cycle, it continuously monitors the RST signal for a low going edge. If an edge transition is detected, the DS3231 debounces the switch by pulling the RST low. After the internal timer has expired ( $PB_{DB}$ ), the DS3231 continues to monitor the RST line. If the line is still low, the DS3231 continuously monitors the line looking for a rising edge. Upon detecting release, the DS3231 forces the RST pin low and holds it low for  $t_{RST}$ .

RST is also used to indicate a power-fail condition. When  $V_{CC}$  is lower than  $V_{PF}$ , an internal power-fail signal is generated, which forces the RST pin low. When  $V_{CC}$  returns to a level above  $V_{PF}$ , the RST pin is held low for approximately 250ms ( $t_{REC}$ ) to allow the power supply to stabilize. If the oscillator is not running (see the *Power Control* section) when  $V_{CC}$  is applied,  $t_{REC}$  is bypassed and RST immediately goes high. Assertion of the RST output, whether by pushbutton or power-fail detection, does not affect the internal operation of the DS3231.

### Real-Time Clock

With the clock source from the TCXO, the RTC provides seconds, minutes, hours, day, date, month, and year information. The date at the end of the month is automatically adjusted for months with fewer than 31 days, including corrections for leap year. The clock operates in either the 24-hour or 12-hour format with an AM/PM indicator.

The clock provides two programmable time-of-day alarms and a programmable square-wave output. The INT/SQW pin either generates an interrupt due to alarm condition or outputs a square-wave signal and the selection is controlled by the bit  $INTCN$ .

| ADDRESS | BIT 7<br>MSB | BIT 6      | BIT 5            | BIT 4    | BIT 3   | BIT 2 | BIT 1 | BIT 0<br>LSB | FUNCTION          | RANGE                 |
|---------|--------------|------------|------------------|----------|---------|-------|-------|--------------|-------------------|-----------------------|
| 00h     | 0            | 10 Seconds |                  |          | Seconds |       |       |              | Seconds           | 00–59                 |
| 01h     | 0            | 10 Minutes |                  |          | Minutes |       |       |              | Minutes           | 00–59                 |
| 02h     | 0            | 12/24      | AM/PM<br>20 Hour | 10 Hour  | Hour    |       |       |              | Hours             | 1–12 + AM/PM<br>00–23 |
| 03h     | 0            | 0          | 0                | 0        | 0       | Day   |       |              | Day               | 1–7                   |
| 04h     | 0            | 0          | 10 Date          |          | Date    |       |       |              | Date              | 01–31                 |
| 05h     | Century      | 0          | 0                | 10 Month | Month   |       |       |              | Month/<br>Century | 01–12 + Century       |
| 06h     | 10 Year      |            |                  |          | Year    |       |       |              | Year              | 00–99                 |
| 07h     | A1M1         | 10 Seconds |                  |          | Seconds |       |       |              | Alarm 1 Seconds   | 00–59                 |
| 08h     | A1M2         | 10 Minutes |                  |          | Minutes |       |       |              | Alarm 1 Minutes   | 00–59                 |
| 09h     | A1M3         | 12/24      | AM/PM<br>20 Hour | 10 Hour  | Hour    |       |       |              | Alarm 1 Hours     | 1–12 + AM/PM<br>00–23 |
| 0Ah     | A1M4         | DY/DT      | 10 Date          |          | Day     |       |       |              | Alarm 1 Day       | 1–7                   |
|         |              |            |                  |          | Date    |       |       |              | Alarm 1 Date      | 1–31                  |
| 0Bh     | A2M2         | 10 Minutes |                  |          | Minutes |       |       |              | Alarm 2 Minutes   | 00–59                 |
| 0Ch     | A2M3         | 12/24      | AM/PM<br>20 Hour | 10 Hour  | Hour    |       |       |              | Alarm 2 Hours     | 1–12 + AM/PM<br>00–23 |
| 0Dh     | A2M4         | DY/DT      | 10 Date          |          | Day     |       |       |              | Alarm 2 Day       | 1–7                   |
|         |              |            |                  |          | Date    |       |       |              | Alarm 2 Date      | 1–31                  |
| 0Eh     | EOSC         | BBSQW      | CONV             | RS2      | RS1     | INTCN | A2IE  | A1IE         | Control           | —                     |
| 0Fh     | OSF          | 0          | 0                | 0        | EN32kHz | BSY   | A2F   | A1F          | Control/Status    | —                     |
| 10h     | SIGN         | DATA       | DATA             | DATA     | DATA    | DATA  | DATA  | DATA         | Aging Offset      | —                     |
| 11h     | SIGN         | DATA       | DATA             | DATA     | DATA    | DATA  | DATA  | DATA         | MSB of Temp       | —                     |
| 12h     | DATA         | DATA       | 0                | 0        | 0       | 0     | 0     | 0            | LSB of Temp       | —                     |

Figure 1. Timekeeping Registers

**Note:** Unless otherwise specified, the registers' state is not defined when power is first applied.

## Address Map

Figure 1 shows the address map for the DS3231 timekeeping registers. During a multibyte access, when the address pointer reaches the end of the register space (12h), it wraps around to location 00h. On an I2C START or address pointer incrementing to location 00h, the current time is transferred to a second set of registers. The time information is read from these secondary registers, while the clock may continue to run. This eliminates the need to reread the registers in case the main registers update during a read.

## I2C Interface

The I2C interface is accessible whenever either  $V_{CC}$  or  $V_{BAT}$  is at a valid level. If a microcontroller connected

to the DS3231 resets because of a loss of  $V_{CC}$  or other event, it is possible that the microcontroller and DS3231 I2C communications could become unsynchronized, e.g., the microcontroller resets while reading data from the DS3231. When the microcontroller resets, the DS3231 I2C interface may be placed into a known state by toggling SCL until SDA is observed to be at a high level. At that point the microcontroller should pull SDA low while SCL is high, generating a START condition.

## Clock and Calendar

The time and calendar information is obtained by reading the appropriate register bytes. Figure 1 illustrates the RTC registers. The time and calendar data are set or initialized by writing the appropriate register bytes. The contents of the time and calendar registers are in the binary-coded

decimal (BCD) format. The DS3231 can be run in either 12-hour or 24-hour mode. Bit 6 of the hours register is defined as the 12- or 24-hour mode select bit. When high, the 12-hour mode is selected. In the 12-hour mode, bit 5 is the AM/PM bit with logic-high being PM. In the 24-hour mode, bit 5 is the 20-hour bit (20–23 hours). The century bit (bit 7 of the month register) is toggled when the years register overflows from 99 to 00.

The day-of-week register increments at midnight. Values that correspond to the day of week are user-defined but must be sequential (i.e., if 1 equals Sunday, then 2 equals Monday, and so on). Illogical time and date entries result in undefined operation.

When reading or writing the time and date registers, secondary (user) buffers are used to prevent errors when the internal registers update. When reading the time and date registers, the user buffers are synchronized to the internal registers on any START and when the register pointer rolls over to zero. The time information is read from these secondary registers, while the clock continues to run. This eliminates the need to reread the registers in case the main registers update during a read.

The countdown chain is reset whenever the seconds register is written. Write transfers occur on the acknowledge from the DS3231. Once the countdown chain is reset, to avoid rollover issues the remaining time and date registers must be written within 1 second. The 1Hz square-wave output, if enabled, transitions high 500ms after the seconds data transfer, provided the oscillator is already running.

## Alarms

The DS3231 contains two time-of-day/date alarms. Alarm 1 can be set by writing to registers 07h to 0Ah. Alarm 2 can be set by writing to registers 0Bh to 0Dh. The alarms can be programmed (by the alarm enable and INTCN bits of the control register) to activate the INT/SQW output on an alarm match condition. Bit 7 of each of the time-of-day/date alarm registers are mask bits (Table 2). When all the mask bits for each alarm are logic 0, an alarm only occurs when the values in the timekeeping registers match the corresponding values stored in the time-of-day/date alarm registers. The alarms can also be programmed to repeat every second, minute, hour, day, or date. Table 2 shows the possible settings. Configurations not listed in the table will result in illogical operation.

The DY/DT bits (bit 6 of the alarm day/date registers) control whether the alarm value stored in bits 0 to 5 of that register reflects the day of the week or the date of the month. If DY/DT is written to logic 0, the alarm will be the result of a match with date of the month. If DY/DT is written to logic 1, the alarm will be the result of a match with day of the week.

When the RTC register values match alarm register settings, the corresponding Alarm Flag 'A1F' or 'A2F' bit is set to logic 1. If the corresponding Alarm Interrupt Enable 'A1IE' or 'A2IE' is also set to logic 1 and the INTCN bit is set to logic 1, the alarm condition will activate the INT/SQW signal. The match is tested on the once-per-second update of the time and date registers.

**Table 2. Alarm Mask Bits**

| DY/DT | ALARM 1 REGISTER MASK BITS (BIT 7) |      |      |                                                    | ALARM RATE                                         |
|-------|------------------------------------|------|------|----------------------------------------------------|----------------------------------------------------|
|       | A1M4                               | A1M3 | A1M2 | A1M1                                               |                                                    |
| X     | 1                                  | 1    | 1    | 1                                                  | Alarm once per second                              |
| X     | 1                                  | 1    | 1    | 0                                                  | Alarm when seconds match                           |
| X     | 1                                  | 1    | 0    | 0                                                  | Alarm when minutes and seconds match               |
| X     | 1                                  | 0    | 0    | 0                                                  | Alarm when hours, minutes, and seconds match       |
| 0     | 0                                  | 0    | 0    | 0                                                  | Alarm when date, hours, minutes, and seconds match |
| 1     | 0                                  | 0    | 0    | 0                                                  | Alarm when day, hours, minutes, and seconds match  |
|       |                                    |      |      |                                                    |                                                    |
| DY/DT | ALARM 2 REGISTER MASK BITS (BIT 7) |      |      | ALARM RATE                                         |                                                    |
|       | A2M4                               | A2M3 | A2M2 |                                                    |                                                    |
| X     | 1                                  | 1    | 1    | Alarm once per minute (00 seconds of every minute) |                                                    |
| X     | 1                                  | 1    | 0    | Alarm when minutes match                           |                                                    |
| X     | 1                                  | 0    | 0    | Alarm when hours and minutes match                 |                                                    |
| 0     | 0                                  | 0    | 0    | Alarm when date, hours, and minutes match          |                                                    |
| 1     | 0                                  | 0    | 0    | Alarm when day, hours, and minutes match           |                                                    |

**Control Register (0Eh)**

|       | BIT 7 | BIT 6 | BIT 5 | BIT 4 | BIT 3 | BIT 2 | BIT 1 | BIT 0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| NAME: | EOSC  | BBSQW | CONV  | RS2   | RS1   | INTCN | A2IE  | A1IE  |
| POR:  | 0     | 0     | 0     | 1     | 1     | 1     | 0     | 0     |

**Special-Purpose Registers**

The DS3231 has two additional registers (control and status) that control the real-time clock, alarms, and square-wave output.

**Control Register (0Eh)**

**Bit 7: Enable Oscillator (T<sub>OSC</sub>).** When set to logic 0, the oscillator is started. When set to logic 1, the oscillator is stopped when the DS3231 switches to V<sub>BAT</sub>. This bit is clear (logic 0) when power is first applied. When the DS3231 is powered by V<sub>CC</sub>, the oscillator is always on regardless of the status of the EOSC bit. When EOSC is disabled, all register data is static.

**Bit 6: Battery-Backed Square-Wave Enable (BBSQW).** When set to logic 1 with INTCN = 0 and V<sub>CC</sub> < V<sub>PF</sub>, this bit enables the square wave. When BBSQW is logic 0, the INT/SQW pin goes high impedance when V<sub>CC</sub> < V<sub>PF</sub>. This bit is disabled (logic 0) when power is first applied.

**Bit 5: Convert Temperature (CONV).** Setting this bit to 1 forces the temperature sensor to convert the temperature into digital code and execute the TCXO algorithm to update the capacitance array to the oscillator. This can only happen when a conversion is not already in progress. The user should check the status bit BSY before forcing the controller to start a new TCXO execution. A user-initiated temperature conversion does not affect the internal 64-second update cycle.

A user-initiated temperature conversion does not affect the BSY bit for approximately 2ms. The CONV bit remains at a 1 from the time it is written until the conversion is finished, at which time both CONV and BSY go to 0. The CONV bit should be used when monitoring the status of a user-initiated conversion.

**Bits 4 and 3: Rate Select (RS2 and RS1).** These bits control the frequency of the square-wave output when

the square wave has been enabled. The following table shows the square-wave frequencies that can be selected with the RS bits. These bits are both set to logic 1 (8.192kHz) when power is first applied.

**SQUARE-WAVE OUTPUT FREQUENCY**

| RS2 | RS1 | SQUARE-WAVE OUTPUT FREQUENCY |
|-----|-----|------------------------------|
| 0   | 0   | 1Hz                          |
| 0   | 1   | 1.024kHz                     |
| 1   | 0   | 4.096kHz                     |
| 1   | 1   | 8.192kHz                     |

**Bit 2: Interrupt Control (INTCN).** This bit controls the INT/SQW signal. When the INTCN bit is set to logic 0, a square wave is output on the INT/SQW pin. When the INTCN bit is set to logic 1, then a match between the time-keeping registers and either of the alarm registers activates the INT/SQW output (if the alarm is also enabled). The corresponding alarm flag is always set regardless of the state of the INTCN bit. The INTCN bit is set to logic 1 when power is first applied.

**Bit 1: Alarm 2 Interrupt Enable (A2IE).** When set to logic 1, this bit permits the alarm 2 flag (A2F) bit in the status register to assert INT/SQW (when INTCN = 1). When the A2IE bit is set to logic 0 or INTCN is set to logic 0, the A2F bit does not initiate an interrupt signal. The A2IE bit is disabled (logic 0) when power is first applied.

**Bit 0: Alarm 1 Interrupt Enable (A1IE).** When set to logic 1, this bit permits the alarm 1 flag (A1F) bit in the status register to assert INT/SQW (when INTCN = 1). When the A1IE bit is set to logic 0 or INTCN is set to logic 0, the A1F bit does not initiate the INT/SQW signal. The A1IE bit is disabled (logic 0) when power is first applied.

**Status Register (0Fh)**

|       | BIT 7 | BIT 6 | BIT 5 | BIT 4 | BIT 3   | BIT 2 | BIT 1 | BIT 0 |
|-------|-------|-------|-------|-------|---------|-------|-------|-------|
| NAME: | OSF   | 0     | 0     | 0     | EN32kHz | BSY   | A2F   | A1F   |
| POR:  | 1     | 0     | 0     | 0     | 1       | X     | X     | X     |

**Status Register (0Fh)**

**Bit 7: Oscillator Stop Flag (OSF).** A logic 1 in this bit indicates that the oscillator either is stopped or was stopped for some period and may be used to judge the validity of the timekeeping data. This bit is set to logic 1 any time that the oscillator stops. The following are examples of conditions that can cause the OSF bit to be set:

- 1) The first time power is applied.
- 2) The voltages present on both V<sub>CC</sub> and V<sub>BAT</sub> are insufficient to support oscillation.
- 3) The EOSC bit is turned off in battery-backed mode.
- 4) External influences on the crystal (i.e., noise, leakage, etc.).

This bit remains at logic 1 until written to logic 0.

**Bit 3: Enable 32kHz Output (EN32kHz).** This bit controls the status of the 32kHz pin. When set to logic 1, the 32kHz pin is enabled and outputs a 32.768kHz square-wave signal. When set to logic 0, the 32kHz pin goes to a high-impedance state. The initial power-up state of this bit is logic 1, and a 32.768kHz square-wave signal appears at the 32kHz pin after a power source is applied to the DS3231 (if the oscillator is running).

**Bit 2: Busy (BSY).** This bit indicates the device is busy executing TCXO functions. It goes to logic 1 when the conversion signal to the temperature sensor is asserted and then is cleared when the device is in the 1-minute idle state.

**Bit 1: Alarm 2 Flag (A2F).** A logic 1 in the alarm 2 flag bit indicates that the time matched the alarm 2 registers. If the A2IE bit is logic 1 and the INTCN bit is set to logic 1, the INT/SQW pin is also asserted. A2F is cleared when written to logic 0. This bit can only be written to logic 0. Attempting to write to logic 1 leaves the value unchanged.

**Bit 0: Alarm 1 Flag (A1F).** A logic 1 in the alarm 1 flag bit indicates that the time matched the alarm 1 registers. If the

A1IE bit is logic 1 and the INTCN bit is set to logic 1, the INT/SQW pin is also asserted. A1F is cleared when written to logic 0. This bit can only be written to logic 0. Attempting to write to logic 1 leaves the value unchanged.

**Aging Offset**

The aging offset register takes a user-provided value to add to or subtract from the codes in the capacitance array registers. The code is encoded in two's complement, with bit 7 representing the sign bit. One LSB represents one small capacitor to be switched in or out of the capacitance array at the crystal pins. The aging offset register capacitance value is added or subtracted from the capacitance value that the device calculates for each temperature compensation. The offset register is added to the capacitance array during a normal temperature conversion, if the temperature changes from the previous conversion, or during a manual user conversion (setting the CONV bit). To see the effects of the aging register on the 32kHz output frequency immediately, a manual conversion should be started after each aging register change.

Positive aging values add capacitance to the array, slowing the oscillator frequency. Negative values remove capacitance from the array, increasing the oscillator frequency.

The change in ppm per LSB is different at different temperatures. The frequency vs. temperature curve is shifted by the values used in this register. At +25°C, one LSB typically provides about 0.1ppm change in frequency.

Use of the aging register is not needed to achieve the accuracy as defined in the EC tables, but could be used to help compensate for aging at a given temperature. See the *Typical Operating Characteristics* section for a graph showing the effect of the register on accuracy over temperature.

**Aging Offset (10h)**

|       | BIT 7 | BIT 6 | BIT 5 | BIT 4 | BIT 3 | BIT 2 | BIT 1 | BIT 0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| NAME: | Sign  | Data  | Data  | Data  | Data  | Data  | Data  | Data  |
| POR:  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |



**Temperature Register (Upper Byte) (11h)**

|       | BIT 7 | BIT 6 | BIT 5 | BIT 4 | BIT 3 | BIT 2 | BIT 1 | BIT 0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| NAME: | Sign  | Data  | Data  | Data  | Data  | Data  | Data  | Data  |
| POR:  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

**Temperature Register (Lower Byte) (12h)**

|       | BIT 7 | BIT 6 | BIT 5 | BIT 4 | BIT 3 | BIT 2 | BIT 1 | BIT 0 |
|-------|-------|-------|-------|-------|-------|-------|-------|-------|
| NAME: | Data  | Data  | 0     | 0     | 0     | 0     | 0     | 0     |
| POR:  | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |

**Temperature Registers (11h–12h)**

Temperature is represented as a 10-bit code with a resolution of 0.25°C and is accessible at location 11h and 12h. The temperature is encoded in two's complement format. The upper 8 bits, the integer portion, are at location 11h and the lower 2 bits, the fractional portion, are in the upper nibble at location 12h. For example, 00011001 01b = +25.25°C. Upon power reset, the registers are set to a default temperature of 0°C and the controller starts a temperature conversion. The temperature is read on initial application of V<sub>CC</sub> or I2C access on V<sub>BAT</sub> and once every 64 seconds afterwards. The temperature registers are updated after each user-initiated conversion and on every 64-second conversion. The temperature registers are read-only.

**I2C Serial Data Bus**

The DS3231 supports a bidirectional I2C bus and data transmission protocol. A device that sends data onto the bus is defined as a transmitter and a device receiving data is defined as a receiver. The device that controls the message is called a master. The devices that are controlled by the master are slaves. The bus must be controlled by a master device that generates the serial clock (SCL), controls the bus access, and generates the START and STOP conditions. The DS3231 operates as a slave on the I2C bus. Connections to the bus are made through the SCL input and open-drain SDA I/O lines. Within the bus specifications, a standard mode (100kHz maximum clock rate) and a fast mode (400kHz maximum clock rate) are defined. The DS3231 works in both modes.

The following bus protocol has been defined (Figure 2):

- Data transfer may be initiated only when the bus is not busy.
- During data transfer, the data line must remain stable whenever the clock line is high. Changes in the data

line while the clock line is high are interpreted as control signals.

Accordingly, the following bus conditions have been defined:

**Bus not busy:** Both data and clock lines remain high.

**START data transfer:** A change in the state of the data line from high to low, while the clock line is high, defines a START condition.

**STOP data transfer:** A change in the state of the data line from low to high, while the clock line is high, defines a STOP condition.

**Data valid:** The state of the data line represents valid data when, after a START condition, the data line is stable for the duration of the high period of the clock signal. The data on the line must be changed during the low period of the clock signal. There is one clock pulse per bit of data.

Each data transfer is initiated with a START condition and terminated with a STOP condition. The number of data bytes transferred between the START and the STOP conditions is not limited, and is determined by the master device. The information is transferred byte-wise and each receiver acknowledges with a ninth bit.

**Acknowledge:** Each receiving device, when addressed, is obliged to generate an acknowledge after the reception of each byte. The master device must generate an extra clock pulse, which is associated with this acknowledge bit.

A device that acknowledges must pull down the SDA line during the acknowledge clock pulse in such a way that the SDA line is stable low during the high period of the acknowledge-related clock pulse. Of course, setup and hold times must be taken into account. A master must signal an end of data to the slave by not generat-

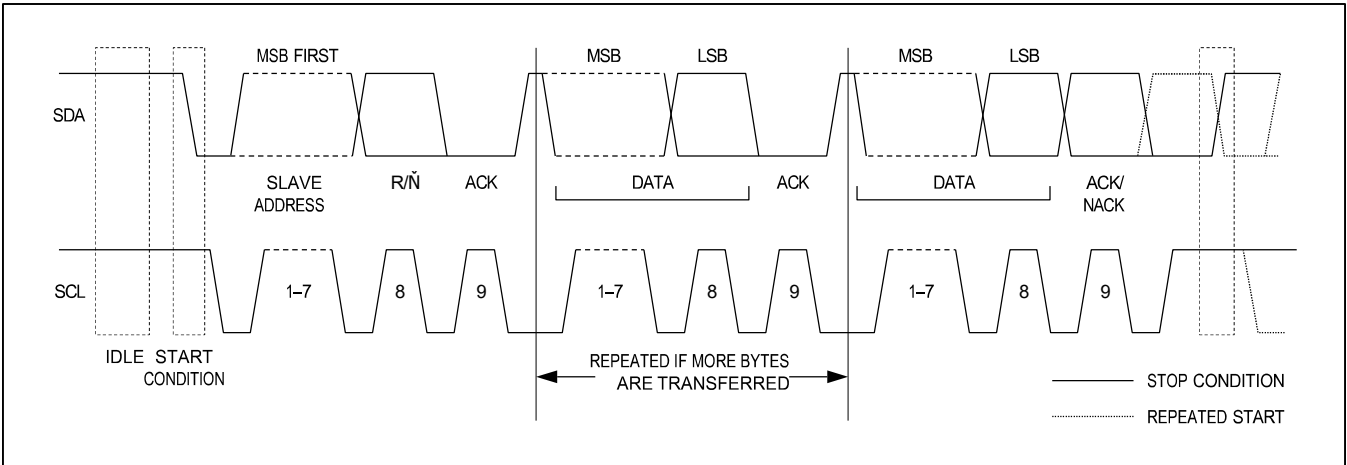


Figure 2. I2C Data Transfer Overview

ing an acknowledge bit on the last byte that has been clocked out of the slave. In this case, the slave must leave the data line high to enable the master to generate the STOP condition.

Figures 3 and 4 detail how data transfer is accomplished on the I2C bus. Depending upon the state of the R/Ž bit, two types of data transfer are possible:

**Data transfer from a master transmitter to a slave receiver.** The first byte transmitted by the master

is the slave address. Next follows a number of data bytes. The slave returns an acknowledge bit after each received byte. Data is transferred with the most significant bit (MSB) first.

**Data transfer from a slave transmitter to a master receiver.** The first byte (the slave address) is transmitted by the master. The slave then returns an acknowledge bit. Next follows a number of data bytes transmitted by the slave to the master. The master returns an acknowledge bit after all received bytes other than the

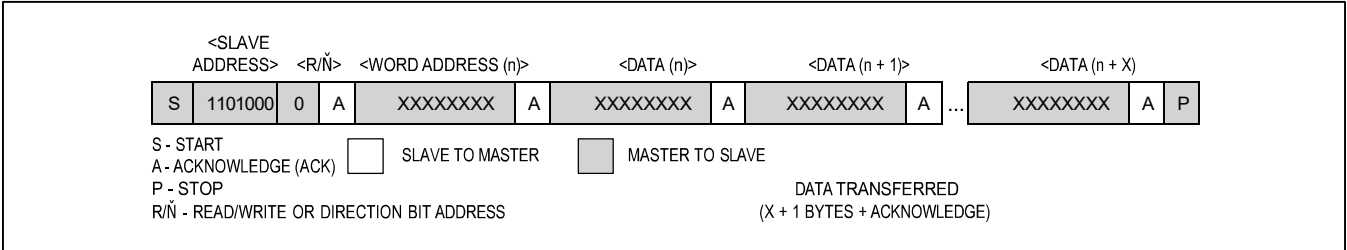


Figure 3. Data Write—Slave Receiver Mode

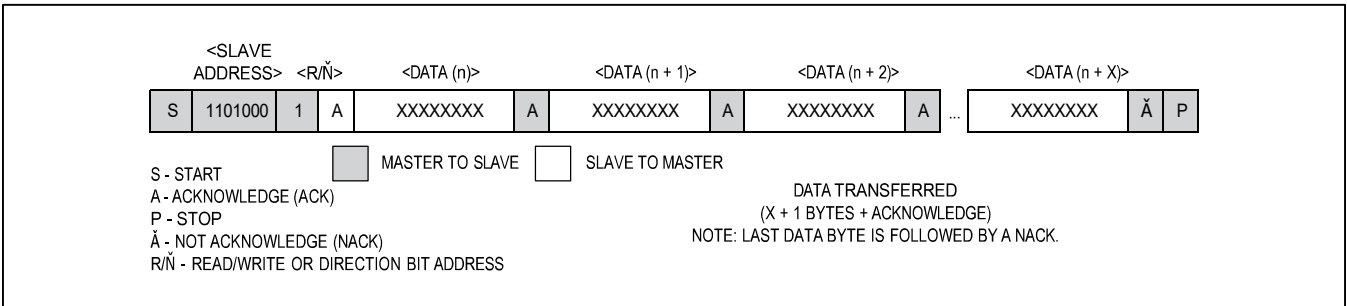


Figure 4. Data Read—Slave Transmitter Mode

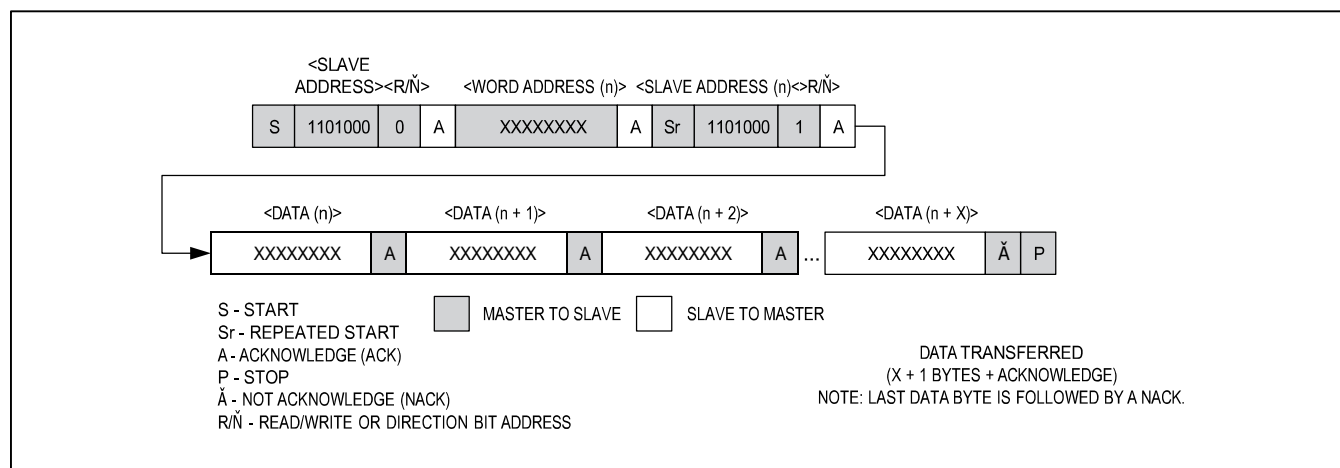


Figure 5. Data Write/Read (Write Pointer, Then Read)—Slave Receive and Transmit

last byte. At the end of the last received byte, a not acknowledge is returned.

The master device generates all the serial clock pulses and the START and STOP conditions. A transfer is ended with a STOP condition or with a repeated START condition. Since a repeated START condition is also the beginning of the next serial transfer, the bus will not be released. Data is transferred with the most significant bit (MSB) first.

The DS3231 can operate in the following two modes:

**Slave receiver mode (DS3231 write mode):** Serial data and clock are received through SDA and SCL. After each byte is received, an acknowledge bit is transmitted. START and STOP conditions are recognized as the beginning and end of a serial transfer. Address recognition is performed by hardware after reception of the slave address and direction bit. The slave address byte is the first byte received after the master generates the START condition. The slave address byte contains the 7-bit DS3231 address, which is 1101000, followed by the direction bit (R/Ñ), which is 0 for a write. After receiving and decoding the slave address byte, the DS3231 outputs an acknowledge on SDA. After the DS3231 acknowledges the slave address + write bit, the master transmits a word address to the DS3231. This sets the register pointer on the DS3231, with the DS3231 acknowledging the

transfer. The master may then transmit zero or more bytes of data, with the DS3231 acknowledging each byte received. The register pointer increments after each data byte is transferred. The master generates a STOP condition to terminate the data write.

**Slave transmitter mode (DS3231 read mode):** The first byte is received and handled as in the slave receiver mode. However, in this mode, the direction bit indicates that the transfer direction is reversed. Serial data is transmitted on SDA by the DS3231 while the serial clock is input on SCL. START and STOP conditions are recognized as the beginning and end of a serial transfer. Address recognition is performed by hardware after reception of the slave address and direction bit. The slave address byte is the first byte received after the master generates a START condition. The slave address byte contains the 7-bit DS3231 address, which is 1101000, followed by the direction bit (R/Ñ), which is 1 for a read. After receiving and decoding the slave address byte, the DS3231 outputs an acknowledge on SDA. The DS3231 then begins to transmit data starting with the register address pointed to by the register pointer. If the register pointer is not written to before the initiation of a read mode, the first address that is read is the last one stored in the register pointer. The DS3231 must receive a not acknowledge to end a read.



# DS3231

# Extremely Accurate I2C-Integrated RTC/TCXO/Crystal

## Handling, PCB Layout, and Assembly

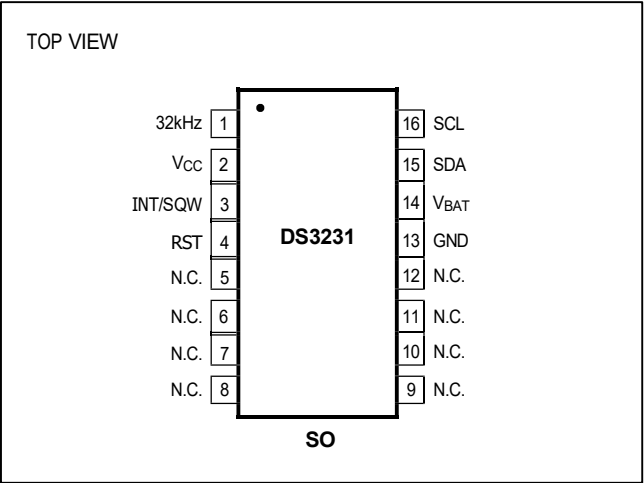
The DS3231 package contains a quartz tuning-fork crystal. Pick-and-place equipment can be used, but precautions should be taken to ensure that excessive shocks are avoided. Ultrasonic cleaning should be avoided to prevent damage to the crystal.

Avoid running signal traces under the package, unless a ground plane is placed between the package and the

signal line. All N.C. (no connect) pins must be connected to ground.

Moisture-sensitive packages are shipped from the factory dry packed. Handling instructions listed on the package label must be followed to prevent damage during reflow. Refer to the IPC/JEDEC J-STD-020 standard for moisture-sensitive device (MSD) classifications and reflow profiles. Exposure to reflow is limited to 2 times maximum.

## Pin Configuration



## Chip Information

SUBSTRATE CONNECTED TO GROUND

PROCESS: CMOS

## Ordering Information

| PART      | TEMP RANGE     | PIN-PACKAGE |
|-----------|----------------|-------------|
| DS3231S#  | 0°C to +70°C   | 16 SO       |
| DS3231SN# | -40°C to +85°C | 16 SO       |

#Denotes an RoHS-compliant device that may include lead (Pb) that is exempt under RoHS requirements. The lead finish is JESD97 category e3, and is compatible with both lead-based and lead-free soldering processes. A “#” anywhere on the top mark denotes an RoHS-compliant device.

## Package Information

For the latest package outline information and land patterns (footprints), go to [www.maximintegrated.com/packages](http://www.maximintegrated.com/packages). Note that a “+”, “#”, or “-” in the package code indicates RoHS status only. Package drawings may show a different suffix character, but the drawing pertains to the package regardless of RoHS status.

| PACKAGE TYPE | PACKAGE CODE | OUTLINE NO.             | LAND PATTERN NO.        |
|--------------|--------------|-------------------------|-------------------------|
| 16 SO        | W16#H2       | <a href="#">21-0042</a> | <a href="#">90-0107</a> |

## Revision History

| REVISION NUMBER | REVISION DATE | DESCRIPTION                                                                                                                                                                                                           | PAGES CHANGED |
|-----------------|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 0               | 1/05          | Initial release.                                                                                                                                                                                                      | —             |
| 1               | 2/05          | Changed Digital Temp Sensor Output from $\pm 2^{\circ}\text{C}$ to $\pm 3^{\circ}\text{C}$ .                                                                                                                          | 1, 3          |
|                 |               | Updated <i>Typical Operating Circuit</i> .                                                                                                                                                                            | 1             |
|                 |               | Changed $T_A = -40^{\circ}\text{C}$ to $+85^{\circ}\text{C}$ to $T_A = T_{\text{MIN}}$ to $T_{\text{MAX}}$ .                                                                                                          | 2, 3, 4       |
|                 |               | Updated <i>Block Diagram</i> .                                                                                                                                                                                        | 8             |
| 2               | 6/05          | Added “UL Recognized” to Features; added lead-free packages and removed S from top mark info in <i>Ordering Information</i> table; added ground connections to the N.C. pin in the <i>Typical Operating Circuit</i> . | 1             |
|                 |               | Added “noncondensing” to operating temperature range; changed $V_{\text{PF}}$ MIN from 2.35V to 2.45V.                                                                                                                | 2             |
|                 |               | Added aging offset specification.                                                                                                                                                                                     | 3             |
|                 |               | Relabeled TOC4.                                                                                                                                                                                                       | 7             |
|                 |               | Added arrow showing input on X1 in the <i>Block Diagram</i> .                                                                                                                                                         | 8             |
|                 |               | Updated pin descriptions for $V_{\text{CC}}$ and $V_{\text{BAT}}$ .                                                                                                                                                   | 9             |
|                 |               | Added the I <sup>2</sup> C Interface section.                                                                                                                                                                         | 10            |
|                 |               | <i>Figure 1</i> : Added sign bit to aging and temperature registers; added MSB and LSB.                                                                                                                               | 11            |
|                 |               | Corrected title for rate select bits frequency table.                                                                                                                                                                 | 13            |
|                 |               | Added note that frequency stability over temperature spec is with aging offset register = 00h; changed bit 7 from Data to Sign (Crystal Aging Offset Register).                                                       | 14            |
|                 |               | Changed bit 7 from Data to Sign (Temperature Register); correct pin definitions in <i>I<sup>2</sup>C Serial Data Bus</i> section.                                                                                     | 15            |
|                 |               | Modified the <i>Handling</i> , <i>PC Board Layout</i> , and <i>Assembly</i> section to refer to J-STD-020 for reflow profiles for lead-free and leaded packages.                                                      | 17            |
| 3               | 11/05         | Changed lead-free packages to RoHS-compliant packages.                                                                                                                                                                | 1             |
| 4               | 10/06         | Changed RST and UL bullets in <i>Features</i> .                                                                                                                                                                       | 1             |
|                 |               | Changed EC condition “ $V_{\text{CC}} > V_{\text{BAT}}$ ” to “ $V_{\text{CC}} = \text{Active Supply}$ (see Table 1).”                                                                                                 | 2, 3          |
|                 |               | Modified Note 12 to correct $t_{\text{REC}}$ operation.                                                                                                                                                               | 6             |
|                 |               | Added various conditions text to TOCs 1, 2, and 3.                                                                                                                                                                    | 7             |
|                 |               | Added text to pin descriptions for 32kHz, $V_{\text{CC}}$ , and RST.                                                                                                                                                  | 9             |
|                 |               | Table 1: Changed column heading “Powered By” to “Active Supply”; changed “applied” to “exceeds $V_{\text{PF}}$ ” in the <i>Power Control</i> section.                                                                 | 10            |
|                 |               | Indicated BBSQW applies to both SQW and interrupts; simplified temp convert description (bit 5); added “output” to INT/SQW (bit 2).                                                                                   | 13            |
|                 |               | Changed the <i>Crystal Aging</i> section to the <i>Aging Offset</i> section; changed “this bit indicates” to “this bit controls” for the enable 32kHz output bit.                                                     | 14            |
| 5               | 4/08          | Added Warning note to EC table notes; updated Note 12.                                                                                                                                                                | 6             |
|                 |               | Updated the <i>Typical Operating Characteristics</i> graphs.                                                                                                                                                          | 7             |
|                 |               | In the <i>Power Control</i> section, added information about the POR state of the time and date registers; in the <i>Real-Time Clock</i> section, added to the description of the RST function.                       | 10            |
|                 |               | In Figure 1, corrected the months date range for 04h from 00–31 to 01–31.                                                                                                                                             | 11            |

## Revision History (continued)

| REVISION NUMBER | REVISION DATE | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | PAGES CHANGED                    |
|-----------------|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| 6               | 10/08         | Updated the <i>Typical Operating Circuit</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1                                |
|                 |               | Removed the $V_{PU}$ parameter from the <i>Recommended DC Operating Conditions</i> table and added verbiage about the pullup to the <i>Pin Description</i> table for INT/SQW, SDA, and SCL.                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2, 9                             |
|                 |               | Added the Delta Time and Frequency vs. Temperature graph in the <i>Typical Operating Characteristics</i> section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7                                |
|                 |               | Updated the <i>Block Diagram</i> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 8                                |
|                 |               | Added the <i>V<sub>BAT</sub> Operation</i> section, improved some sections of text for the <i>32kHz TCXO</i> and <i>Pushbutton Reset Function</i> sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 10                               |
|                 |               | Added the register bit POR values to the register tables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 13, 14, 15                       |
|                 |               | Updated the <i>Aging Offset</i> and <i>Temperature Registers (11h–12h)</i> sections.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 14, 15                           |
|                 |               | Updated the I <sup>2</sup> C timing diagrams (Figures 3, 4, and 5).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 16, 17                           |
| 7               | 3/10          | Removed the “S” from the top mark in the <i>Ordering Information</i> table and the <i>Pin Configuration</i> to match the packaging engineering marking specification.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1, 18                            |
| 8               | 7/10          | Updated the <i>Typical Operating Circuit</i> ; removed the “Top Mark” column from the <i>Ordering Information</i> ; in the <i>Absolute Maximum Ratings</i> section, added the theta-JA and theta-JC thermal resistances and Note 1, and changed the soldering temperature to +260°C (lead(Pb)-free) and +240°C (leaded); updated the functional description of the V <sub>BAT</sub> pin in the <i>Pin Description</i> ; changed the timekeeping registers 02h, 09h, and 0Ch to “20 Hour” in Bit 5 of Figure 1; updated the BBSQW bit description in the <i>Control Register (0Eh)</i> section; added the land pattern no. to the <i>Package Information</i> table. | 1, 2, 3, 4, 6, 9, 11, 12, 13, 18 |
| 9               | 1/13          | Updated <i>Absolute Maximum Ratings</i> , and last paragraph in <i>Power Control</i> section                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2, 10                            |
| 10              | 3/15          | Revised <i>Benefits and Features</i> section.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                |

For pricing, delivery, and ordering information, please contact Maxim Direct at 1-888-629-4642, or visit Maxim Integrated's website at [www.maximintegrated.com](http://www.maximintegrated.com).

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# ANNEXURE II

## SUMMARY OF RESEARCH PAPERS

The reviewed research papers provide a comprehensive understanding of modern thermal management systems and their critical role in handling increasing heat loads in high-performance environments such as AI data centers, industrial systems, and advanced electronic devices. The studies collectively emphasize that traditional cooling methods, particularly air cooling, are becoming insufficient due to their lower heat transfer efficiency and inability to manage compact high-density thermal loads effectively.

Several papers focused on **liquid cooling technologies**, showing that they offer significantly higher thermal efficiency compared to air cooling by directly absorbing heat from critical components. Research on **dual-loop and multi-stage cooling systems** demonstrated that separating heat absorption and heat rejection into independent loops improves temperature stability, operational flexibility, and fault isolation. These architectures are widely adopted in industrial cooling systems and AI server infrastructures because they reduce thermal bottlenecks and improve reliability.

A major area of study was **heat exchanger optimization**, where researchers highlighted the importance of surface area, coolant flow rate, material properties, and channel design in improving heat transfer efficiency. Experimental studies on coolant flow in multi-loop systems showed that balanced flow distribution is essential for maintaining stable thermal performance.

Several papers also explored **embedded monitoring and IoT-based thermal management**, validating the integration of sensors, microcontrollers, and real-time monitoring systems for predictive maintenance and automated fault detection. This directly supports the use of ESP32-S3, thermocouples, and flow sensors in the project.

Research on **sustainable cooling systems** further showed that efficient liquid cooling architectures can significantly reduce power consumption in data centers, improving overall Power Usage Effectiveness (PUE) and reducing environmental impact. Additional studies on nanofluids and advanced coolant technologies introduced future opportunities for improving thermal conductivity and system efficiency.

Overall, the literature strongly validates the technical feasibility of the proposed **Dual-Loop Controlled Cooling System**, proving that liquid-based multi-stage cooling is a practical, scalable, and energy-efficient solution. These studies provide the theoretical foundation, industrial relevance, and design guidance for developing a low-cost, modular, and intelligent thermal management prototype for educational and experimental applications.

# ANNEXURE III

## AI- INTEGRATION

## Use of AI tools at various stages of project:

| Stage                                                      | AI tool        | Purpose                                                 | Generate output                                                           | Team verification                                                                                              |
|------------------------------------------------------------|----------------|---------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>User research questionnaire</b>                         | <b>ChatGPT</b> | <b>Finding relevant type of questions to ask people</b> | Prepared a user questionnaire and collected user requirements.            | Questionnaire content reviewed, user requirements validated, and questions approved by the project guide/team. |
| <b>Identifying the user (primary, secondary, tertiary)</b> | <b>ChatGPT</b> | <b>Finding the type of people in that sector</b>        | Classified users into primary, secondary, and tertiary categories         | User categories verified by comparing with the target audience and project scope.                              |
| <b>Literature survey</b>                                   | <b>ChatGPT</b> | <b>Find Research papers</b>                             | Compiled relevant research papers and summarized key findings.            | Research papers checked for authenticity, relevance, and consistency with project objectives.                  |
| <b>Prior art search</b>                                    | <b>Gemini</b>  | <b>Image Generation</b>                                 | Generated a conceptual image based on patent analysis and project design. | Patent references and conceptual design verified to ensure originality and avoid duplication.                  |
| <b>Ideation</b>                                            | <b>ChatGPT</b> | <b>Improving the initial idea</b>                       | Developed a refined project idea with improved features.                  | Refined project idea evaluated for feasibility, innovation, and alignment with user requirements.              |
| <b>System design</b>                                       | <b>ChatGPT</b> | <b>Design Efficient System</b>                          | Generated the circuit schematic with required components and connections. | System architecture and component selection reviewed for technical feasibility.                                |

|                       |                |                            |                                                                           |                                                                                                                          |
|-----------------------|----------------|----------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Circuit design</b> | <b>ChatGPT</b> | <b>Design Circuits</b>     | Generated the circuit schematic with required components and connections. | Circuit schematic, component selection, and system architecture verified for correctness and feasibility.                |
| <b>Documentation</b>  | <b>ChatGPT</b> | <b>Document Everything</b> | Produced the complete project documentation in a structured format.       | Project documentation reviewed for technical accuracy, completeness, formatting, and compliance with project guidelines. |

### List of the AI Tools and the prompt

#### 1. AI tool used: ChatGPT

Prompt:

Design and engineer a complete Dual-Loop Controlled Cooling System prototype inspired by industrial thermal architectures used in AI data centers, high-performance computing, and advanced thermal management systems.

Core Problem

Traditional cooling systems (air cooling and single-loop cooling) struggle to manage increasing thermal loads in high-density systems. This causes overheating, lower efficiency, increased operational cost, and reduced component life. The goal is to create a low-cost modular prototype that demonstrates how industrial dual-loop cooling architectures solve this problem.

System Objective

Build a real-time monitored dual-loop thermal management system that:

generates controlled heat

transfers heat efficiently

separates heat absorption and heat rejection

provides fault isolation



allows real-time sensor monitoring

demonstrates industrial-scale cooling principles at lab scale

Final Physical Architecture

Heat Generation Unit

Use:

Stainless steel cylindrical chamber

External nichrome wire heating jacket directly wound around cylinder

Fiberglass tape as insulation between nichrome and steel

Outer fiberglass/high-temp tape layer for heater retention

RTV silicone spot anchoring only

NO aluminum sphere.

Direct cylinder heating only.

Thermal stack:

Steel cylinder → fiberglass insulation → nichrome heater → fiberglass retention wrap

Cooling Architecture

Primary Loop (Heat Absorption Loop)

Flow:

Cylinder chamber → Pump 1 (D385) → Flow sensor (ZJ-S201) → Liquid-to-liquid heat exchanger  
→ back to chamber

Purpose:

Absorb heat directly from heated cylinder.

Coolant:

Silicone oil / water / machine coolant (experimental)

Secondary Loop (Heat Rejection Loop)

Flow:

Reservoir → Pump 2 (D385) → Heat exchanger secondary side → Cooling block/radiator → 12V fan → Reservoir

Purpose:

Reject heat into ambient.

Electronics Architecture

Controller:

ESP32-S3 (Olimex)

Power:

12V 50A SMPS

Buck:

LM2596 (12V → 5V)

Sensors

3x MAX6675 + K-Type thermocouples:

T1 = Cylinder wall temp

T2 = Primary loop temp

T3 = Secondary loop temp

1x ZJ-S201 flow sensor

1x DS3231 RTC

1x 16x2 I2C LCD

Actuators

Relay 1 → Primary pump

Relay 2 → Secondary pump

Relay 3 → Cooling fan

Relay 4 → Heater master cut-off

MOSFET:

IRFZ44N for nichrome heater switching

Buzzer:

Fault indication

RGB LED:

System status indication

Push button:

Reset / acknowledge

Control Logic

Phase 1:

Initialize:

ESP32

Sensors

LCD

RTC

Relays

Pumps

Fan

Phase 2:

Activate nichrome heater

Read T1 continuously

Phase 3:

Start primary pump

Check flow sensor

Heat absorbed

Phase 4:

Start secondary pump

Activate cooling fan

Heat exchanger transfers thermal energy

Phase 5:

Monitor:

T1

T2

T3

Flow rate

Display all values on LCD

Phase 6:

Safety decision:

IF:

Overtemperature

No flow

Pump failure

THEN:

Heater OFF

Buzzer ON

RGB RED

Maintain pumps

ELSE:

continue cycle

Deliverables Required

Generate:

Complete block diagram

System architecture diagram

Full wiring table

Power distribution table

Flowchart diagram

Feasibility study

Market analysis

Patent benchmarking

Literature review (20 papers IEEE format)

Literature explanation (100 words each)

Literature summary

Customer research

Empathy map

Customer journey map

Risk assessment

Cost estimation

Revenue model

Target market analysis

Regulatory compliance list

Full component documentation

SDG mapping

Poster presentation content

A3 technical poster

LinkedIn article content

Professional abstract

Methodology writeup

Conclusion

Engineering Constraints

Heater must not short to cylinder

Fiberglass insulation mandatory

No M-seal permanent thermal contact

RTV only as anchor

Type-C power input ONLY through regulated 5V (never direct 12V)

Common grounding across all modules

Pumps and fan powered directly from 12V relay-switched lines

ESP32 only handles logic

### **Core Innovation Positioning**

This is NOT inventing dual-loop cooling.

#### **Position it as:**

“Low-cost modular educational and experimental implementation of industrial dual-loop thermal architecture for thermal studies, AI data center cooling simulation, and embedded thermal control research.”

That is the actual novelty.

Not the physics.